# PHYS6751: Nuclear Graduate Lab
# Lab manual v24.5

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Do not print this manual all at once, as its content is likely to be updated regularly. Only print the chapter relevant to your work. Most of the text that appears in green in this document is a hyper-reference (you can click on it).

## Contents

**1 Alpha Spectroscopy** **3**
1.1 Physical Concepts . . . 3
1.2 Experimental setup . . . 8
1.3 Experimental goals and Physical law under study . . . 9
1.4 Preliminary questions . . . 12

**2 Cosmic muons** **13**
2.1 Muon from cosmic rays . . . 13
2.2 Experimental setup . . . 16
2.3 Experimental goals and law under study . . . 20
2.4 Preliminary questions . . . 20

**3 Gamma-ray Spectroscopy with Scintillators** **23**
3.1 Interaction of Gamma Rays and Charged Particles with Nuclear Matter . . . 23
3.2 Experimental setup . . . 27
3.3 Experimental goals . . . 34
3.4 Preliminary questions . . . 34

**4 Gamma-ray spectroscopy with Ge(Li) and Abundance of Uranium Isotopes** **37**
4.1 Physical concept . . . 37

4.2 Experimental setup . . . 39
4.3 Experimental goals and law under study . . . 44
4.4 Preliminary questions . . . 45

**5 Simulation of Particle Interactions for Radiotherapy 47**
5.1 Physical concept . . . 47
5.2 Experimental Setup . . . 48
5.3 Experiment goals and law under study . . . 52
5.4 Preliminary questions . . . 53

**6 Rutherford Scattering 55**
6.1 Physical concept . . . 55
6.2 Experimental setup . . . 62
6.3 Experiment goals and law under study . . . 68
6.4 Preliminary questions . . . 68

**A Edwards Accelerator Data Acquisition System – Background Information 71**

**B General Characteristic of Detectors 73**
B.1 Sensitivity . . . 73
B.2 Detector Response Function . . . 73
B.3 Energy resolution . . . 74
B.4 Detector Efficiency . . . 74
B.5 Response Time . . . 75
B.6 Dead time . . . 75

**C Counting statistics 76**
C.1 Random events . . . 76
C.2 Statistical distributions . . . 76

**D Silicon Detectors 79**

**E Germanium Detectors 81**

**F Scintillators 84**

**G Photomultipliers 85**

# 1 Alpha Spectroscopy

In this experiment, you will learn about $\alpha$-particle spectroscopy techniques using a solid-state detector. You will study $\alpha$ particles from a source that was originally pure $^{226}$Ra. You will estimate when the source was made.

## 1.1 Physical Concepts

### 1.1.1 Decay chains

*This section is copied from the Wikipedia page on "Radioactive Decay Chains."*

In nuclear science, the decay chain refers to a series of radioactive decays of different radioactive decay products as a sequential series of transformations. It is also known as a "radioactive cascade." Most radioisotopes do not decay directly to a stable state but rather undergo a series of decays until, eventually, a stable isotope is reached.

Decay stages are referred to by their relationship to previous or subsequent stages. A parent isotope undergoes decay to form a daughter isotope. One example is uranium (atomic number 92) decaying into thorium (atomic number 90). The daughter isotope may be stable or decay to form a daughter isotope of its own. The daughter of a daughter isotope is sometimes called a granddaughter isotope.

The time it takes for a single parent atom to decay to an atom of its daughter isotope can vary widely, not only between different parent-daughter pairs but also randomly between identical pairings of parent and daughter isotopes. The decay of every single atom occurs spontaneously, and the decay of an initial population of identical atoms over time t follows a decaying exponential distribution, $\exp(-\lambda t)$, where $\lambda$ is called a decay constant. One of the properties of an isotope is its half-life, the time by which half of an initial number of identical parent radioisotopes have decayed to their daughters, which is inversely related to $\lambda$. Half-lives have been determined in laboratories for many radioisotopes (or radionuclides). These can range from nearly instantaneous (less than $10^{-21}$ seconds) to more than $10^{19}$ years.

The intermediate stages each emit the same amount of radioactivity as the original radioisotope (i.e., there is a one-to-one relationship between the numbers of decay in successive stages), but each stage releases a different quantity of energy. If and when equilibrium is achieved, each successive daughter isotope is present in direct proportion to its half-life; but since its activity is inversely proportional to its half-life, each nuclide in the decay chain finally contributes as many individual transformations as the head of the chain, though not the same energy. For example, uranium-238 is weakly radioactive, but pitchblende, a uranium ore, is 13 times more radioactive than pure uranium metal because of the radium and other daughter isotopes it contains. Not only are unstable radium isotopes significant radioactivity emitters, but as the next stage in the decay chain, they also generate radon, a heavy, inert, naturally occurring radioactive gas. Rock containing thorium and/or uranium (such as some granites) emits radon gas that can accumulate in enclosed places such as

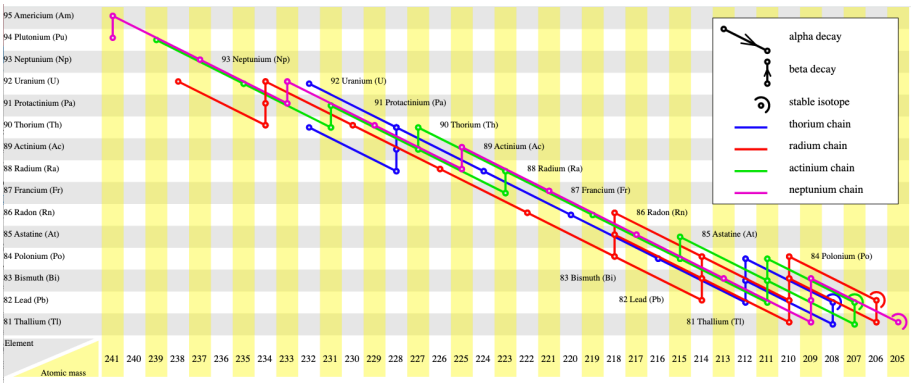



Figure 1: *Decay chains discussed in the text: thorium (4n, in blue), neptunium (4n+1, in pink), radium (4n+2, in red), and actinium (4n+3, in green).*

basements or underground mines. The quantity of isotopes in the decay chains at a certain time is calculated with the Bateman equation (see further in the text).

The four most common modes of radioactive decay are alpha decay, beta decay, inverse beta decay (considered as both positron emission and electron capture), and isomeric transition. Of these decay processes, only alpha decay changes the atomic mass number (A) of the nucleus and always decreases it by four. Because of this, almost any decay will result in a nucleus whose atomic mass number has the same residue mod 4, dividing all nuclides into four chains. The members of any possible decay chain must be drawn entirely from one of these classes. All four chains also produce helium-4 (alpha particles are helium-4 nuclei).

Three main decay chains (or families) are observed in nature, commonly called the thorium series, the radium or uranium series, and the actinium series (see figure 1) representing three of these four classes, and ending in three different, stable isotopes of lead. The mass number of every isotope in these chains can be represented as A = 4n, A = 4n + 2, and A = 4n + 3, respectively. The long-lived starting isotopes of these three isotopes, respectively thorium-232, uranium-238, and uranium-235, have existed since the earth's formation, ignoring the artificial isotopes and their decays since the 1940s.

Due to the relatively short half-life of its starting isotope neptunium-237 (2.14 million years), the fourth chain, the neptunium series with A = 4n + 1, is already extinct, except for the final rate-limiting step, decay of bismuth-209. Traces of 237Np and its decay products still occur in nature, however, due to neutron capture in uranium ore. The ending isotope of this chain is now known to be thallium-205. Some older sources give the final isotope as bismuth-209, but it was recently discovered that it is very slightly radioactive, with a half-life of 2.01×1019 years.

There are also non-transuranic decay chains of unstable isotopes of light elements, for example, those of magnesium-28 and chlorine-39. On Earth, most of the starting isotopes of these chains before 1945 were generated by cosmic radiation. Since 1945, the testing and use of nuclear weapons have also released numerous radioactive fission products. Almost all such isotopes decay by either $\beta-$ or $\beta+$ decay modes, changing from one element to another

without changing atomic mass. These later daughter products, being closer to stability, generally have longer half-lives until they finally decay into stability.

### 1.1.2 Alpha decay

*This section is copied from "Introductory Nuclear Physics 3rd Edition" by K. Krane, ISBN-13:978-0471805533*
Alpha($\alpha$) particles were first identified as the least penetrating of the radiations emitted by naturally occurring materials. The spontaneous emission of an $\alpha$ particle can be represented by the following process:

$$ {}^{A}_{Z}X_N \rightarrow {}^{A-4}_{Z-2}X_{N-2} + \alpha \tag{1} $$

The $\alpha$ particle, as was shown by Rutherford, is a nucleus of $^4$He, consisting of two neutrons and two protons. In 1903, Rutherford measured their charge-to-mass ratio by deflecting particles from the decay of radium in electric and magnetic fields. Despite the difficulty of these early experiments, Rutherford's result was only about 25% higher than the presently accepted value. In 1909 Rutherford showed that, as suspected, the $\alpha$ particles were, in fact, helium nuclei; in his experiments, the particles entered an evacuated thin-walled chamber by penetrating its walls and after several days of collecting atomic spectroscopy revealed the presence of helium gas inside the chamber. Many heavy nuclei, especially those of the naturally occurring radioactive series, decay through $\alpha$ emission. Only exceedingly rarely does any other spontaneous radioactive process result in the emission of nucleons; we do not, for example, observe deuteron emission as a natural decay process.

### 1.1.3 Determining the 'age' of a $^{226}$Ra source

*This section is copied from a document written by Prof. Brune*
Assuming that one starts with pure $^{226}$Ra, it is possible to determine its "age" (defined as the time since it started its life in a pure state) by observing the decay of its daughter products. First note that all the decays half-life are very short (less than one week), except for two: the initial $^{226}$Ra decay ($t_{1/2}$ =1600 years) and the $^{210}$Pb decay ($t_{1/2}$=22.3 years). Therefore when a $^{226}$Ra isotope decays it "shortly" becomes a $^{210}$Pb. This paragraph will show how by measuring the decay rate of $^{210}$Pb, one can measure the age of $^{226}$Ra. Figure 2 shows that $^{226}$Ra is a daughter of $^{238}$U. In the experiment you will perform, you will be able to measure most of the alpha particles noted in figure 2.

The number of $^{226}$Ra atoms $N_A$ present in the source is a function of time $t$ and it is described by

$$ N_A = N_0 \exp(-t/\tau_A) \tag{2} $$

where $N_0 = N_A(0)$ and $\tau_A$ is the mean lifetime of $^{226}$Ra given by $t_{1/2}/\ln 2$. The decay rate of $^{226}$Ra is given by

$$ -\frac{dN_A}{dt} = N_A/\tau_A \tag{3} $$

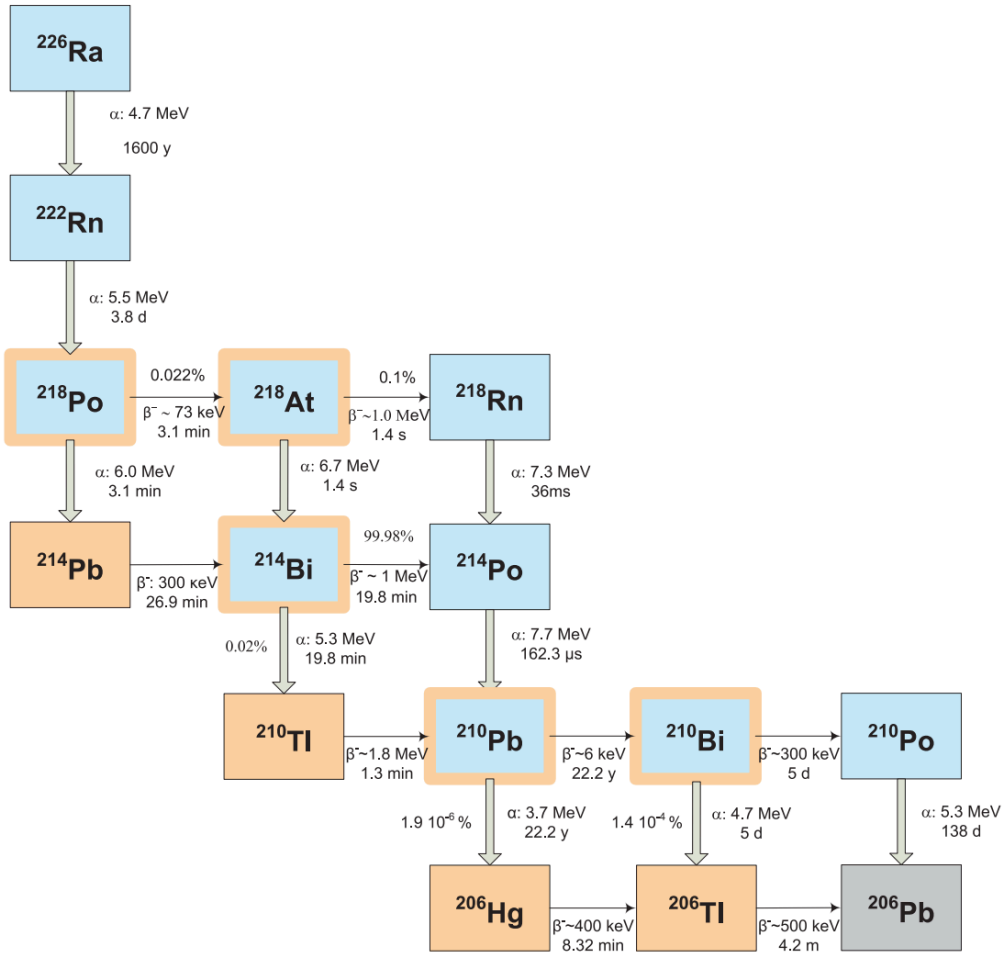



Figure 2: *The decay series of* $^{226}Ra$ *showing kinetic energies of decay product, the half-life of the transitions, and the probability of each decay. Assume that all possible decays of a specific nucleus are shown on this graph. The isotopes in blue boxes undergo* $\alpha$ *decays. The isotopes in orange boxes undergo* $\beta$ *decay. The isotopes in blue boxes with orange borders are* $\alpha$ *and* $\beta$ *emitters. This figure is copied from 10.1016/j.apradiso.2017.04.029*

Assume $t = 0$ when the source is made. Note that $\frac{dN_a}{dt}$ is a quantity that you measure in the laboratory and can be used to determine $N_A$.

Once the source is several weeks/months or years old, an equilibrium is established. Consider for example $^{222}$Rn. In equilibrium, this isotope is produced at essentially the same rate as which it decays - this must be the case because every $^{222}$Rn which is created decays shortly after. This same argument can be applied to all of the other isotopes in the decay chain which have short half-lives (i.e. less than a week). Unfortunately, these decays don't tell us anything about the age of the source.
For $^{210}$Pb and $^{206}$Pb, we have to be a little bit more careful. Once the source is several weeks old, $^{210}$Pb is produced at the same rate as $^{226}$Ra decays. However, $^{210}$Pb decays with a 22.3-year half-life. We can describe the number of $^{210}$Pb atoms $N_B$ as a function of time via:

$$\frac{dN_B}{dt} = -\frac{dN_A}{dt} - \frac{N_B}{\tau_B} = \frac{N_A}{\tau_A}\exp(-t/\tau_A) - \frac{N_B}{\tau_B} \tag{4}$$

The first term in this equation describes the creation of $N_B$ by the decay of parent nuclei. The second term describes the decay of $N_B$ due to its own half-life ($\tau_B$ is the mean lifetime of $^{210}$Pb). This differential equation can be solved by "guessing" the answer:

$$N_B = C\exp(-t/\tau_A) + D\exp(-t/\tau_B) \tag{5}$$

where $C$ and $D$ are arbitrary constants. If we start with a pure $^{226}$Ra source that means we have zero $^{210}$Pb initially so we have $N_B(t = 0) = 0$, i.e. $C = -D$. We can determine $C$ by rewriting Equation 4 as :

$$N_B = \frac{N_0}{\tau_A/\tau_B - 1}[\exp(-t/\tau_A) - exp(-t/\tau_B)]. \tag{6}$$

The quantity $N_B$ *does* tell us something about the age of the source. Unfortunately, neither $N_B$ nor its decay rate can be measured directly with our setup because $^{210}$Pb decays by $\beta$ emission. We can apply the equilibrium argument to decays following $^{210}$Pb. In particular, the decay rate of $^{210}$Po is equal to the decay rate of $^{210}$Pb once the source is several months old. The nucleus $^{210}$Po decays by $\alpha$ emission - something we can measure. In addition the $^{210}$Po decay rate is just given by $N_B/\tau_B$ - something we can calculate from Equation 6.

Taking the ratio of equations 3 and 6, one obtains:

$$R = \frac{N_B\tau_A}{N_A\tau_B} = \frac{\tau_A}{\tau_A - \tau_B}[1 - \exp(-t/\tau_B + t/\tau_A)] \tag{7}$$

where A is for the initial isotope and B is the daughter isotope.This expression is a simple expression of the Bateman equation. **When using this equation, you need to remember that according to the definition of Equation 3, $N_X/\tau_X = -\frac{dN_X}{dt}$ is the decay rate of atom $X$. $N_X$ is the number of atom $X$ in your sample.**

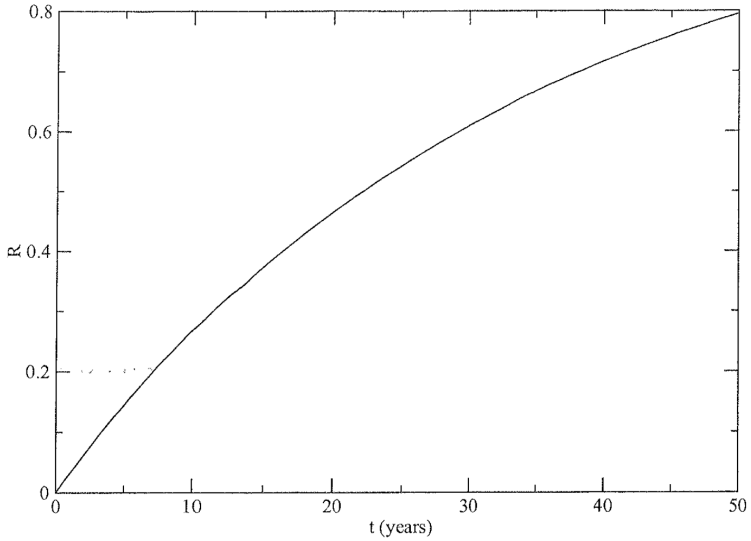



Figure 3: *Ratio $R$ of the decay rates of $^{210}Pb$ atoms over $^{226}Ra$ atoms (see equation 7) as a function of time.*

The ratio $R$ between $^{210}$Pb (B) and $^{226}$Ra (A) as a function of the age $t$ is shown in Figure 3. Note that the ratio $R$ is also equal to the measured ratio of $^{210}$Po (B) and $^{226}$Ra (A) $\alpha$ decays. Equation 7 can be inverted to find the age $t$:

$$t = -\frac{\tau_A \tau_B}{\tau_A - \tau_B} \ln(1 - R\frac{\tau_A - \tau_B}{\tau_A}) \tag{8}$$

### 1.1.4 To go further

- The "p", "r" and the "s" processes are astro-nuclear chains responsible for the creation of heavy nuclei. What are they?

- What are everyday applications of $\alpha$ decay?

## 1.2 Experimental setup

In this experiment, a Ra source is placed in an evacuated chamber facing a Silicon detector. You may find Background Information on Silicon Detectors in Appendix D. The signal is amplified and then digitized by an Analog to Digital Converter (ADC) and readout by an OHIO University-developed Data AcQuisition (DAQ) system (see figure 4). Figures 5 and 6 are copies of the technical sheets for the Radium source and the Silicon detector.

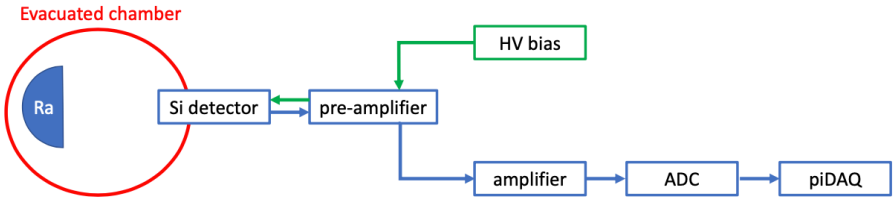



Figure 4: *Experimental Setup for the $\alpha$ Spectroscopy experiment.*

### 1.2.1 Operation notes

- The chamber in which the source and the detector are located must be evacuated before starting the measurements. Do not leave the high-voltage (HV) bias on when either pumping out or venting the chamber. It is OK to turn on the HV when the pump is in use to keep a good vacuum. The high-voltage bias also needs to be off when the Si detector is exposed to light.

- The HV supply is on the shelf below the table on which the detector stands. Increase the HV slowly (a few tens of volts at a time) while keeping an eye on the current drawn through the detector. As you increase the high voltage in steps, the current will rise and then settle at a lower value. Keep the current below 1 $\mu$ A. The nominal HV is $\sim$ 75V but you should choose an HV value that maintains the current below 1 $\mu$ A.

- Do not touch the Si part of the detector with your bare hands or you risk damaging it.

- The Radium source is attached to the lid of the vacuum chamber. When opening the chamber make sure to put the lid in a safe place away from you and other physicists in the room. Practice the ALARA principle.

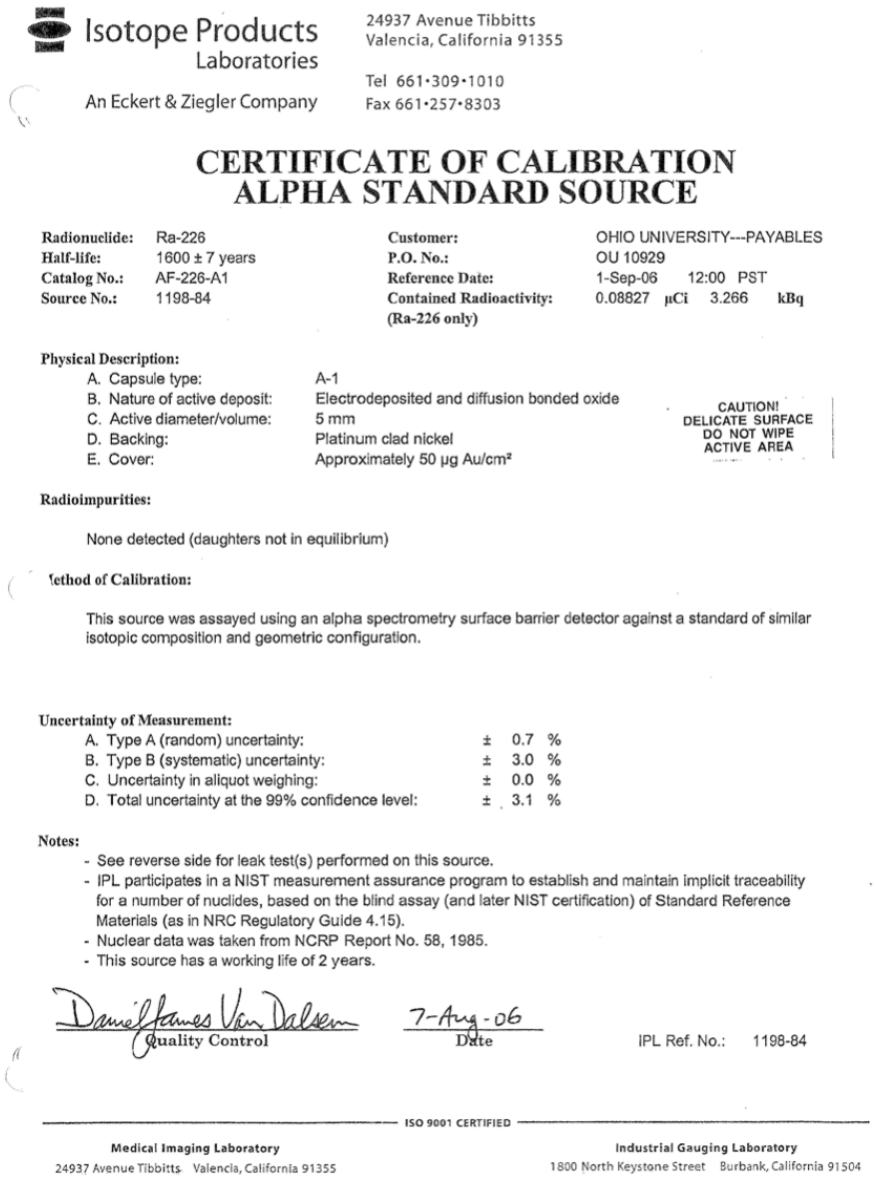
## 1.3 Experimental goals and Physical law under study

The physical law under study is "nuclear decay chains".
The goals of this experiment are

- to determine the activity of the source, and to estimate the"age" of the Radium source using the technique described in Section 1.1.3.

- to save a scope picture of the relative timing of the strobe and the energy signal.

- to confirm that the statistical precision on a measurement of N counts is $\sqrt{N}$. For this goal, use repeated measurements of your spectrum. Evaluate the RMS of the variation of the number of counts for one peak and for your whole spectrum.



Isotope Products Laboratories
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24937 Avenue Tibbitts
Valencia, California 91355

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Fax 661•257•8303

**CERTIFICATE OF CALIBRATION**
**ALPHA STANDARD SOURCE**

| | | | |
|---|---|---|---|
| **Radionuclide:** | Ra-226 | **Customer:** | OHIO UNIVERSITY---PAYABLES |
| **Half-life:** | 1600 ± 7 years | **P.O. No.:** | OU 10929 |
| **Catalog No.:** | AF-226-A1 | **Reference Date:** | 1-Sep-06 12:00 PST |
| **Source No.:** | 1198-84 | **Contained Radioactivity: (Ra-226 only)** | 0.08827 µCi 3.266 kBq |

**Physical Description:**

| | | |
|---|---|---|
| A. Capsule type: | A-1 | |
| B. Nature of active deposit: | Electrodeposited and diffusion bonded oxide | |
| C. Active diameter/volume: | 5 mm | |
| D. Backing: | Platinum clad nickel | |
| E. Cover: | Approximately 50 µg Au/cm² | |

CAUTION!
DELICATE SURFACE
DO NOT WIPE
ACTIVE AREA

**Radioimpurities:**

None detected (daughters not in equilibrium)

**Method of Calibration:**

This source was assayed using an alpha spectrometry surface barrier detector against a standard of similar isotopic composition and geometric configuration.

**Uncertainty of Measurement:**

| | | |
|---|---|---|
| A. Type A (random) uncertainty: | ± | 0.7 % |
| B. Type B (systematic) uncertainty: | ± | 3.0 % |
| C. Uncertainty in aliquot weighing: | ± | 0.0 % |
| D. Total uncertainty at the 99% confidence level: | ± | 3.1 % |

**Notes:**

- See reverse side for leak test(s) performed on this source.
- IPL participates in a NIST measurement assurance program to establish and maintain implicit traceability for a number of nuclides, based on the blind assay (and later NIST certification) of Standard Reference Materials (as in NRC Regulatory Guide 4.15).
- Nuclear data was taken from NCRP Report No. 58, 1985.
- This source has a working life of 2 years.

Daniel James Van Dalsem
Quality Control

7-Aug-06
Date

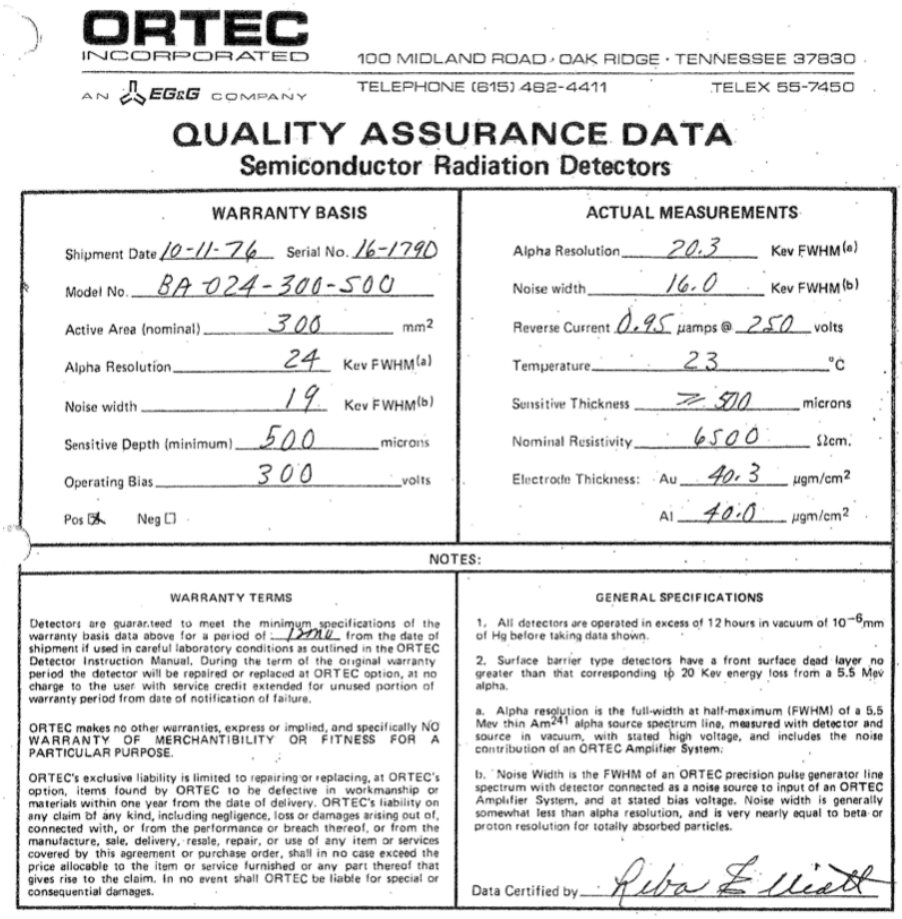
IPL Ref. No.: 1198-84

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**Medical Imaging Laboratory**
24937 Avenue Tibbitts Valencia, California 91355

**Industrial Gauging Laboratory**
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Figure 5: *Technical specifications of the Ra source used in the $\alpha$ Spectroscopy experiment.*



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## QUALITY ASSURANCE DATA
### Semiconductor Radiation Detectors

**WARRANTY BASIS**

Shipment Date 10-11-76 Serial No. 16-1790
Model No. BA-024-300-500
Active Area (nominal) 300 mm²
Alpha Resolution 24 Kev FWHM(a)
Noise width 19 Kev FWHM(b)
Sensitive Depth (minimum) 500 microns
Operating Bias 300 volts
Pos ☒ Neg ☐

**ACTUAL MEASUREMENTS**

Alpha Resolution 20.3 Kev FWHM(a)
Noise width 16.0 Kev FWHM(b)
Reverse Current 0.95 μamps @ 250 volts
Temperature 23 °C
Sensitive Thickness ≥ 500 microns
Nominal Resistivity 6500 Ωcm.
Electrode Thickness: Au 40.3 μgm/cm²
Al 40.0 μgm/cm²

NOTES:

**WARRANTY TERMS**

Detectors are guaranteed to meet the minimum specifications of the warranty basis data above for a period of one year from the date of shipment if used in careful laboratory conditions as outlined in the ORTEC Detector Instruction Manual. During the term of the original warranty period the detector will be repaired or replaced at ORTEC option, at no charge to the user with service credit extended for unused portion of warranty period from date of notification of failure.

ORTEC makes no other warranties, express or implied, and specifically NO WARRANTY OF MERCHANTIBILITY OR FITNESS FOR A PARTICULAR PURPOSE.

ORTEC's exclusive liability is limited to repairing or replacing, at ORTEC's option, items found by ORTEC to be defective in workmanship or materials within one year from the date of delivery. ORTEC's liability on any claim of any kind, including negligence, loss or damages arising out of, connected with, or from the performance or breach thereof, or from the manufacture, sale, delivery, resale, repair, or use of any item or services covered by this agreement or purchase order, shall in no case exceed the price allocable to the item or service furnished or any part thereof that gives rise to the claim. In no event shall ORTEC be liable for special or consequential damages.

**GENERAL SPECIFICATIONS**

1. All detectors are operated in excess of 12 hours in vacuum of $10^{-6}$ mm of Hg before taking data shown.

2. Surface barrier type detectors have a front surface dead layer no greater than that corresponding to 20 Kev energy loss from a 5.5 Mev alpha.

a. Alpha resolution is the full-width at half-maximum (FWHM) of a 5.5 Mev thin $Am^{241}$ alpha source spectrum line, measured with detector and source in vacuum, with stated high voltage, and includes the noise contribution of an ORTEC Amplifier System.

b. Noise Width is the FWHM of an ORTEC precision pulse generator line spectrum with detector connected as a noise source to input of an ORTEC Amplifier System, and at stated bias voltage. Noise width is generally somewhat less than alpha resolution, and is very nearly equal to beta or proton resolution for totally absorbed particles.

Data Certified by Reba E. Heath

Figure 6: *Technical specifications of the Si detector used in the α Spectroscopy experiment*

## 1.4 Preliminary questions

The following questions should help you prepare for this lab

1. The physical principle under study is nuclear decay chains. Write a brief 100 to 200 words description of that principle and how it relates to your measurements. This is in preparation for the introductory part of your report. Show your word count.

2. What are the five different kinetic energies of the $\alpha$ particles you are most likely to measure in this experiment? Information in figure 2 are useful to answer this question. Which alpha peaks do you need to evaluate to find the age of the Radium source?

3. In section 1.1.3, you read : *We can apply the equilibrium argument to decays following* $^{210}Pb$. *In particular, the decay rate of* $^{210}Po$ *is equal to the decay rate of* $^{210}Pb$ *once the source is a few years.* Using a computer graphing program, plot the ratio R (see equation 7) of $^{210}$Po to $^{210}$Pb versus time and use your plot to explain why the decay rate of $^{210}$Po is equal to the decay rate of $^{210}$Pb once the source is several years old. The half-life times of those two nuclei are listed in figure 2.

4. Suppose you measure 140 decays of element $^{226}Ra$ (mother) and 70 decays of element $^{210}Po$ (daughter) in one minute. What is the value of $R$? What is the precision $\Delta R$ on R? Using figure 3 as a guide, determine the age of the source based on your value of $R$. Using figure 3 as a guide, what is the precision of your determination of the age of the target? Hint: Remember that if you are measuring a quantity $C = A/B$, then $(\Delta C/C)^2 = (\Delta A/A)^2 + (\Delta B/B)^2$.

5. The alpha source you use in this measurement emits particles isotropically in 3D (like most sources). The Si detector you use doesn't detect $\alpha$ in 3D but only over a small surface $S$. Assume the detector is located at a distance $R$ from the source and you detected $N$ $\alpha$-particles in 1 second. What is the source activity? What is the unit of activity (in radioactive processes)? Hint: you might find the information in section B.4 useful to answer this question.

# 2 Cosmic muons

In this experiment, you will use a scintillator telescope to determine the angular distribution of muon cosmic rays in the Edwards laboratory and compare it to expectations from the literature. Then you will estimate the mean lifetime of the muons by measuring the time distribution between muons and the signal produced by their daughter electrons.

## 2.1 Muon from cosmic rays

*This section is copied from the 2016 Particle Data Group Review*

The cosmic radiation incident at the top of the terrestrial atmosphere includes all stable charged particles and nuclei with lifetimes of order $10^6$ years or longer. Technically, "primary" cosmic rays are those particles accelerated at astrophysical sources, and "secondaries" are those particles produced in the interaction of the primaries with interstellar gas (see Figure 7). Thus electrons, protons, and helium, as well as carbon, oxygen, iron, and other nuclei synthesized in stars, are primaries. Nuclei such as lithium, beryllium, and boron (which are not abundant end-products of stellar nucleosynthesis) are secondaries. Anti-protons and positrons are also in large part secondary. Whether a small fraction of these particles may be primary is a question of current interest. Apart from particles associated with solar flares, the cosmic radiation comes from outside the solar system. The incoming charged particles are "modulated" by the solar wind, the expanding magnetized plasma generated by the Sun, which decelerates and partially excludes the lower energy galactic cosmic rays from the inner solar system. There is a significant anti-correlation between solar activity (which has an alternating eleven-year cycle) and the intensity of the cosmic rays with energies below about 10 GeV. In addition, the lower-energy cosmic rays are affected by the geomagnetic field, which they must penetrate to reach the top of the atmosphere. Thus the intensity of any component of the cosmic radiation in the GeV range depends both on the location and time.

### 2.1.1 Angular distribution of cosmic muon at ground level

The flux of cosmic rays through the atmosphere is described by a set of coupled cascade equations with boundary conditions at the top of the atmosphere to match the primary spectrum. Numerical or Monte Carlo calculations are needed to account accurately for decay and energy-loss processes, and the energy dependencies of the primary spectrum and the interactions. Models need to reproduce measurements made at ground level, near the top of the atmosphere, and those made from airplanes and balloons.
On the ground, the rate of cosmic particles reaches a maximum value at the zenith, and continually decreases at increasing zenith angles. At $90^0$, the rate is minimal. As it can be found in "Review of Particle Data", the distribution of the measured values can be described using a $\cos^2$ function. Different effects are responsible for this, the former having the greatest influence.

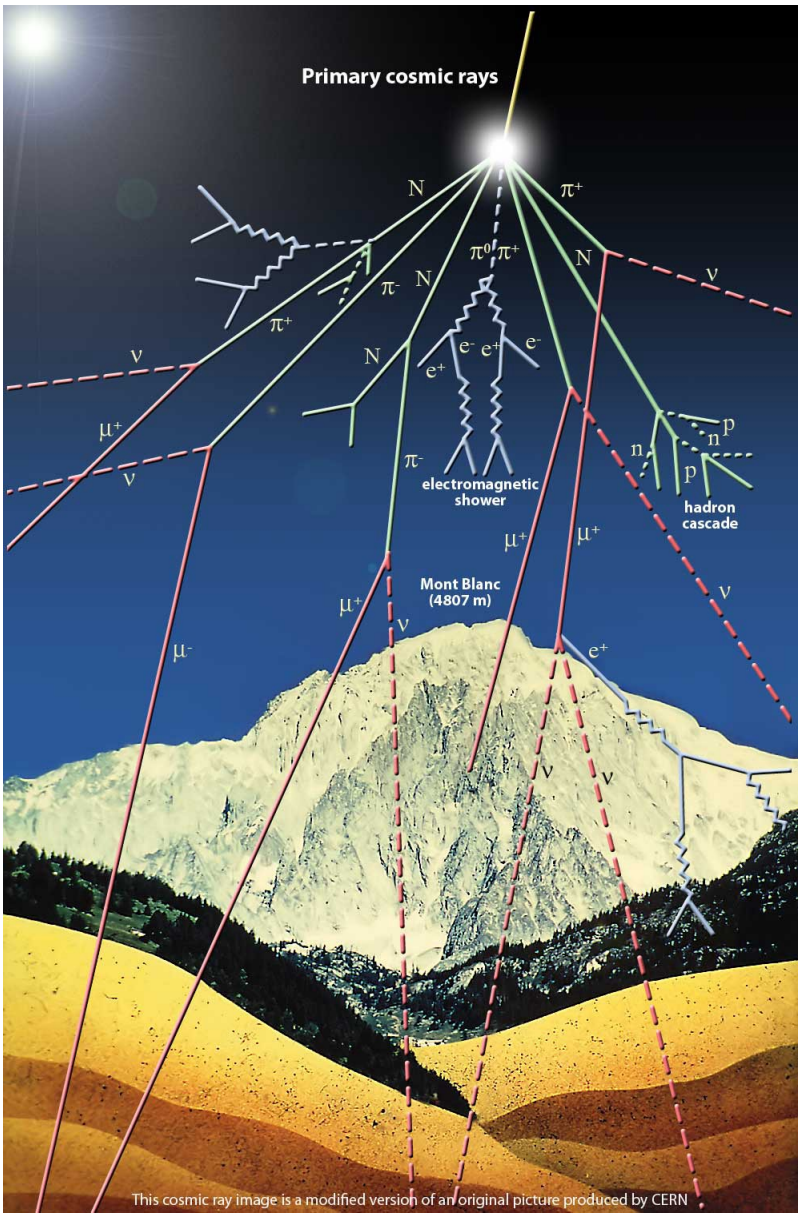



Figure 7: *The primary cosmic rays consist of protons, alpha particles, and heavier nuclei that continuously bombard the top of the Earth's atmosphere. The most abundant particles emerging from the energetic hadronic collisions are pions. Particles such as kaons, hyperons, charmed particles, and nucleon/anti-nucleon pairs are also produced. However, most of these particles interact further with the air or decay to other lighter particles. The charged pions and kaons decay to muons and neutrinos as shown below:*

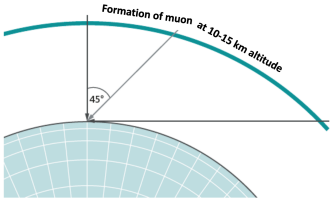



Figure 8: *Path of muon through the atmosphere*

**Path through the atmosphere** If secondary cosmic particles like pions decay at an altitude of about 15 km, muons are produced. These are unstable particles and have an average life of 2.2 $\mu s$. With classical mechanics, we can calculate using $v = c$ that muons should travel only about 660 m before decaying. However, since muons move at nearly the speed of light, the special theory of relativity must also be considered. As a result of "time-dilation" arising from high velocity, the time for a near-speed-of-light muon passes more slowly, so that it can cover the distance of 15 km. The time dilation/length contraction allows the muons to cover an average distance of 33 km. With a larger angle of incidence, the distance from the place of origin of the muons in the atmosphere to the detector on the earth's surface is extended. At an angle of incidence of $90^0$, the distance the muon has to travel through the atmosphere to the surface of the earth is about 400 km (see Figure 8). Only a few high-energy muons manage to cover this path.

**Dependence on the direction of incidence of the pions** Due to the conservation of momentum, muons continue to move in the same direction as their parent pions. The probability of decay or interaction of a pion depends on the mean free path. When a pion hits the atmosphere head-on, it quickly reaches the denser lower atmosphere, where there are many more particles to interact with. The likelihood of an interaction is then high, and more pions are produced. The energy of one pion is divided among the resulting pions. If their energy is so low that a decay becomes more likely than another interaction, they decay into muons. When pions enter the atmosphere at a larger angle, an interaction becomes less likely due to the lower density and thus fever available interaction partners. If a charged pion does not interact, it decays into a muon and a neutrino after just $0.028\mu s$. Due to energy conservation, it transfers all of its energy to the muon and the neutrino. These muons receive more energy than those produced from pions that interact several times.

### 2.1.2 The decay of the muon

*This section is copied from the Chicago U. "Mean Life of the Muon" manual*

The muon ($\mu^-$) is an unstable lepton particle with charge $-e$ which decays into an electron, a neutrino, and an anti-neutrino. The anti-muon (also called positive muon or $\mu^+$) has a charge $+e$ and similarly decays into a positron, a neutrino, and an anti-neutrino. Both

processes can be symbolically expressed as

$$\mu^{\pm} \rightarrow e^{\pm} + \bar{\nu}_e + \nu_\mu \tag{9}$$

Consider a large ensemble of unstable particles - either muons (or anti-muons) like those just discussed or the more familiar radioactive nuclei. One can measure the number of decays per second within that ensemble as a function of time and find that -since the count rate is proportional to the number of remaining unstable particles - the decay rate falls exponentially as time increases:

$$R(t) = R(t=0) \times e^{(-t/\tau)} \tag{10}$$

here $\tau$ is the time for the decay rate to fall by $1/e$ of its initial value. $R(t=0)$ is the rate at the reference time. One calls $\tau$ the *mean lifetime* since $\tau$ is the mean or expectation value of the exponential distribution. Interestingly $\tau$ is also the standard deviation of such a distribution.

In the second part of the experiment, you will collect a large number of muon decays to determine the probability function. While one measurement of a muon decay cannot tell us much, a large number of decays is expected to yield a distribution of values as prescribed by Equation 10.

The decay of the muon is one of the few processes in nature that involves only the Weak interaction. The other three forces are ElectroMagnetic, Strong, and Gravitational forces. As no other forces are involved, the lifetime of the muon (along with its mass $m_\mu$) is a directed measurement of the "strength" of the weak interaction. This strength can be characterized by the Fermi coupling constant given (approximately) by

$$\frac{G_F}{(\bar{h}c)^3} = \sqrt{\frac{192\,\pi^3 \bar{h}}{(m_\mu c^2)^5 \tau}} \tag{11}$$

where $m_\mu\, c^2 = 105.7$ MeV is the rest energy of the muon and $\tau = 2.2\mu s$. The literature value according to the National Institute of Standards and Technology is $\frac{G_F}{(\bar{h}c)^3} = 1.166\,3787(6) \times 10^{-5}$ GeV$^{-2}$.

### 2.1.3 To go further

- How did daring physicist Victor Hess and his balloons contribute to the beginning of muon physics?

- What is the anomalous magnetic of the muon and what do we hope to learn from it?

## 2.2 Experimental setup

Muon and their decay electrons are charged particles. If they have kinetic energy they will deposit energy in any bulk matter, including scintillator material, through the Coulomb interaction. Some of that deposited energy will produce a flash of light in the scintillator. If

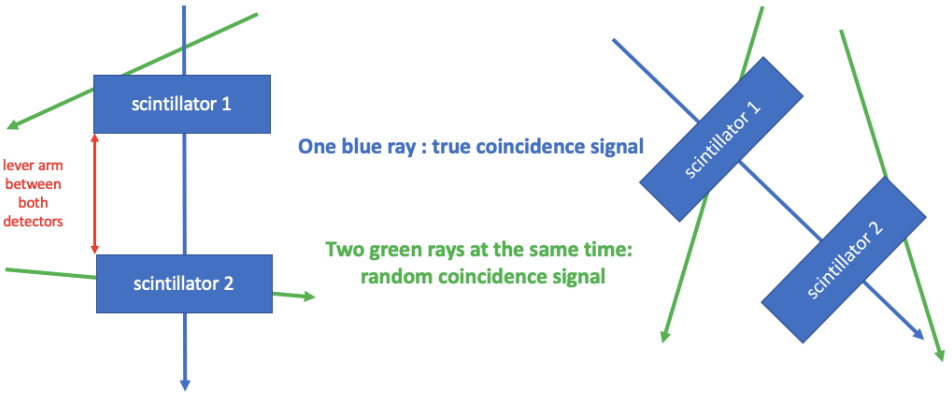



Figure 9: *A scintillator telescope is made up of two scintillators separated by a "large" lever arm. The assembly can be rotated (compare left and right sketch) to select the mean direction of the muons going through both detectors. Looking at either the left or the right sketch: a blue ray goes through both detectors at the same time, produces signals in coincidence, and has a defined direction. Two green rays go through both detectors at the same time in which case they will produce signals in random coincidence.*

the muon decays while in the scintillator, its decay electron will produce a second flash as it travels through the scintillator. Measuring the distribution of times between the two flashes for a large number of decays enables us to determine the mean lifetime of the muon If the muon decays while moving in the scintillator, the two flashes of light are smeared together as one continuous flash.
One way to measure the direction of travel of a particle is to build a telescope. Telescopes are made up of at least two detectors separated by a "large" lever arm. The telescope can be rotated to choose a direction (see Figure 9). When a particle goes through both detectors it produces what is called a true coincidence. When two particles go through each detector at the same time, they will produce a random coincidence.

### 2.2.1 Electronic chain

The electronic chain used to measure the muon lifetime is shown in Figure 10. In this setup, one will measure muons that travel through scintillator 1 and stop and later on decay in scintillator 2.

**Discriminator** If an input analog signal has an amplitude larger than a threshold, a discriminator outputs a digital signal. The discriminator threshold can be adjusted by a rather delicate potentiometer. To check the value of the discriminator measure the voltage between the small white jack and ground (see Figure 11). The jack usually reads $\times 10$ the actual threshold. The value of the threshold may be changed by sticking a small screwdriver in the hole labeled THRESH. Remember this is a delicate piece of equipment.

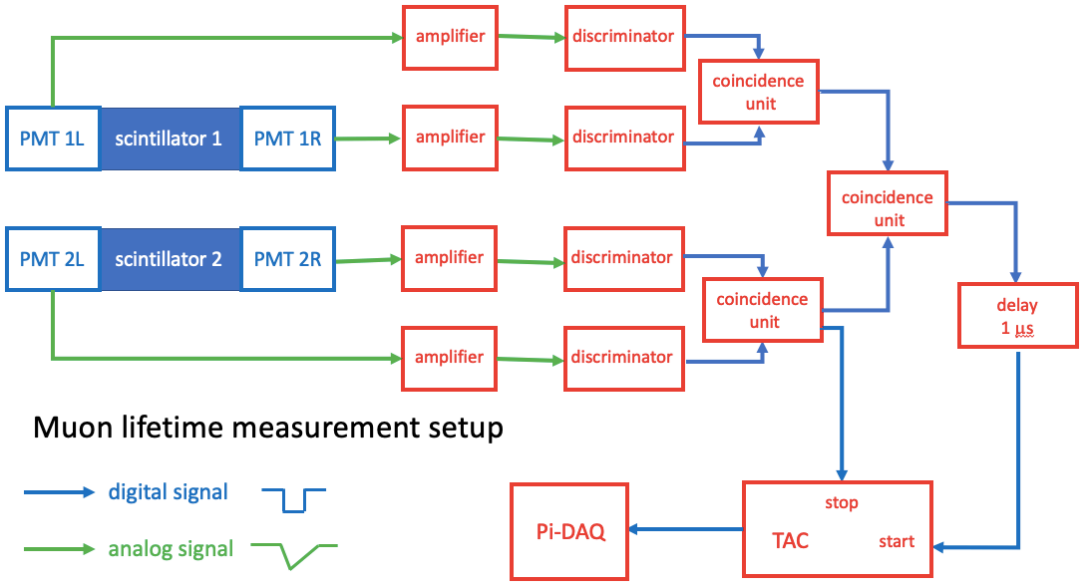



Figure 10: *Electronic chain for the cosmic muon lifetime measurement. Each scintillator of the telescope is read on the left and right by a PhotoMultiplier Tube (PMT). If the PMT analog signal is above the threshold of the discriminator a logic signal is produced. A coincidence unit produces an output signal if two logic signals are present at its input at the same time. A Time-to-Analog-Converter (TAC) produces a logic signal whose amplitude is proportional to the time between the start and the stop signals.*




Figure 11: *Front plate of a typical discriminator. There are little screws inside the tubes labeled THRESH and WIDTH. They allow for adjusting the output signal. The screws are delicate. Be subtle or you will break this expensive piece of equipment.*

### 2.2.2 Operating notes

- Both thick signal cables (aka BNC) and thin signal cables (aka LEMO) have latch connectors that ensure the cables are securely connected to the electronic modules. Don't pull the cable by the flexible wires part, instead pull while holding the connectors. The LEMO cable has a spring that needs to be pulled back. The BNC has a rotating cuff.

- The potentiometer that adjusts the outputs of the electronic (see Figure 11) is FRAGILE. Make sure to insert the screwdriver straight. Don't force the motion. Don't push too hard.

- There is one common power supply for the four PMTs. The operating voltage of the PMTs is -1320V.

### 2.2.3 Time calibration

The TAC outputs a voltage that is proportional to the time between the start and the stop signal. The TAC signal is measured by the computer and presented as a histogram. The horizontal axis of the histogram is time in arbitrary units. To calibrate the horizontal axis, one needs to supply the TAC with separations between the start and the stop that can be controlled and are known. In that way, the known time delay can be related to a specific channel of the histogram. To perform the time calibration: (1) set up the coincidence chain shown in Figure 12, (2) supply delays between 1 and 10 $\mu s$ as an input to the TAC stop. For each set delay, measure the time difference between the start and the stop both on the oscilloscope and the histogram.

### 2.2.4 Measuring count rates

*This section is largely copied from the Wikipedia entry on Poisson statistic*
In this lab, you will measure the output rate from the discriminators using a visual counter (TENNELEC TC531 COUNTER) and a clock. In probability theory and statistics, the Poisson distribution, named after French mathematician Siméon Denis Poisson, is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space if these events occur with a known constant mean rate and independently of the time since the last event. The Poisson distribution can also be used for the number of events in other specified interval types such as distance, area, or volume. If a rate average is $N$, the variance of this number of counts is $N$. In the limit when $N$ is large, then the precision on $N$ is $\sqrt{N}$.
Note that when you measure a rate $N$ that results from a coincidence, you measure both true $N_{true}$ and random $N_{rand}$ coincidences (see Figure 9) such that :

$$N = N_{true} + N_{rand} \tag{12}$$

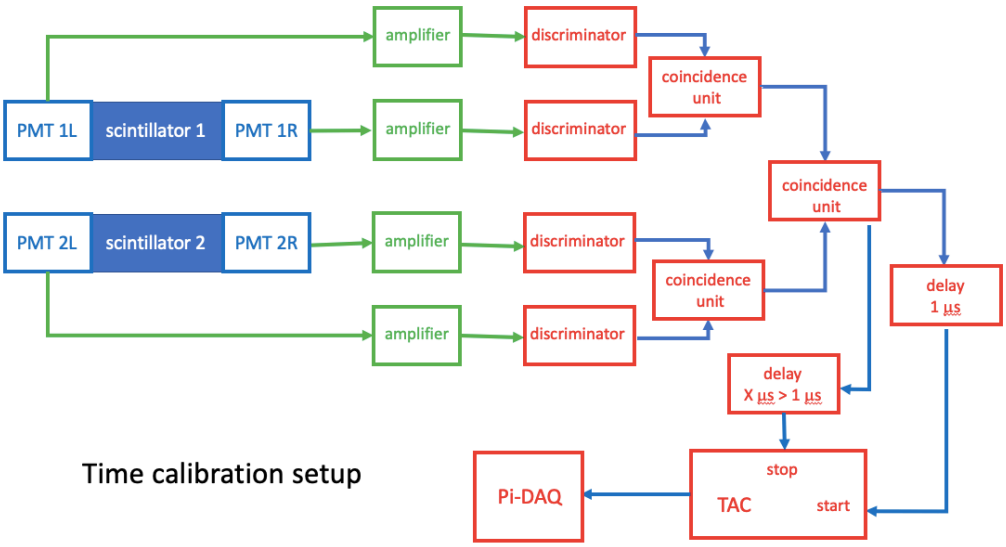



Figure 12: *Setup to calibrate the TAC output. For each set delay of the stop signal, measure the time difference between the start and the stop both on the oscilloscope and the histogram. The connections between the stop and start signal of the TAC and the oscilloscope are not shown on this graph.*

## 2.3 Experimental goals and law under study

The goals of this experiment are to:

- Look at the signals from each PMTs on the oscilloscope and determine the minimum threshold that triggers the discriminator above the noise. Set the threshold of the discriminator 10 mV or so above the noise.
- Measure the counting rates of each PMTs, 2-fold coincidences for detector 1 and detector 2, and the counting rate of the coincidence between detector 1 and detector 2.
- Confirm the $\cos^2$ angular distribution of muon cosmic rates,
- Calibrate the TAC. Measure the peak position in the TAC for different values of delay. Measure the peak position for different values of the full range of the TAC for a given delay.

## 2.4 Preliminary questions

The following questions should help you prepare for this lab:

1. The physical principle under study is cosmic muon rays. Write a 200 to 300 words description of this phenomenon and how it relates to your measurements. This is preparation for your report, for which you will be expected to discuss those issues. Show your word count.

2. When measuring the angular distribution of cosmic muon, you will measure coincidences between the two paddles of the telescope. Suppose that when the telescope is vertical (see Figure 9-left), you measure a true muon rate through your detectors of 1200 muons per second and a random coincidence rate of 500 muons per second. Assume that the random coincidence rate doesn't depend on the telescope angle. Evaluate the total rate you will measure by tilting your telescope in steps of 10 degrees from vertical to horizontal. Use a computer to graph your measurements first using the angle of the telescope as the horizontal axis and second using $\cos^2$ of the angle as the horizontal axis. Do not forget to include the error bars in your plot.

3. A Peak-sensing Analog to Digital Converter converts a voltage into an integer. For example, a 12-bit ADC with a full range of 12 V returns 0 when it senses 0 V and $2^{12} - 1 = 4095$ when it measures 12 V. In this experiment, you will use a Time to Analog (TAC) to measure the time difference between a start and a stop signal. The TAC also has a range (the maximal duration it can measure) and a maximal voltage out at full range. The TAC voltage output will be read by an ADC. What is the integer output of the 12-bit ADC with a full range of 12V if:

   (a) the duration between the start and the stop signal is $4\mu s$ and the TAC has a full range of $6\mu s$ and an output voltage at full range of 8V.

   (b) the duration between the start and the stop signal is $4\mu s$ and the TAC has a full range of $12\mu s$ and an output voltage at full range of 8V.

   (c) the duration between the start and the stop signal is $4\mu s$ and the TAC has a full range of $16\mu s$ and an output voltage at full range of 4V.

   (d) the duration between the start and the stop signal is $2\mu s$ and the TAC has a full range of $6\mu s$ and an output voltage at full range of 8V.

4. What are the differences between Figure 10 and Figure 12. What are the reasons for these differences?

# 3 Gamma-ray Spectroscopy with Scintillators

*Many of the details in this chapter were extracted from notes for the Advanced Physics Laboratory at the University of Michigan*

The purpose of this experiment is to acquaint you with some of the basic techniques used for measuring gamma rays and the characteristics of a typical gamma-ray spectrum. Most isotopes used for gamma-ray measurement also have beta emission(electron or positron) in their decay schemes. The typical decay scheme for the isotope will include a beta decay to a particular level followed by gamma-ray emissions to the ground state of the daughter isotope. The beta will usually be absorbed in the surrounding material and not enter the detector. However, these betas will frequently interact with bound atomic electrons in the material and produce other gamma rays clearly seen in the detector. Understanding the characteristic of the measured gamma-ray spectra is an important goal of this experiment.

## 3.1 Interaction of Gamma Rays and Charged Particles with Nuclear Matter

Henri Becquerel first detected gamma rays with a film placed behind "solid" objects, proving that although they were very penetrating, they ultimately reacted with matter like light. The measured flux rate[1] of gamma rays from the source changes if a piece of material is inserted between the source and the detector. For material with a linear absorption coefficient $\mu$, the flux-rate of the detected radiation $I$ is related to the incident intensity $I_0$ by the exponential relation:

$$I = I_0 e^{-\mu x} \tag{13}$$

where $x$ is the perpendicular thickness of the material traversed by the photon. The absorption coefficient is usually expressed in units of $\text{cm}^{-1}$. Figure 13 show the linear absorption coefficient of photons through NaI as a function of the photon energy. This figure shows that photons get absorbed through one of 3 main mechanisms: Photo-electric, Compton, or Pair production. It also shows that the most likely process depends on the photon's incoming energy. To understand any technique for gamma detection, we have to start with the quantum interaction of light with matter, and the ways photo-energy is transferred to the detector. In all cases, the photon transfers energy to electrons, and detection relies on sensing the ionization created as the high-energy electron moves through matter. In the next section, we review the energy loss mechanisms and the techniques for sensing ionization.

### 3.1.1 The Photoelectric Effect

In the photoelectric effect, the photon transfers all of its energy to an electron and disappears. In our case, the electron gains energy equal to that of the gamma-ray minus the

[1] Number of gamma rays reaching a detector per unit of time.

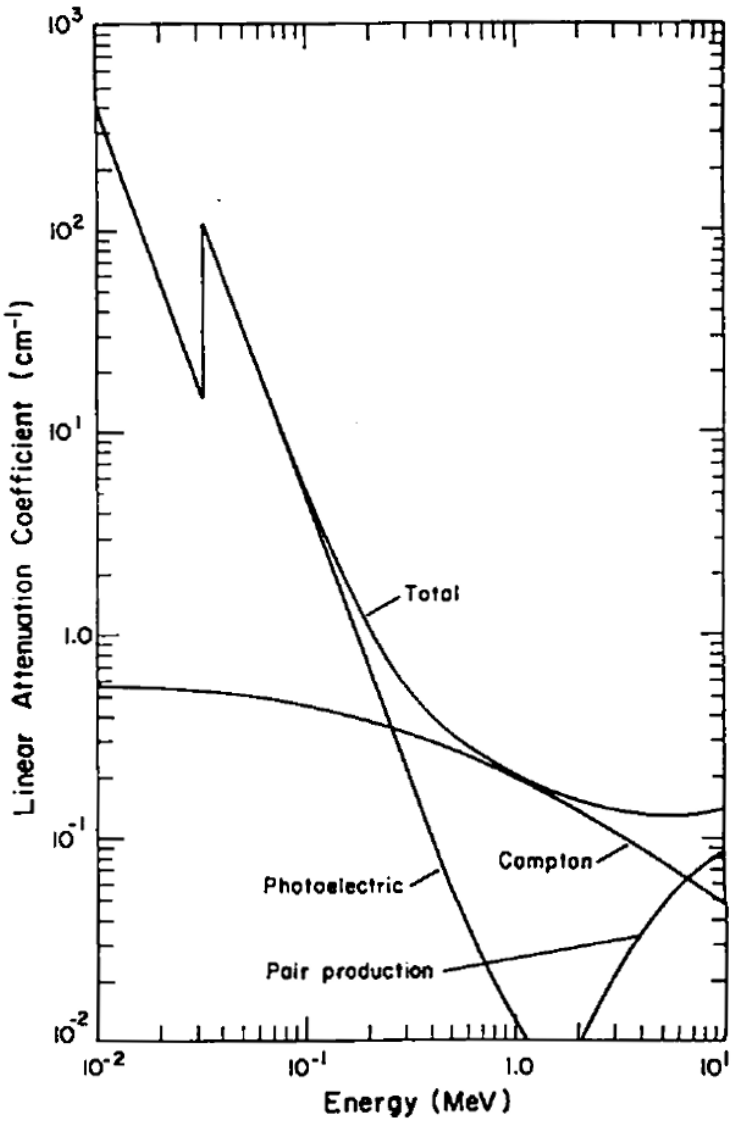



Figure 13: The linear attenuation coefficient $\mu$ for NaI as a function of gamma-ray energy. This coefficient is defined such that a beam of gamma rays of well-defined energy traversing a thickness $x$ of material will be attenuated by $\exp(-\mu x)$. The separate contributions due to the underlying processes of the photoelectric, Compton, and pair-production effects are also shown.

ionization energy of the atom of the target. Since inner shell electron ionization potentials, typically keV, are $\mathcal{O}(10^{-3})$ of the gamma-ray energy, we may consider that the electron energy measures the gamma-ray energy. The cross-section for the ejection of one electron from the $K$-shell of an atom with a nuclear charge $Z$ is:

$$\sigma_{photoel} \propto \sigma_T Z^5 \left[\frac{h\nu}{mc^2}\right]^{-7/2}, \tag{14}$$

where $\sigma_T$ is the classical Thomson (after J.J. Thomson) cross section for an electromagnetic wave incident on a free electron, which is independent of frequency. Therefore, we see that the probability of a photoelectric interaction is steeply falling with gamma-ray energy above 511 keV, and increases very strongly with the charge of the target nucleus. The above expression is valid for energies above the $K$-shell ionization level.

### 3.1.2 Compton Scattering

In Compton scattering (named after Arthur Compton), the photon is not absorbed but instead scatters elastically with the electron, like a collision between two billiard balls. The energy and momentum of the photon and electron after the scattering are exactly as if this is a collision between particles, and, as first measured by Compton, this was one of the early pieces of convincing evidence that light could behave as a "particle". The detailed calculation shows that if a photon of energy $E$ is deflected from its original direction through an angle $\theta$, the new photon energy $E'$ is given by:

$$E' = \frac{E}{1 + \gamma(1 - \cos\theta)}, \tag{15}$$

where $\gamma = E/m_e c^2$. In our case, the scintillator records the energy of the recoiling electron, which obviously acquires the energy lost by the photon, $E_e = E - E'$. Since there is a continuum of scattering angles, there is, therefore, a continuum of scattered electron energies, up to a maximum angle at $\theta = \pi$, when the photons bounces straight backward, and the electron recoils in the original direction of the photon.

The probability of Compton interaction is given by the so-called Klein-Nishina formula. It peaks forward and is proportional to the atomic number $Z$ of the target nucleus (which should be evident as the probability of interaction must depend on the density of target electrons). Fig. 14 shows the probability for Compton scattering as a function of the recoil electron energy. The result shows that the spectrum of scattered electrons is flat at low energies and then rises to a peak at the maximum energy transfer at $\theta = \pi$. This feature is called the Compton edge.

### 3.1.3 Pair Production

In pair production, a photon with energy greater than $2m_e c^2$ converts into an electron-positron pair. This does not happen in free space as a single mass-less photon turning into

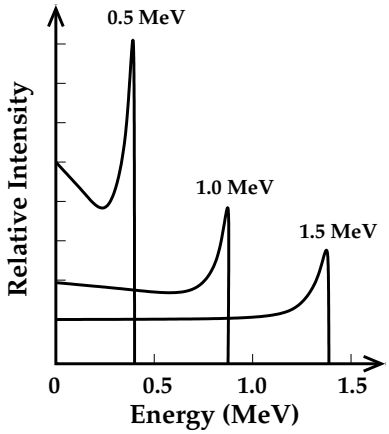



Figure 14: The relative probability of interaction vs. the energy of Compton recoil electrons for three characteristic gamma-ray energies (0.5, 1.0, and 1.5 MeV).

two massive particles would violate the conservation of energy and momentum. However, in matter, a nearby nucleus can participate in the recoil, and the pair production process can take place. The pair production rises from zero at threshold and eventually becomes dominant as the photon energy becomes greater than a few MeV.

### 3.1.4 Experimental Detection of Gamma Rays: general considerations

The detection of gamma rays relies on the transfer of energy to the electrons of the scintillator material and the measurement of the ionization created as the high-energy electrons move through matter. Appendices F and G review how this is done with a scintillator and photomultiplier (PMT).

**Electronics** At this point in the signal chain, for a given gamma-ray interaction, we have at the PMT anode output a small charge pulse proportional to the energy deposited in the scintillator. If there is more than a single mono-chromatic energy to measure, then we want to convert the charge signal to a voltage signal, digitize the voltage pulse heights, one by one, and plot a histogram of the number of occurrences of each pulse height. Knowing the conversion from pulse height into energy will then turn this histogram into a plot of the energy spectrum produced by a given source of gamma rays.

The PMT anode charge signal conversion to a voltage signal is done with a charge-integrating preamplifier. This signal is then sent to a linear amplifier. The signal is differentiated to remove baseline shifts and then integrated to remove high-frequency noise. Finally, the amplifier output is sent to an analog-to-digital converter (ADC), whose output is read out by the computer. The computer runs the data acquisition (DAQ) system that is used to histogram and plot the results.

**Spectral Analysis** The gamma-ray spectrum obtained for a given isotope directly reflects its decay scheme. As an example, the decay scheme of $^{137}$Cs is shown in Fig. 15. The corresponding gamma-ray spectrum that is typical for this isotope obtained using a NaI(Tl) detector is shown in Fig. 16. The spectrum shows several features characteristic of a gamma-ray spectrum. Here we see the photo peak at 0.662 MeV and the Compton edge at 0.478 MeV. Another typical feature is the so-called *back-scatter* peak. Back-scatter occurs when gamma rays make Compton interactions in the material that surrounds the detector. Back-scattered gamma rays from these interactions make photoelectric interactions in the NaI(Tl) detector when they enter the crystal. The energy of the back-scattered peak can be found by solving Equation (15).

$^{22}$Na that you will use in this laboratory is excellent for a simple gamma-gamma coincidence experiment. Fig. 17 shows the decay scheme for this isotope. From this decay scheme, it can be seen that 90% of the time, the decay of $^{22}$Na occurs by positron emission and by electron capture to the 1.275 MeV excited state of $^{22}$Ne. Ninety percent of these decay events occur with positron emission, which then annihilates and produces a pair of 0.511 MeV gamma rays that appear in the gamma spectrum. Fig. 18 is a time-lapse representation of this decay. Fig. 19 shows a gamma-ray spectrum for $^{22}$Na that was obtained with a NaI(Tl) detector in our laboratory. The 0.511-MeV peak will usually be quite a bit more intense than the 1.275-MeV peak, because there are two 0.511-MeV photons created for each positron and the photo-peak efficiency is higher at 0.511 MeV than 1.275 MeV. Also, note that, in our laboratory, the detectors register a large number of counts below about 0.4 MeV due to gamma rays scattering from the aluminum plate.

## 3.2 Experimental setup

### 3.2.1 Electronics

Figure 20 shows the electronic setup used for this experiment. The electronics modules used in the OHIO U. setup are listed below. Note that equivalent modules may, at times, be substituted.

- **Ortec 113 Preamplifier**: This preamplifier is designed for use with dynode or anode signals from photomultiplier tubes. The charge in the photomultiplier output pulse is integrated into the input capacitance of the preamplifier to produce a voltage pulse.

- **Tennelec TC 241 Amplifier** : This high performance spectroscopy amplifier is designed for low noise and high count rate operations that are ideally suited for scintillation detectors.

- **Ortec 427a Delay Amplifier**: This module is designed for general-purpose variable delay of linear or logic signals. The amplifier ensures the output amplitude matches the input amplitude.

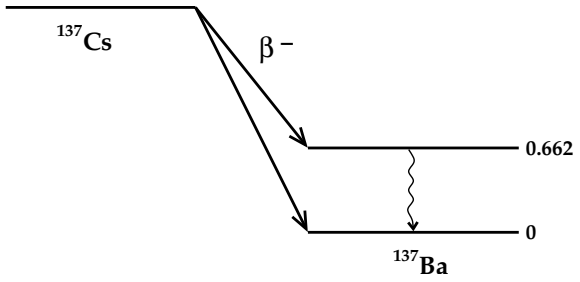



Figure 15: The decay scheme for $^{137}$Cs to $^{137}$Ba. Energies are given in MeV.

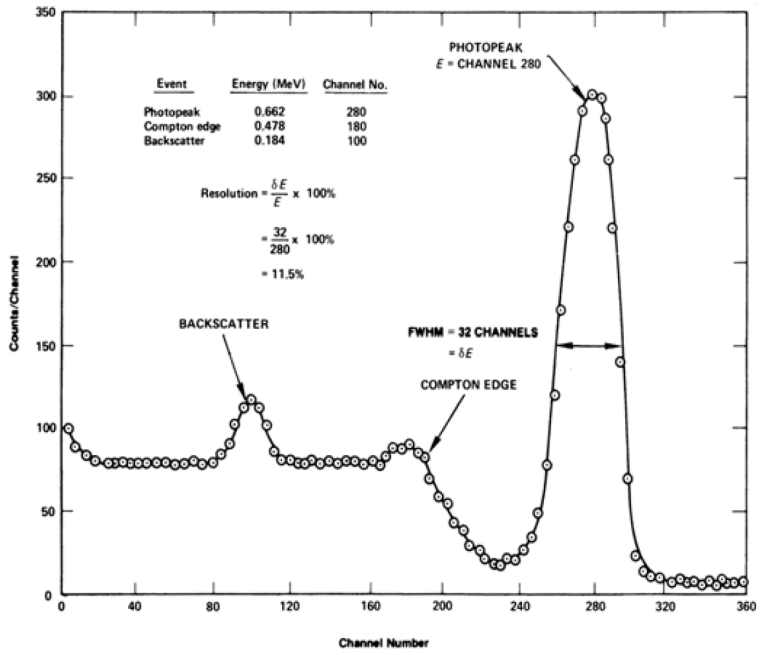



Figure 16: NaI(Tl) spectrum for $^{137}$Cs identifying the different features. Note that photons are not detected directly. The photons transmit energy to electrons that are themselves detected by NaI(Tl).

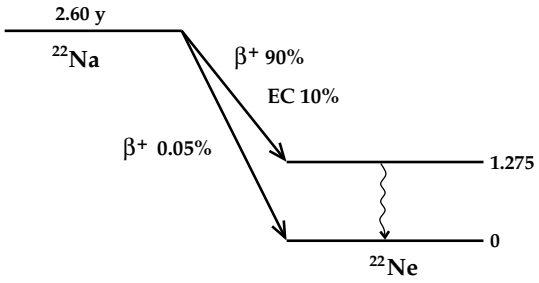



Figure 17: The decay scheme of $^{22}$Na to $^{22}$Ne. Energies are given in MeV. Sodium22 mostly decays to Neon's 1.275 MeV excited state and a tiny bit to the ground state of Neon (0.05%). Most of the decays to the excited state of Neon happens through $\beta^+$ decay (90% of the time) and sometimes (about 10%) through Electron Capture.

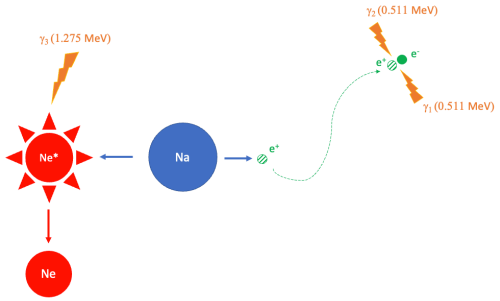



Figure 18: "Time-lapse" representation of the Na decay to Neon. First, Na decays into an excited Neon and a positron. As the positron travels through matter, it loses energy to a point where it annihilates with an electron resulting in two back-to-back photons. The excited Neon radiates a large energy photon.

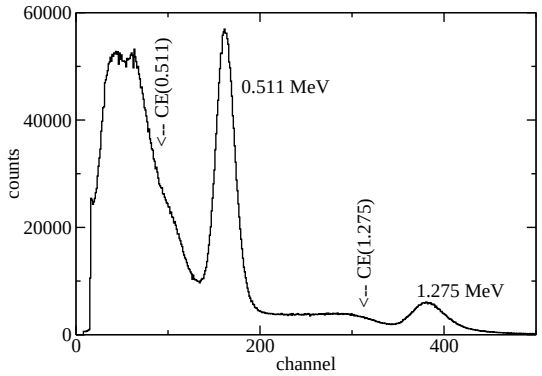



Figure 19: NaI(Tl) spectrum for $^{22}$Na identifying the different features. Note that CE stands for Compton Edge. The 0.511 MeV peak results from the annihilation of the position ($\beta^+$) emitted during the first step of the Na22 decay.

- **Ortec 552 PSA/T-SCA**: This Pulse-Shape Analyzer and Timing Single Channel Analyzer is designed for performing pulse height analysis, timing signal derivation, and determination of pulse shapes.
- **Tennelec TC 861 TAC** : This Time-to-Amplitude converter is designed to produce an analog output pulse proportional to the time interval between the START and STOP inputs.

### 3.2.2 Data Acquisition

When a strobe is issued, the energy deposited in Detector 1, the energy deposited in Detector 2, and the time between a signal in Detector 1 and a signal in Detector 2 are saved on disk by the Data Acquisition (DAQ) program. To issue the strobe, the size of the signal is evaluated by the ORTEC552 discriminator module (see Figure 20). If the input signal amplitude is larger than a given threshold, the discriminator module outputs a logical signal. The logical signal produced by Detector 1 or Detector 2 is OR-ed to generate a strobe signal that triggers the acquisition of information by the computer. When a strobe signal is received by the computer, the energy deposited in Detector 1 (Detector 2 resp.) is registered in the variable "gam1" ("gam2" resp.) The time difference between detecting a large signal in detector 1 and detecting a large signal in detector 2 is registered by the variable "gtac". Suppose the strobe is issued by a signal in detector 1 (respectively in detector 2); most of the time, there is no signal in detector 2 (respectively in detector 1); in that case, gtac is 0.

When a strobe is issued, gam1, gam2, and gtac are recorded by the computer and stored in histograms. The histograms filled by the computer are listed in Table 1.

- The simplest products of the DAQ are the histograms "gam1", "gam2" and "gtac". Every time the computer receives a strobe, the energy in detector 1 is histogramed in "gam1".
- The DAQ also looks at correlations between those quantities. For example, the DAQ histograms "gam1 versus gam2" to look at the energy detected in Detector 2 as a function of the energy detected in Detector 1 event. That histogram is called "g1_g2".
- Finally, the DAQ fills conditional histograms. For example, histogram "g11" shows the energy deposited in Detector 1 when there is an event in Detector 1 and Detector 2 and when the energy is in Detector 2 as a specific value.

To enable conditional histograms, gates need to be defined. The gates used by the DAQ are defined as:

**Gate 0** : The coincidence time overlap of the two detector signals from the output of the time-to-amplitude converter (TAC). This spectrum is called `gtac`.

**Gate 1** : Set on the `gam2` spectrum to identify the 0.511 MeV gamma-ray peak of $^{22}$Na.

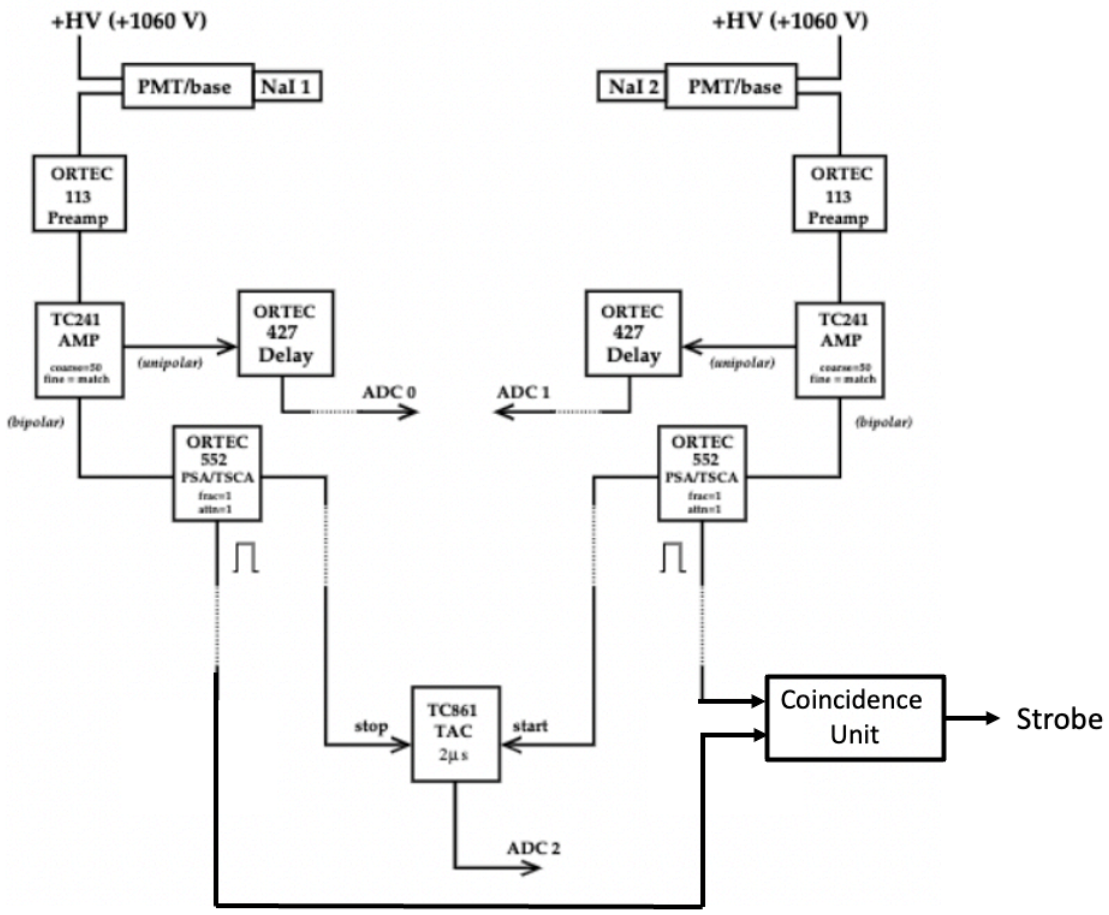



Figure 20: Block diagram of the electronics used for the Gamma-ray experiment. The ADC0, ADC1, and ADC2 signals will be acquired by the Data Acquisition system when a strobe is issued.

**Gate 2** : Set on the `gam2` spectrum to identify the 1.275 MeV gamma-ray peak of $^{22}$Na.

They can be set by using the command

*"gate i limlo limhi"*

where $i$ is the gate number (0,1,2) and the lower and upper limits (in channels) are given by *limlo* and *limhi*, respectively. Table 1 lists the histograms filled by the DAQ. One should check the definition of the gate before interpreting the gated histograms.

| histogram name | type | comment |
|---|---|---|
| gam1 | 1D | energy of NaI 1 |
| gam2 | 1D | energy of NaI 2 |
| gtac | 1D | time between NaI 1 and NaI 2 |
| g1_t | 2D | time vs. gam1 |
| g2_t | 2D | time vs. gam2 |
| g1_g2 | 2D | gam2 vs. gam1 |
| g11 | 1D | gam1 when Gate 0 AND Gate 1 are satisfied |
| g21 | 1D | gam2 when Gate 0 AND Gate 1 are satisfied |
| g12 | 1D | gam1 when Gate 0 AND Gate 2 are satisfied |
| g22 | 1D | gam2 when Gate 0 AND Gate 2 are satisfied |

Table 1: Pre-Defined Histograms for the Gamma-ray experiment. See the definition of the gates in paragraph 3.2.2.

### 3.2.3 Deadtime

The Sodium source you will be using for this experiment is new and very active. For this experiment, you will work with two detectors on arms rotating around a pivot. When the source is placed on the pivot between the two detectors, the source activity is such that the setup suffers from deadtime (see section B.6). This means that the measured rates don't match the rate of particles going through the detectors: this effect is rate dependent (see Figure 21-right). This is a problem when doing experiments that depend on counting rates in various conditions, as you will do for objective 2 of this lab. One way to mitigate dead time is to reduce the overall particle rates for example, by moving the source away from the detectors. It is also useful to use a detector to check that dead time is no longer an issue. For example, when you place the absorbers in front of detector 1 (but not in front of detector 2), you expect the rates in detector 1 to decrease but the rates in detector 2 to remain stable (if deadtime is not an issue.

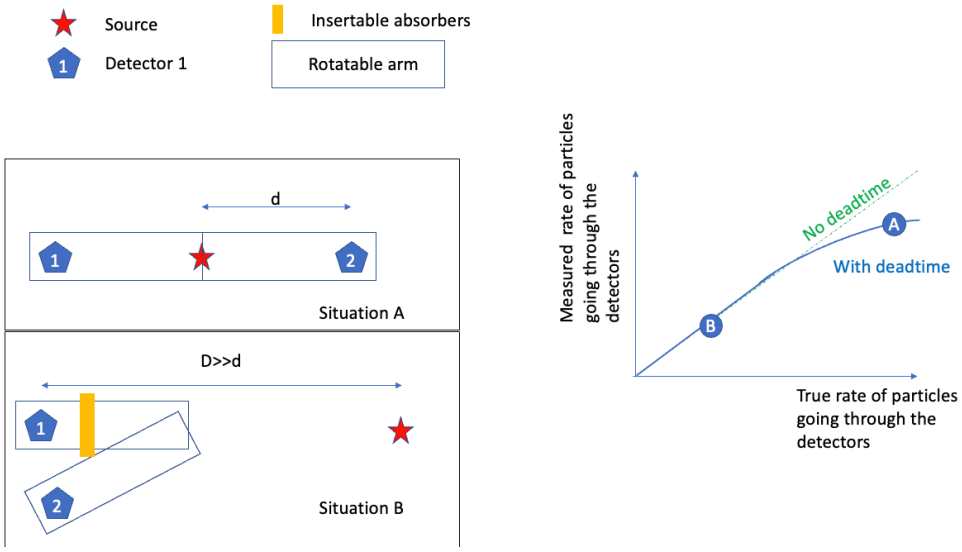



Figure 21: *When the current sodium source is placed in between the two detectors (situation A), rates of particles going through either detector are high enough that the data acquisition system suffers from dead time. One way to reduce dead time is to locate the source away from both detectors. That second configuration (situation B) will be used for objective 2 of this experiment.*

## 3.3 Experimental goals

The goals of this experiment are:

1. to show that Sodium simultaneously emits two 0.511 MeV gamma-rays back to back. Also, show that the 1.275 MeV gamma-ray emitted by Sodium has no angular or temporal correlation with the 0.511 MeV rays. This means the 1.275 MeV gamma ray is emitted at all angles and random times compared to the 0.511 MeV. Hint on how to proceed:

    - Find the coincidence unit that triggers the computer. This module lets you choose to trigger the computer when either Detector 1 or Detector 2 produces a large signal. In that case, the knob is set to 1. This module also lets you choose to trigger the computer when both detector 1 and detector 2 produce a large signal simultaneously: in that case, the knob is set to 2. Observe the difference on the histogram gam1 and gam2 between the knob set on 1 or 2 and conclude on the temporal correlation between the 0.511 and 1.275 MeV rays.
    - When the computer receives a trigger, it can act as a coincidence module in the software. It can sort in different histograms the signals measured by Detector 1 and Detector 2 and the timing between the two signals event by event. Read section 3.2.2 for how this works.

2. to measure the attenuation of gamma rays through Aluminum and Copper. With your data, estimate the absorption coefficient for those two materials. Is the coefficient energy dependent? Compare your results to literature values found in this NIST tabulation: https://www.nist.gov/pml/x-ray-mass-attenuation-coefficients

3. To save a scope picture showing the relative timing of the DAQ strobe signal, gam1, and gam2. Save a scope picture showing the relative timing of the DAQ strobe signal and the gtac signal.

## 3.4 Preliminary questions

The following questions should help you prepare for this lab.

1. The physical principle understudy is the Interaction of gamma-ray with matter. Write a brief 200 to 300 words description of that principle and how it relates to your measurements. This paragraph may be used in the Introductory/Background section part of your report. Show your word count.

2. To measure the linear coefficient of lead, you count how many 0.511 MeV photons you detect in 1 minute when various lead plates are inserted between the source and your detector. You measure 1016 gamma when no plates are inserted, 580 when a 1cm plate is inserted, 318 for 2cm, 202 for 3cm, and 109 for 4 cm. Using a computer graphing

software, plot your counting rates on a semi-logarithmic scale[2] (log in vertical). Don't forget your error bars. In this semi-log scale, equation 13 is a line. Use linear regression technique to evaluate $\mu$. What value of $\mu$ do you extract? What is the precision of the $\mu$ your measure?

3. Suppose you detect gamma rays emitted by a $^{22}$Na source in a scintillator material. Remember that you detect those gamma rays indirectly: the gamma ray transfers energy to electrons; the electrons are detected. The full absorption of the 1.275 MeV gamma ray is measured at channel 956, and the full absorption peak of the 0.511 MeV ray at channel 383. At which channels will you measure the Compton edge of the 1.275 MeV rays and the 0.511 MeV rays? Your answer will have three steps: first, you need to figure out the calibration of your ADC. Next, you need to figure out the energy of the Compton edges for both photon energies (remember you don't detect the Compton photon, you detect the Compton electron). Finally, you use your ADC calibration to figure out where the Compton edges appear in the ADC.

4. The two 0.511 MeV photons resulting from Na-22 decays are emitted back-to-back, while the 1.275 MeV is emitted isotropically with respect to the 0.511 MeV photons. Why?

5. Read about PET scan in this Hyperphysics article
http://hyperphysics.phy-astr.gsu.edu/hbase/NucEne/nucmed.html.
What is the relationship between the experiment you will perform and PET scans?

[2] see log scale Wikipedia

# 4 Gamma-ray spectroscopy with Ge(Li) and Abundance of Uranium Isotopes

This experiment aims to acquaint you with some of the basic techniques used for measuring gamma rays with high resolution, called gamma-ray spectroscopy. It is based on using a High-Purity Germanium detector drifted with Lithium HPGe(Li).

In this experiment, a High-Purity Germanium (HPGe) detector will be used to study the gamma rays that are given off during the radioactive decay of various nuclei. The detector is designed to provide excellent energy resolution, and thus it will be possible to determine with high precision the gamma-ray energies present in a decay spectrum. Understanding the characteristics of the measured gamma-ray spectra is an important goal of this experiment. Necessary background information for this laboratory is given in section 3.1.

## 4.1 Physical concept

*This section is copied from the section on enrichment of the U.S. Nuclear Regulatory Commission website and the Wikipedia page on "Enriched Uranium" (this page).*

When uranium is mined, it consists of approximately 99.3% uranium-238 ($U^{238}$), 0.7% uranium-235 ($U^{235}$), and $\leq$ 0.01% uranium-234 ($U^{234}$). These are the different uranium isotopes. Isotopes of uranium contain 92 protons in the atom's center or nucleus. The $U^{238}$ atoms contain 146 neutrons, the $U^{235}$ atoms contain 143 neutrons, and the $U^{234}$ atoms contain only 142 neutrons. The total number of protons plus neutrons gives the atomic mass of each isotope — that is 238, 235, or 234, respectively. On an atomic level, the size and weight of these isotopes are slightly different. This implies that with the right equipment and under the right conditions, the isotopes can be separated, creating enriched or depleted uranium.

Enriched uranium is a critical component for both civil nuclear power generation and military nuclear weapons. The International Atomic Energy Agency attempts to monitor and control enriched uranium supplies and processes in its efforts to ensure nuclear power generation safety and curb nuclear weapons proliferation. There are about 2,000 tonnes of highly enriched uranium in the world produced mostly for nuclear power, nuclear weapons, naval propulsion, and smaller quantities for research reactors.

The uranium-238 remaining after enrichment is known as depleted uranium (DU) and is considerably less radioactive than even natural uranium, though still very dense. Depleted uranium is used as a radiation-shielding material and for armor-penetrating weapons.

Enriching Uranium

The nuclear fuel used in a nuclear reactor needs to have a higher concentration of the uranium-235 isotope than that which exists in natural uranium ore. $U^{235}$, when concentrated (or "enriched"), is fissionable in light-water reactors (the most common reactor design in the

USA). During fission, the atom's nucleus splits apart, producing heat and extra neutrons. Under controlled conditions, these extra neutrons can cause additional nearby atoms to fission, and a nuclear reaction can be sustained. The heat energy released by the controlled nuclear reaction within the nuclear reactor can be harnessed to produce electricity. Commercially, the $U^{235}$ isotope is enriched to 3 to 5% (from the natural state of 0.7%) and is then further processed to create nuclear fuel.

At the conversion plant, uranium oxide is converted to the chemical form of uranium hexafluoride (UF6) to be usable in an enrichment facility. UF6 is used for a couple of reasons; 1) The element fluorine has only one naturally-occurring isotope, which is a benefit during the enrichment process (e.g., while separating $U^{235}$ from $U^{238}$ the fluorine does not contribute to the weight difference), and 2) UF6 exists as a gas at a suitable operating temperature. The two primary hazards at enrichment facilities include chemical hazards that could be created from a UF6 release and criticality hazards associated with enriched uranium. There are several enrichment processes utilized worldwide. They are Gaseous Diffusion, Gas Centrifuge, and Laser Separation.

### 4.1.1 Uranium series

*This section is copied from this article.*

The $\alpha$ decay chain of uranium-238 is called the "uranium series" or "radium series". Beginning with naturally occurring uranium-238, this series includes the following elements: astatine, bismuth, lead, polonium, protactinium, radium, radon, thallium, and thorium. All are present, at least transiently, in any natural uranium-containing sample, whether metal, compound or mineral. The series terminates with lead-206. The total energy released from uranium-238 to lead-206, including the energy lost to neutrinos, is 51.7 MeV. Figure 22 shows the decay chain of U238. Figure[3] 23 shows a typical gamma-ray spectrum of a uranium-238 source measured with a Ge detector: it shows many peaks from uranium-238 but also its daughters. Table 24 list the different rays and their relative intensities.

### 4.1.2 Actinium series

*This section is copied from this article.*
The $\alpha$ decay chain of uranium-235 is called the "actinium series" or "actinium cascade." Beginning with the naturally-occurring isotope U-235, this decay series includes the following elements: actinium, astatine, bismuth, francium, lead, polonium, protactinium, radium, radon, thallium, and thorium. All are present, at least transiently, in any sample containing uranium-235, whether metal, compound, ore, or mineral. This series terminates with the stable isotope lead-207. The total energy released from uranium-235 to lead-207, including the energy lost to neutrinos, is 46.4 MeV. Figure 22 shows the decay chain of uranium-235.

[3]This figure and the associated tables are extracted from the "Gamma-Ray Spectrum catalog. Ge and Si detector system" from the Idaho National Engineering & Environmental Laboratory. Publication od March 23, 1999.

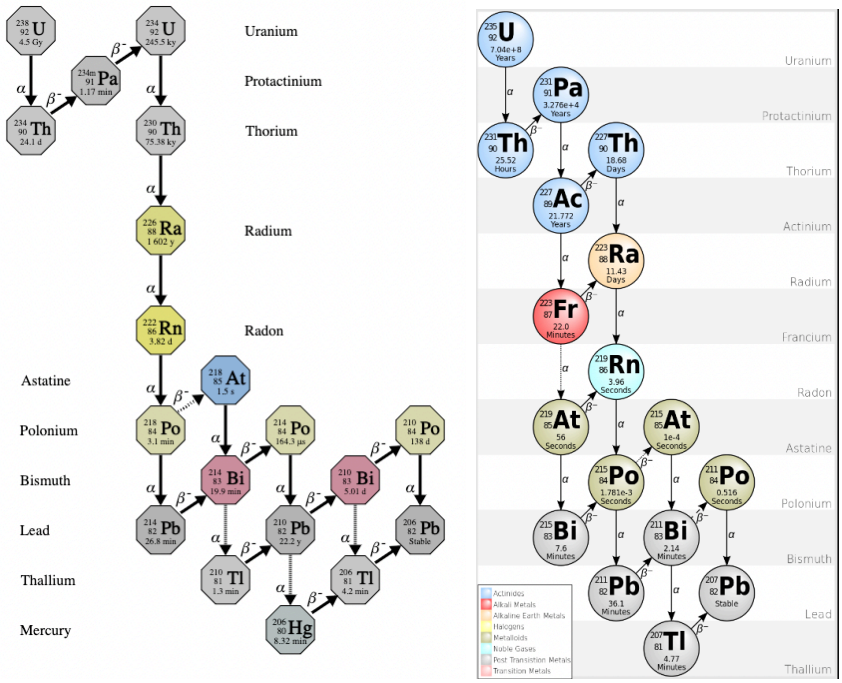



Figure 22: *Decay chains for the Uranium series (left) and the Actinium series (right).*

Figures 25 show a typical gamma-ray spectrum measured with a Ge detector. The most unique and, therefore, useful lines produced by this series are highlighted in red. Tables 26 list the different rays and their relative intensities.

## 4.2 Experimental setup

### 4.2.1 High-Resolution Gamma-Ray Spectroscopy with HPGe Detectors

The use of semiconductor detectors for gamma-ray detection has provided tremendous gains in measurement capability. The gamma-ray peaks obtained with NaI detectors are very broad by comparison, such that two peaks close to each other cannot be resolved, and low-energy peaks may not be easily observed. Semiconductor detectors made of germanium or silicon compensated with lithium provide significantly better energy resolution. Learn more about high-purity Germanium detectors in Appendix E.

### 4.2.2 Setup in EDWARDS

The basic layout of the gamma-ray experiment with the Ge detector is shown in Figure 27.

**Operation notes**

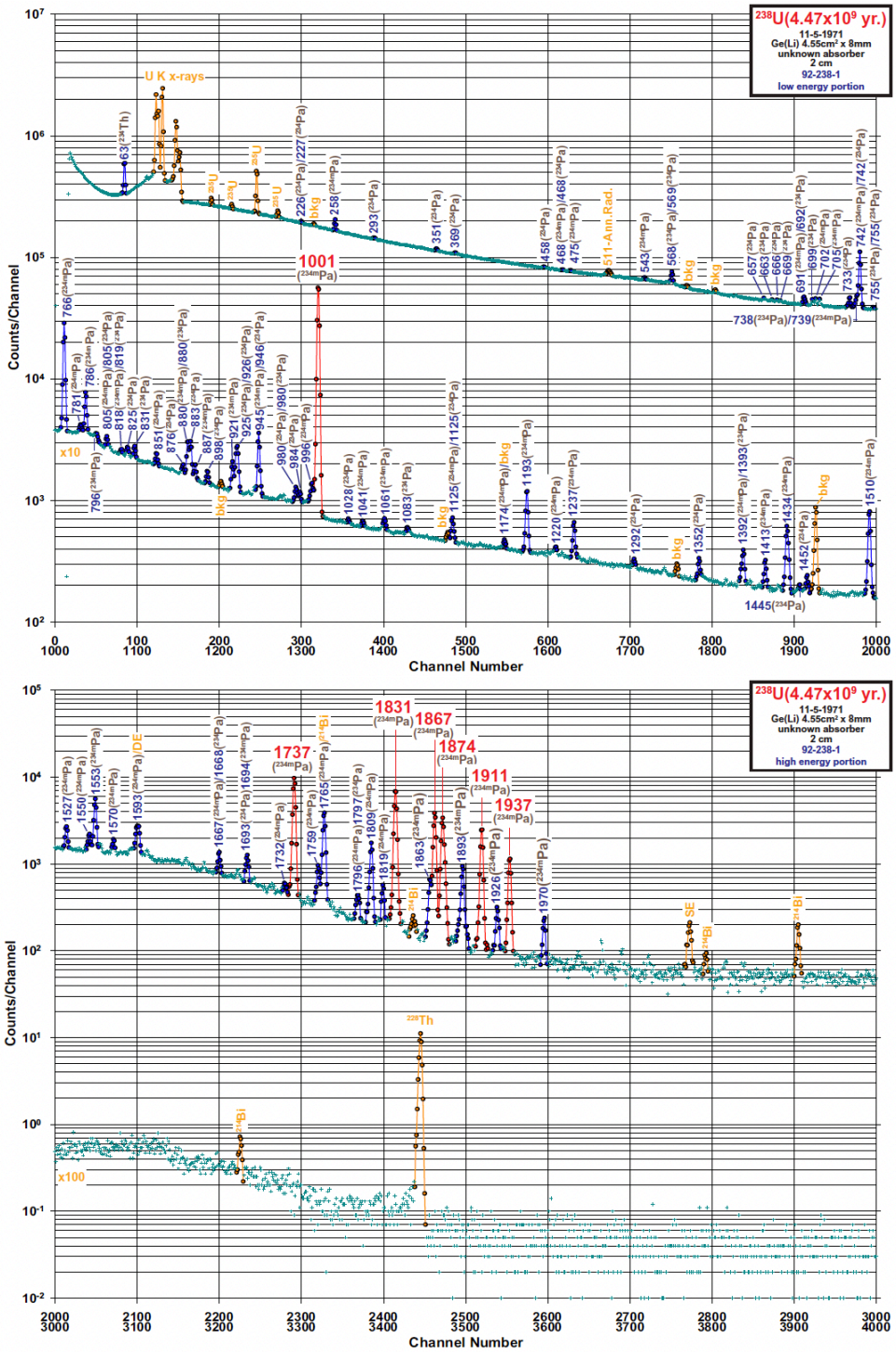



Figure 23: *Gamma-ray spectrum of uranium-238 (and its daughters) measured by a Ge detector. The red peaks show the most distinctive peaks that should be used to identify uranium-238.*

**GAMMA-RAY ENERGIES AND INTENSITIES** (page 2 of 2)

Nuclide: $^{234m}$Pa — $E_\gamma$ $\sigma E_\gamma$, $I_\gamma$ $\sigma I_\gamma$, Levels- from ENSDF Database as of February 8, 2000 — Half Life: 1.17(3) min.

| $E_\gamma$ (keV) | $\sigma E_\gamma$ | [1] $I_\gamma$ | [2] $\sigma I_\gamma$ | Level | |
|---|---|---|---|---|---|
| 786.27 | 0.03 | 0.048 5 | 0.001 9 | 786.29 | β– |
| 791.94 | 0.05 | 0.000 010 | 0.000 003 | 1,601.8 | β– |
| 805.74 | 0.10 | 0.004 3 | 0.000 5 | 849.3 | β– |
| 808.2 | 0.5 | 0.003 0 | 0.000 3 | 851.7 | β– |
| 811 | | EO | | 809.88 | β– |
| 818.2 | 0.5 | 0.001 0 | 0.000 3 | 1,667.4 | β– |
| 825.6 | 0.5 | 0.001 4 | 0.000 3 | 1,911.17 | β– |
| 844.1 | 0.8 | 0.001 08 | 0.000 23 | 1,693.8 | β– |
| 851.57 | 0.10 | 0.006 2 | 0.000 6 | 851.7 | β– |
| 866.8 | 1.0 | 0.001 06 | 0.000 22 | 1,911.17 | β– |
| 880.9 | 0.5 | 0.003 8 | 0.000 5 | 1,667.4 | β– |
| 883.24 | 0.04 | 0.001 8 | 0.000 3 | 926.74 | β– |
| 883.24 | 0.04 | 0.001 7 | 0.000 5 | 1,809 | β– |
| 887.28 | 0.10 | 0.007 08 | 0.000 13 | | β– |
| 921.70 | 0.10 | 0.012 69 | 0.000 14 | 1,911.17 | β– |
| 926.61 | 0.10 | 0.001 23 | 0.000 13 | 926.74 | β– |
| 936.3 | 1.0 | 0.001 8 | 0.000 5 | 1,863.14 | β– |
| 941.94 | 0.10 | 0.002 49 | 0.000 11 | 1,085.3 | β– |
| 945.90 | 0.10 | 0.009 9 | 0.001 0 | 989.45 | β– |
| 960.0 | 1.0 | 0.000 9 | 0.000 3 | 1,809 | β– |
| 996.1 | 2.0 | 0.004 1 | 0.000 7 | 1,781.23 | β– |
| **1,001.03** | **0.03** | **0.837** | **0.010** | **1,044.53** | **β–** |
| 1,041.70 | 0.10 | 0.001 20 | 0.000 10 | 1,085.3 | β– |
| 1,059.4 | 0.8 | 0.001 09 | 0.000 22 | 1,911.17 | β– |
| 1,061.86 | 0.10 | 0.002 29 | 0.000 12 | 1,911.17 | β– |
| 1,081.9 | 1.0 | 0.000 89 | 0.000 19 | 1,126.68 | β– |
| 1,085.7 | 1.0 | 0.000 48 | 0.000 09 | 1,085.3 | β– |
| 1,120.6 | 0.8 | 0.001 70 | 0.000 15 | 1,969.9 | β– |
| 1,125.7 | 0.5 | 0.003 5 | 0.000 6 | 1,911.17 | β– |
| 1,125.7 | 0.5 | 0.000 53 | 0.000 13 | 1,126.68 | β– |
| 1,174.2 | 1.0 | 0.001 90 | 0.000 19 | 1,174.2 | β– |
| 1,193.77 | 0.03 | 0.013 47 | 0.000 13 | 1,237.24 | β– |
| 1,220.37 | 0.10 | 0.000 90 | 0.000 10 | | β– |
| 1,237.24 | 0.10 | 0.005 29 | 0.000 10 | 1,237.24 | β– |
| 1,353.0 | 1.5 | 0.000 62 | 0.000 13 | | β– |
| 1,392.7 | 1.0 | 0.003 44 | 0.000 20 | 1,435.36 | β– |
| 1,413.88 | 0.10 | 0.002 29 | 0.000 10 | 1,457.59 | β– |
| 1,434.13 | 0.10 | 0.009 68 | 0.000 13 | 1,435.36 | β– |
| 1,458.5 | 1.5 | 0.001 8 | 0.000 5 | 1,500 | β– |
| 1,501 | 2 | 0.001 3 | ≈ | 1,500 | β– |
| 1,510.20 | 0.10 | 0.012 87 | 0.000 16 | 1,553.74 | β– |
| 1,527.27 | 0.10 | 0.002 39 | 0.000 09 | 1,570.69 | β– |
| 1,550.0 | 1.0 | 0.001 84 | 0.000 15 | 1,592.3 | β– |
| 1,553.74 | 0.10 | 0.008 08 | 0.000 13 | 1,553.74 | β– |
| 1,558.4 | 1.0 | 0.000 75 | 0.000 16 | 1,601.8 | β– |
| 1,570.67 | 0.10 | 0.001 10 | 0.000 08 | 1,570.69 | β– |
| 1,593.88 | 0.10 | 0.002 69 | 0.000 10 | 1,592.3 | β– |
| 1,601.8 | 1.5 | 0.000 47 | 0.000 22 | 1,601.8 | β– |
| 1,667.6 | 1.0 | 0.000 82 | 0.000 17 | 1,667.4 | β– |
| 1,694.1 | 1.0 | 0.000 45 | 0.000 09 | 1,693.8 | β– |
| 1,720.5 | 1.5 | 0.000 32 | 0.000 15 | | β– |
| 1,732.2 | 1.5 | 0.001 8 | 0.000 3 | | β– |
| **1,737.73** | **0.10** | **0.021 1** | **0.000 3** | **1,781.23** | **β–** |
| 1,759.81 | 0.10 | 0.001 4 | 0.000 7 | | β– |
| 1,765.44 | 0.10 | 0.008 68 | 0.000 13 | 1,809 | β– |
| 1,796.2 | 1.0 | 0.000 31 | 0.000 06 | 1,796.7 | β– |
| 1,809.04 | 0.10 | 0.003 69 | 0.000 08 | 1,809 | β– |
| 1,819.69 | 0.10 | 0.000 90 | 0.000 07 | 1,863.14 | β– |
| 1,831.30 | 0.10 | 0.017 2 | 0.000 3 | 1,875.2 | β– |
| 1,863.09 | 0.10 | 0.001 20 | 0.000 06 | 1,863.14 | β– |
| **1,867.68** | **0.10** | **0.009 18** | **0.000 13** | **1,911.17** | **β–** |
| **1,874.85** | **0.10** | **0.008 18** | **0.000 13** | **1,875.2** | **β–** |
| 1,893.50 | 0.10 | 0.002 18 | 0.000 07 | 1,937 | β– |
| **1,911.17** | **0.10** | **0.006 28** | **0.000 10** | **1,911.17** | **β–** |
| 1,926.5 | 1.0 | 0.000 44 | 0.000 09 | 1,969.9 | β– |
| **1,937.01** | **0.10** | **0.002 89** | **0.000 07** | **1,937** | **β–** |
| 1,970.0 | 1.5 | 0.000 55 | 0.000 12 | 1,969.9 | β– |

① These Iγ are per 100 Decays of $^{234m}$Pa.

② For total uncertainty add systematic component of 0.004% in quadrature, based on the normalization factor 0.000 999 84(4)

Figure 24: *Tabulation of the decay lines of Protactinium (Pa), the daughter of uranium-238. The red lines are the most likely to be observed.*

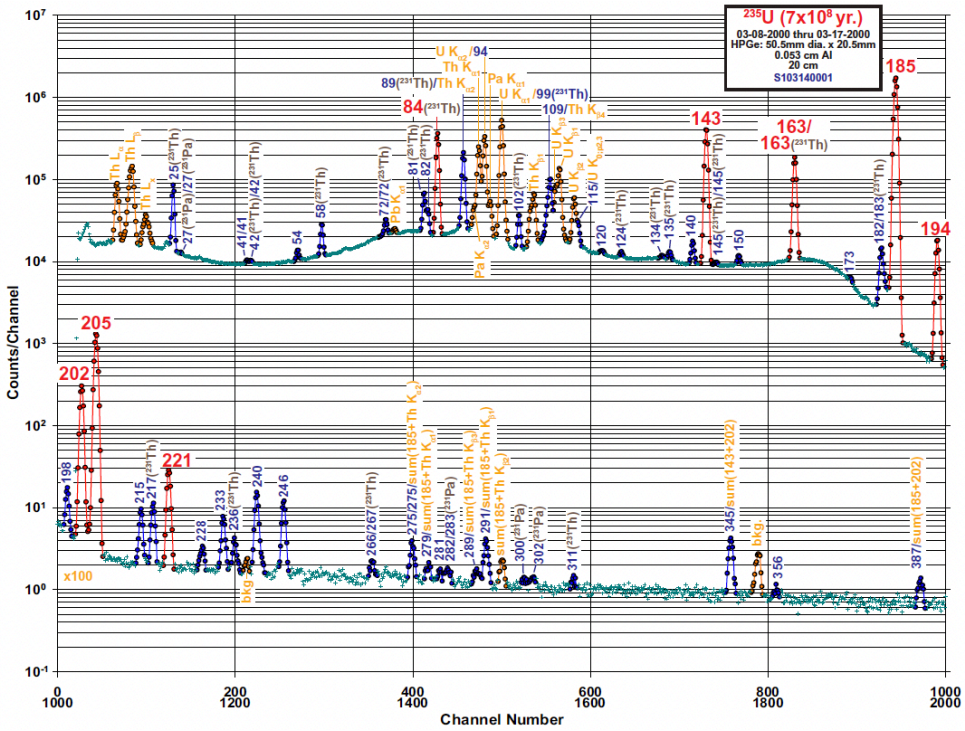
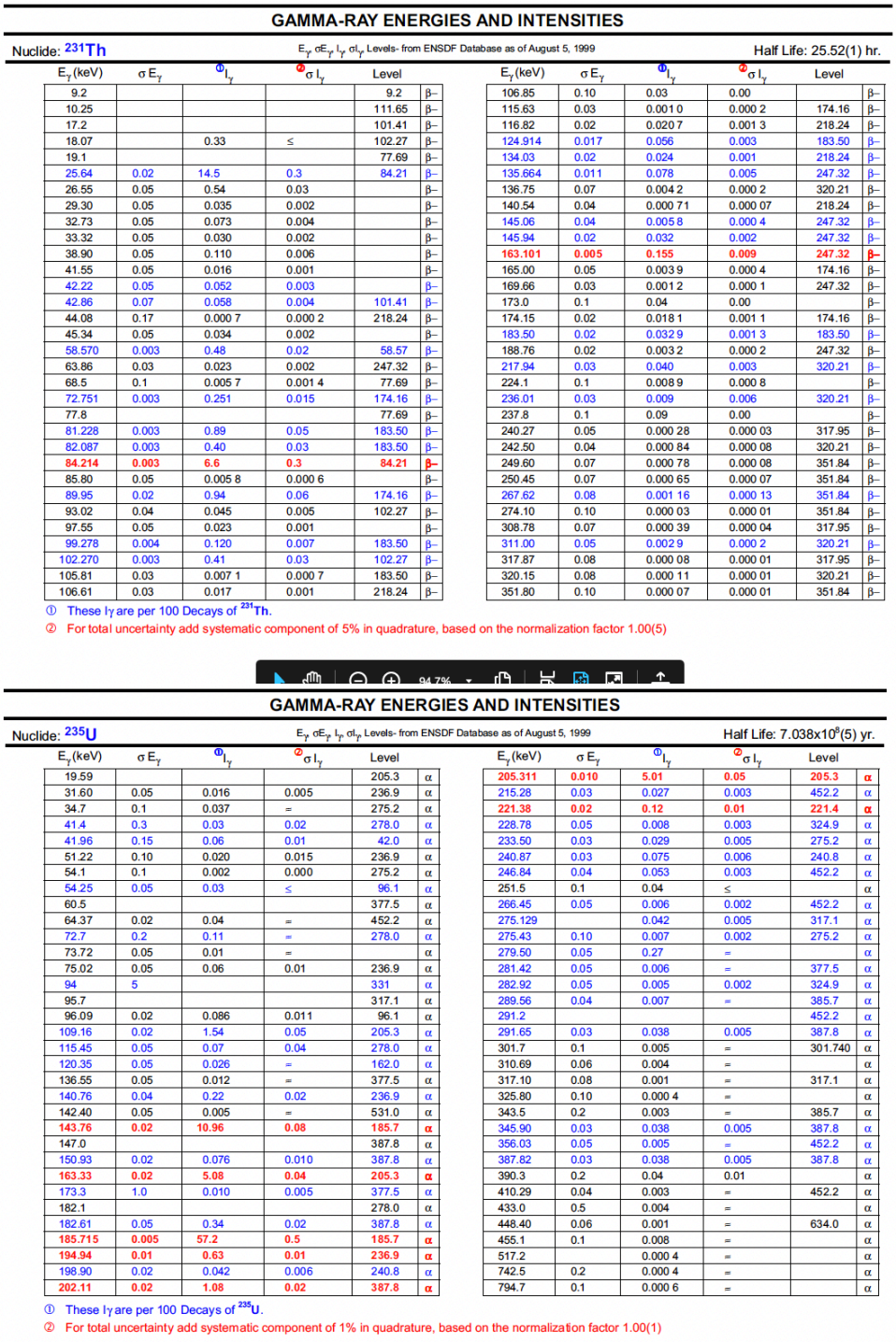



Figure 25: *Gamma-ray spectrum of uranium-235 (and its daughters) measured by a Ge detector. The red peaks show the most distinctive peaks that should be used to identify uranium-235.*



**GAMMA-RAY ENERGIES AND INTENSITIES**

Nuclide: $^{231}$Th — $E_\gamma$ σ$E_\gamma$ $I_\gamma$ σ$I_\gamma$ Levels- from ENSDF Database as of August 5, 1999 — Half Life: 25.52(1) hr.

| $E_\gamma$(keV) | σ $E_\gamma$ | ① $I_\gamma$ | ② σ $I_\gamma$ | Level | |
|---|---|---|---|---|---|
| 9.2 | | | | 9.2 | β– |
| 10.25 | | | | 111.65 | β– |
| 17.2 | | | | 101.41 | β– |
| 18.07 | | 0.33 | ≤ | 102.27 | β– |
| 19.1 | | | | 77.69 | β– |
| 25.64 | 0.02 | 14.5 | 0.3 | 84.21 | β– |
| 26.55 | 0.05 | 0.54 | 0.03 | | β– |
| 29.30 | 0.05 | 0.035 | 0.002 | | β– |
| 32.73 | 0.05 | 0.073 | 0.004 | | β– |
| 33.32 | 0.05 | 0.030 | 0.002 | | β– |
| 38.90 | 0.05 | 0.110 | 0.006 | | β– |
| 41.55 | 0.05 | 0.016 | 0.001 | | β– |
| 42.22 | 0.05 | 0.052 | 0.003 | | β– |
| 42.86 | 0.07 | 0.058 | 0.004 | 101.41 | β– |
| 44.08 | 0.17 | 0.000 7 | 0.000 2 | 218.24 | β– |
| 45.34 | 0.05 | 0.034 | 0.002 | | β– |
| 58.570 | 0.003 | 0.48 | 0.02 | 58.57 | β– |
| 63.86 | 0.03 | 0.023 | 0.002 | 247.32 | β– |
| 68.5 | 0.1 | 0.005 7 | 0.001 4 | 77.69 | β– |
| 72.751 | 0.003 | 0.251 | 0.015 | 174.16 | β– |
| 77.8 | | | | 77.69 | β– |
| 81.228 | 0.003 | 0.89 | 0.05 | 183.50 | β– |
| 82.087 | 0.003 | 0.40 | 0.03 | 183.50 | β– |
| 84.214 | 0.003 | 6.6 | 0.3 | 84.21 | β– |
| 85.80 | 0.05 | 0.005 8 | 0.000 6 | | β– |
| 89.95 | 0.02 | 0.94 | 0.06 | 174.16 | β– |
| 93.02 | 0.04 | 0.045 | 0.005 | 102.27 | β– |
| 97.55 | 0.05 | 0.023 | 0.001 | | β– |
| 99.278 | 0.004 | 0.120 | 0.007 | 183.50 | β– |
| 102.270 | 0.003 | 0.41 | 0.03 | 102.27 | β– |
| 105.81 | 0.03 | 0.007 1 | 0.000 7 | 183.50 | β– |
| 106.61 | 0.03 | 0.017 | 0.001 | 218.24 | β– |
| 106.85 | 0.10 | 0.03 | 0.00 | | β– |
| 115.63 | 0.03 | 0.001 0 | 0.000 2 | 174.16 | β– |
| 116.82 | 0.02 | 0.020 7 | 0.001 3 | 218.24 | β– |
| 124.914 | 0.017 | 0.056 | 0.003 | 183.50 | β– |
| 134.03 | 0.02 | 0.024 | 0.001 | 218.24 | β– |
| 135.664 | 0.011 | 0.078 | 0.005 | 247.32 | β– |
| 136.75 | 0.07 | 0.004 2 | 0.000 2 | 320.21 | β– |
| 140.54 | 0.04 | 0.000 71 | 0.000 07 | 218.24 | β– |
| 145.06 | 0.04 | 0.005 8 | 0.000 4 | 247.32 | β– |
| 145.94 | 0.02 | 0.032 | 0.002 | 247.32 | β– |
| 163.101 | 0.005 | 0.155 | 0.009 | 247.32 | β– |
| 165.00 | 0.05 | 0.003 9 | 0.000 4 | 174.16 | β– |
| 169.66 | 0.03 | 0.001 2 | 0.000 1 | 247.32 | β– |
| 173.0 | 0.1 | 0.04 | 0.00 | | β– |
| 174.15 | 0.02 | 0.018 1 | 0.001 1 | 174.16 | β– |
| 183.50 | 0.02 | 0.032 9 | 0.001 3 | 183.50 | β– |
| 188.76 | 0.02 | 0.003 2 | 0.000 2 | 247.32 | β– |
| 217.94 | 0.03 | 0.040 | 0.003 | 320.21 | β– |
| 224.1 | 0.1 | 0.008 9 | 0.000 8 | | β– |
| 236.01 | 0.03 | 0.009 | 0.006 | 320.21 | β– |
| 237.8 | 0.1 | 0.09 | 0.00 | | β– |
| 240.27 | 0.05 | 0.000 28 | 0.000 03 | 317.95 | β– |
| 242.50 | 0.04 | 0.000 84 | 0.000 08 | 320.21 | β– |
| 249.60 | 0.07 | 0.000 78 | 0.000 08 | 351.84 | β– |
| 250.45 | 0.07 | 0.000 65 | 0.000 07 | 351.84 | β– |
| 267.62 | 0.08 | 0.001 16 | 0.000 13 | 351.84 | β– |
| 274.10 | 0.10 | 0.000 03 | 0.000 01 | 351.84 | β– |
| 308.78 | 0.07 | 0.000 39 | 0.000 04 | 317.95 | β– |
| 311.00 | 0.05 | 0.002 9 | 0.000 2 | 320.21 | β– |
| 317.87 | 0.08 | 0.000 08 | 0.000 01 | 317.95 | β– |
| 320.15 | 0.08 | 0.000 11 | 0.000 01 | 320.21 | β– |
| 351.80 | 0.10 | 0.000 07 | 0.000 01 | 351.84 | β– |

① These Iγ are per 100 Decays of $^{231}$Th.

② For total uncertainty add systematic component of 5% in quadrature, based on the normalization factor 1.00(5)

**GAMMA-RAY ENERGIES AND INTENSITIES**

Nuclide: $^{235}$U — $E_\gamma$ σ$E_\gamma$ $I_\gamma$ σ$I_\gamma$ Levels- from ENSDF Database as of August 5, 1999 — Half Life: 7.038x10$^8$(5) yr.

| $E_\gamma$(keV) | σ $E_\gamma$ | ① $I_\gamma$ | ② σ $I_\gamma$ | Level | |
|---|---|---|---|---|---|
| 19.59 | | | | 205.3 | α |
| 31.60 | 0.05 | 0.016 | 0.005 | 236.9 | α |
| 34.7 | 0.1 | 0.037 | = | 275.2 | α |
| 41.4 | 0.3 | 0.03 | 0.02 | 278.0 | α |
| 41.96 | 0.15 | 0.06 | 0.01 | 42.0 | α |
| 51.22 | 0.10 | 0.020 | 0.015 | 236.9 | α |
| 54.1 | 0.1 | 0.002 | 0.000 | 275.2 | α |
| 54.25 | 0.05 | 0.03 | ≤ | 96.1 | α |
| 60.5 | | | | 377.5 | α |
| 64.37 | 0.02 | 0.04 | = | 452.2 | α |
| 72.7 | 0.2 | 0.11 | = | 278.0 | α |
| 73.72 | 0.05 | 0.01 | = | | α |
| 75.02 | 0.05 | 0.06 | 0.01 | 236.9 | α |
| 94 | 5 | | | 331 | α |
| 95.7 | | | | 317.1 | α |
| 96.09 | 0.02 | 0.086 | 0.011 | 96.1 | α |
| 109.16 | 0.02 | 1.54 | 0.05 | 205.3 | α |
| 115.45 | 0.05 | 0.07 | 0.04 | 278.0 | α |
| 120.35 | 0.05 | 0.026 | = | 162.0 | α |
| 136.55 | 0.05 | 0.012 | = | 377.5 | α |
| 140.76 | 0.04 | 0.22 | 0.02 | 236.9 | α |
| 142.40 | 0.05 | 0.005 | = | 531.0 | α |
| 143.76 | 0.02 | 10.96 | 0.08 | 185.7 | α |
| 147.0 | | | | 387.8 | α |
| 150.93 | 0.02 | 0.076 | 0.010 | 387.8 | α |
| 163.33 | 0.02 | 5.08 | 0.04 | 205.3 | α |
| 173.3 | 1.0 | 0.010 | 0.005 | 377.5 | α |
| 182.1 | | | | 278.0 | α |
| 182.61 | 0.05 | 0.34 | 0.02 | 387.8 | α |
| 185.715 | 0.005 | 57.2 | 0.5 | 185.7 | α |
| 194.94 | 0.01 | 0.63 | 0.01 | 236.9 | α |
| 198.90 | 0.02 | 0.042 | 0.006 | 240.8 | α |
| 202.11 | 0.02 | 1.08 | 0.02 | 387.8 | α |
| 205.311 | 0.010 | 5.01 | 0.05 | 205.3 | α |
| 215.28 | 0.03 | 0.027 | 0.003 | 452.2 | α |
| 221.38 | 0.02 | 0.12 | 0.01 | 221.4 | α |
| 228.78 | 0.05 | 0.008 | 0.003 | 324.9 | α |
| 233.50 | 0.03 | 0.029 | 0.005 | 275.2 | α |
| 240.87 | 0.03 | 0.075 | 0.006 | 240.8 | α |
| 246.84 | 0.04 | 0.053 | 0.003 | 452.2 | α |
| 251.5 | 0.1 | 0.04 | ≤ | | α |
| 266.45 | 0.05 | 0.006 | 0.002 | 452.2 | α |
| 275.129 | | 0.042 | 0.005 | 317.1 | α |
| 275.43 | 0.10 | 0.007 | 0.002 | 275.2 | α |
| 279.50 | 0.05 | 0.27 | = | | α |
| 281.42 | 0.05 | 0.006 | = | 377.5 | α |
| 282.92 | 0.05 | 0.005 | 0.002 | 324.9 | α |
| 289.56 | 0.04 | 0.007 | = | 385.7 | α |
| 291.2 | | | | 452.2 | α |
| 291.65 | 0.03 | 0.038 | 0.005 | 387.8 | α |
| 301.7 | 0.1 | 0.005 | = | 301.740 | α |
| 310.69 | 0.06 | 0.004 | = | | α |
| 317.10 | 0.08 | 0.001 | = | 317.1 | α |
| 325.80 | 0.10 | 0.000 4 | = | | α |
| 343.5 | 0.2 | 0.003 | = | 385.7 | α |
| 345.90 | 0.03 | 0.038 | 0.005 | 387.8 | α |
| 356.03 | 0.05 | 0.005 | = | 452.2 | α |
| 387.82 | 0.03 | 0.038 | 0.005 | 387.8 | α |
| 390.3 | 0.2 | 0.04 | 0.01 | | α |
| 410.29 | 0.04 | 0.003 | = | 452.2 | α |
| 433.0 | 0.5 | 0.004 | = | | α |
| 448.40 | 0.06 | 0.001 | = | 634.0 | α |
| 455.1 | 0.1 | 0.008 | = | | α |
| 517.2 | | 0.000 4 | = | | α |
| 742.5 | 0.2 | 0.000 4 | = | | α |
| 794.7 | 0.1 | 0.000 6 | = | | α |

① These Iγ are per 100 Decays of $^{235}$U.

② For total uncertainty add systematic component of 1% in quadrature, based on the normalization factor 1.00(1)

Figure 26: *Tabulation of the decay lines of Thorium, a daughter of uranium-235 (top) and uranium-235 (bottom). The red lines are the most likely to be observed.*

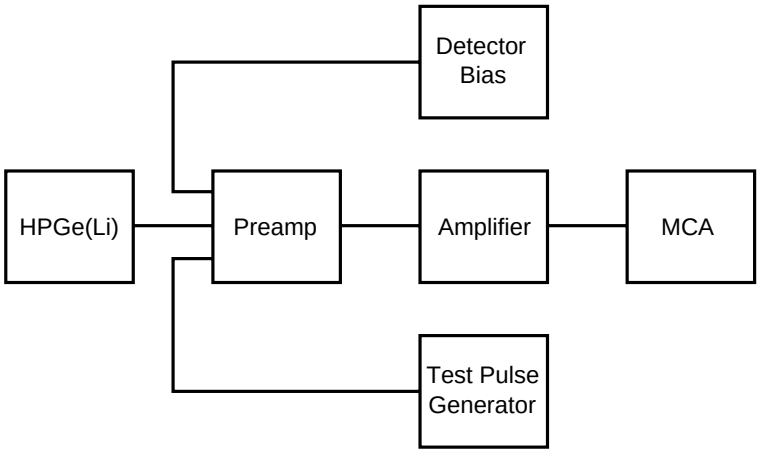



Figure 27: *Major components of the gamma-ray spectroscopy system set up for this experiment.*

- COLD: The Germanium detector should only be powered when cold. The detector is cooled by a Nitrogen bath contained in a large dewar. The dewar needs to be filled about once a week. Once the dewar is filled, cooling the Ge detector fully takes about 6 hours. Ensure with your instructor that your detector can be powered before operating it. Figure 28 shows the nitrogen cooling system for the Ge detector you will use.
- Power up and down SLOWLY: The power supply we are using currently[4] is ORTEC659.
  - Check that the High Voltage (HV) knob is powered all the way down. The HV should be set to positive. Make sure you understand how to read the output HV indicator.
  - Observe the raw signal on the oscilloscope. As you raise the HV, the signal will get large and positive and settle down to zero. Raise the HV by small steps (e.g., 200 V) and wait for the signal to return to zero before taking the next step.
  - Repeat until your reach 1500V. Do not power over 1500 V.
  - After you power the HV all the way down, wait a few minutes to turn the HV off.
- ALARA: Time, distance, and shielding are your best bets when dealing with radioactive sources. Take advantage of the room to temporarily store the sources you are not measuring away from you.

## 4.3 Experimental goals and law under study

The goals of this experiment are to:

[4]Spring 2023

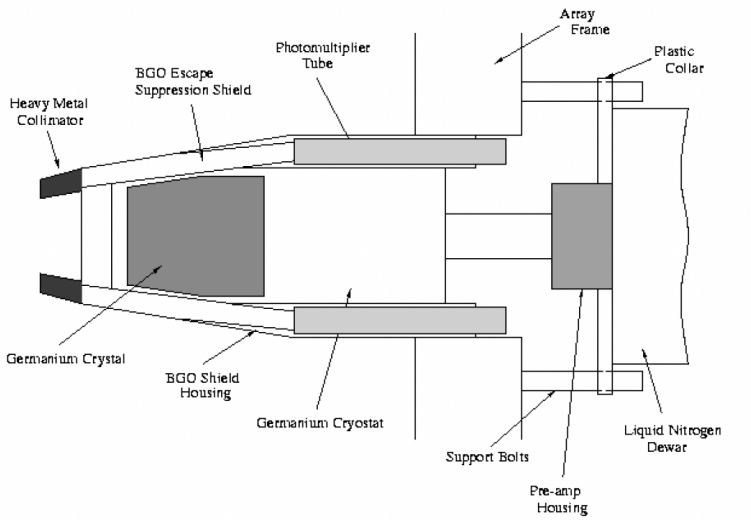



Figure 28: *Schematic diagram of a typical germanium detector showing the germanium crystal, the photomultiplier tube, and the liquid nitrogen cooling system.*

- Use the Cobalt-60, Europium-152, Cesium-137, and Sodium-22 sources to calibrate energy and estimate the energy-dependent detector efficiency.
- Determine the ratio of uranium-235 and uranium-238 in our laboratory uranium samples: a yellow cake and an ingot sample. To carry out this measurement, understand the decay chains of uranium-235 and uranium-238. Radioactive decays from this sample include three radiation types $\alpha$, $\beta$, and $\gamma$. From the measured energy spectrum, decide which decays from a specific isotope and specific level are visible and should be present only once in the decay scheme. Using the determined detector efficiency, determine the ratio N(235)/N(238) to measure the relative uranium isotope abundance of $A = 235$ and $A = 238$. Is it a ratio expected from the normal isotopic abundance found in nature? Are the samples 235 natural, enriched or depleted?

## 4.4 Preliminary questions

The following questions should help you prepare for this lab:

1. The physical principle under a study is isotopic abundance. Write a 200 to 300 words description of this phenomenon and how it relates to your measurements. This is in preparation for your report, for which you will be expected to discuss those issues. Show your word count.
2. Using NuDat or the shorter Firestone tables[5], evaluate the main peaks energy and

[5]You will find both of these references on Blackboard under the "Lab manual..." section and "Nuclear

intensity you will measure from simple nuclei like Cobalt-60, Europium-152, Cesium-137, and Sodium-22.

3. Using tables 24 and 26, list the lines (energy and intensity) you expect to measure for U235 and U238.

4. Suppose you are measuring gamma decays from Sodium-22. Based on the decay scheme of Sodium-22 (see figure 17), how many 0.511 MeV photons do you expect to detect compared to 1.275 MeV photons? Now, look at Figure 19. How many 0.511 MeV photons were detected compared to 1.275 MeV photons? Based on the difference between expectation and measurement, what can you say about the efficiency of the detector used to produce Figure 19?

data" subsection.

# 5 Simulation of Particle Interactions for Radiotherapy

This experiment aims to acquaint the student with the procedures necessary to simulate the interaction of particles in matter, particularly in dose calculations for radiotherapy.

**The following document and exercises are largely copied from a laboratory experiment at the Univeristy of Glasgow.**

## 5.1 Physical concept

### 5.1.1 Radiotherapy for Cancer Treatment

More and more treatment plans for cancer patients involve the use of radiotherapy, in which a beam of high-energy subatomic particles is used to cause catastrophic cell damage in cancerous tumours. The damage is done at the DNA level through microscopic interactions in which the beam particles transfer energy to the tumour tissue causing bonds to be broken and eventually cellular death. In order that the dose to which the tumour is exposed is much larger than the dose received by healthy tissue, treatment plans normally involve the use of shaped beams targeting the tumour area from several different orientations as shown in Figure 29.

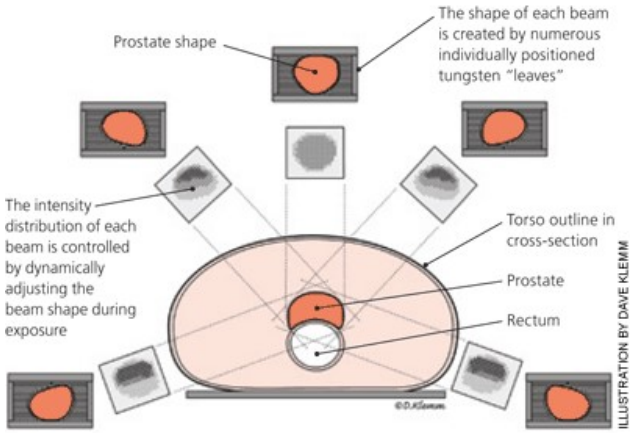



Figure 29: A diagram showing the use of several beam shapes, orientations and intensities to target a tumor. Figure taken from Reference [1]

While the effectiveness of such treatment has been demonstrated in the clinical setting, there is a large international research effort into using different types of beams. Much of the motivation for this research is due to the need to further minimise the damage caused to healthy tissue during these types of treatments. There has therefore been a paradigm shift away from the use of high-energy photon beams to the use of hadron (proton or heavy ion) beams. The shift from gamma to hadron beams is intimately connected to the nature in which these different particles interact and deposit dose in the patient. The motivation for the

current virtual experiment is exploring the relevant particle interactions in this radiotherapy context.

## 5.2 Experimental Setup

### 5.2.1 Accessing GEANT

**Geant4 Virtual Machine, provided by LP2i Bordeaux**

Since 2004, LP2i Bordeaux, CNRS / IN2P3 / Bordeaux University laboratory provides free of charge and licensing to Geant4 users a Geant4 Virtual Machine using either VMware Workstation or VirtualBox for Windows or VMware Fusion for Mac.

See the downloads and instructions at:
https://extra.lp2ib.in2p3.fr/G4/

**Docker container, provided by University of Pavia**

As part of a GEANT4 tutorial workshop, the University of Pavia has a Docker container with GEANT4 installed in it. The downloads and instructions are available at (using these credentials: geant4/pavia2024)
https://geant4.lns.infn.it/pavia2024/docker.html

**Using an installed instance**

You may have access to an installed instance of GEANT on machines you have access through your research, such as at Jefferson Lab.

There may be a locally installed instance at Ohio, but we'll need to discuss that together if you want to use that.

**Installing it yourself**

This is not recommended, but you could look into it here:
https://geant4-userdoc.web.cern.ch/UsersGuides/InstallationGuide/html/index.html

### 5.2.2 Downloading the Glasgow example

Log into your virtual machine instance or remote machine and run the following in a terminal:

```
git clone https://github.com/PHYS5036/Sim.git
cd Sim
tar -xvf Sim.tar.gz
cd PHYS5036_setup/
bash SimSetUp2021.sh
```

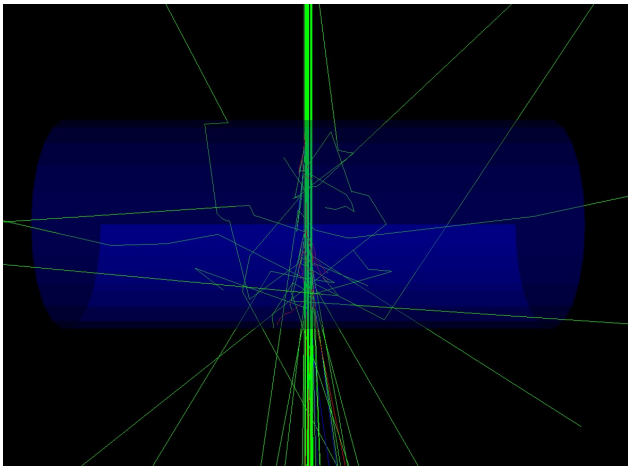


Figure 30: Example visualization of the simulation.

### 5.2.3 Using the simulation

To begin with we will generate a beam of 1 MeV gamma-rays. This is a good starting point as radiotherapy is currently dominated globally by treatment plans and technologies involving high energy gamma-rays.

We can simulate this by running the following commands in a UNIX terminal in your environment with GEANT4 installed.

```
cd ~;
cd DoseProfileMC;
./runsim.sh [beamtype] [energy in MeV]
```

In this case we want to have gamma as [beamtype] and 1 as [energy in MeV] (in general, as we will see later, [beamtype] can be gamma, proton or ion).

Figure 30 shows what you would see in a visualisation of the simulation: the blue cylinder is the water phantom with dimensions representative of a typical patient. The green, blue and red lines represent the tracks of different particle types (by charge) resulting from the gamma beam. We are not running the visualisation on ourselves, so do not expect to see this on your screen.

The simulation outputs three pdf files in your data/ directory:

`G4_gamma_1P.pdf` - dose profile by particle type
`G4_gamma_1L.pdf` - longitudinal dose profile (i.e. parallel to beam direction)
`G4_gamma_1T.pdf` - transverse dose profile (i.e. perpendicular to beam direction)

These filenames have the format: `G4_[beamtype]_[energy]{P,L,T}.pdf`

The first of these shows the calculated total energy deposited in the phantom per incident beam particle in MeV. It also shows a histogram (frequency distribution) of deposited energy for different particle types. The particle type in this case refers to the particle which deposited

the energy and can be either lepton, meson, baryon or ion (in this case, only leptons are relevant). The `L` and `T` files are similar in that they plot the distribution of energy deposit in a central slice through the phantom in the longitudinal (parallel to the beam) and transverse (perpendicular to the beam) direction, respectively.

Many radiotherapy machines that are in use for treatment around the world (especially in developing countries) operate with similar beams to the one we just simulated. The source of gamma-rays for these devices is a rod of Co-60 (with gamma-ray energies of 1.173 and 1.333 MeV). Another common radiotherapy technology is based on the use of compact electron linear accelerators producing higher energy x-ray beams via bremsstrahlung, reaching photon energies up to tens of MeV.

### 5.2.4 Deposited Energy, Absorbed Dose and Equivalent Dose

Up to now, when discussing interactions between the beam and the phantom we have been interchanging the terms deposited energy and dose. In order to understand, in a quantitative sense, the efficacy of a particular radiotherapy technology it is important to define formally what is meant by these terms and the relationship between them.

The stochastic effect in terms of cell damage as a result of exposure to ionising radiation is characterised by the equivalent dose H measured in Sieverts (Sv). The equivalent dose absorbed in tissue is defined such that:

$$H = \Sigma_R W_R D_R \tag{16}$$

where $W_R$ is a weighting factor for radiation type $R$, which is a measure of the relative biological effectiveness of the radiation type. Nominal values of $W_R$ for different particle types as set by the International Committee for Radiation Protection (ICRP) are:

| Radiation Type | $W_R$ |
|---|---|
| lepton | 1.0 |
| meson | 2.0 |
| baryon | 2.0 |
| ion | 20.0 |

The absorbed dose $D_R$ is defined as the mean energy deposited by ionising radiation in matter per unit mass, the unit of which is the Gray (J/kg):

$$D = \frac{E_d}{m} \tag{17}$$

Analysis of the simulated data gives the energy deposited in a given volume element (or voxel) of the phantom. Using the fact that the volume element used in the analysis has the dimensions $20 \times 20 \times 1$ mm$^3$, calculate the total absorbed dose ($D$) and equivalent dose ($H$) per incident beam particle for the four gamma-ray energies you just simulated (use the value for $E_d$ given in the longitudinal dose profile). How many 10 MeV gamma-rays would a patient need to be exposed to achieve a absorbed dose of 50 Gy?

### 5.2.5 Electromagnetic Interactions of Heavy Charged Particles

Electromagnetic energy loss for heavy charged particles is dominated by the ionisation process as described by the famous Bethe-Bloch equation, in which the ionisation density is related to the charge, mass and velocity of the incident particle. As the particle slows, the cross section for interactions with atomic electrons increases leading to higher and higher energy loss until the particle comes to rest. It is this behaviour, characterised by the presence of a Bragg peak in the dose distribution for heavy charged particles, that has led to intense interest in proton and ion radiotherapy in recent years.

In order to demonstrate the significant difference between gamma-ray and hadron radiotherapy, we are going to run the simulation with beams of protons and carbon ions. As before, let's simulate and extract the dose distributions for proton beams with kinetic energies of 100, 500 and 1000 MeV. Now let's do the same for ion beams of 500, 1000 and 2000 MeV.

### 5.2.6 Realistic Radiotherapy Simulations with Hadronic Interactions

The investigations up to now have focused on electromagnetic interactions between the beam particles and the phantom. In order to fully understand the dose profiles in radiotherapy applications, we must also include in the simulation hadronic interactions between the beam and phantom. Take a short time to look into what is meant by hadronic interactions and how they lead to a higher dose being deposited in the phantom.

Before including these interactions, we are first going to make the radiotherapy simulation more realistic by placing a small sphere (1 cm diameter) at the centre of the phantom. By recording the energy deposited in this sphere separately from the rest of the phantom, we can start to quantify the effectiveness of different therapy treatments. To do so, we consider the sphere to be the tumour that is being targeted and the rest of the phantom to represent healthy tissue. This then allows us to simulate and measure the Tumour-to-Normal (TNR) ratio, which as the name suggests is simply:

$$\mathrm{TNR} = \frac{D_{tumour}}{D_{healthy}} \tag{18}$$

The ultimate goal in radiotherapy treatment planning is keeping TNR as high as possible in order to minimise the damage sustained in the patient's normal or healthy tissue.

Edit the first line of the `macros/DetectorSetup.mac` file to include the tumour volume in your simulations:

```
/DoseProfileMC/detector/SetTumourOn 0 - change to 1 to include spherical tumour volume
```

Next, use the data that you analysed in the first part of the experiment to identify the optimum energy for targeting this tumour (i.e. maximising the TNR) for the three different beam types. This will require a bit of trial and error. Once you are happy that you have optimised the beam energies, run the simulations and take note of the results. Pay particular attention to the fraction of energy deposited by different particle types.

Finally, add hadronic interactions into your simulation by changing the following line in the macros/batch.mac file:

```
from: /DoseProfileMC/physics/addPhysics standard_opt3
to: /DoseProfileMC/physics/addPhysics QGSP_BIC_EMY
```

Re-run your three optimised beam scenarios.

## 5.3 Experiment goals and law under study

The goals of this experiment are:

1. Discuss the magnitude and distribution of energy deposited in the phantom for a beam 1 MeV gamma-rays, using the output pdf files from your simulation. How do these observations relate to the underlying gamma-ray interactions?

2. Simulate photon beams with kinetic energies of 1, 5, 10 and 50 MeV to compare how the energy transferred to the phantom varies. How do the nature of the underlying interactions change as the beam energy is increased?

3. Simulate and extract the dose distributions for proton beams with kinetic energies of 100, 500 and 1000 MeV. Do the same for carbon ion beams of 500, 1000 and 2000 MeV.

   What can you conclude about the nature of the energy deposit for hadron beams? Discuss the range of the charged particles in the water phantom, and how this might be exploited in cancer treatment. Comment on the typical length scale (i.e. FWHM) associated with the Bragg peaks for protons and ions of the same energy. What implications does this latter observation have in terms of treatment planning?

4. Use the data that you analysed in the first part of the experiment to identify the optimum energy for targeting this tumour as described in Section 5.2.6 (i.e. maximising the TNR) for the photon, proton, and carbon ion beam types. This will require a bit of trial and error. Once you are happy that you have optimised the beam energies, run the simulations and take note of the results. Pay particular attention to the fraction of energy deposited by different particle types.

5. Comment on the differences in dose distribution with and without hadronic interactions turned on in the simulation as described at the end of Section 5.2.6. Which beam type is affected most by this change and why? The addition of hadronic interactions means that you should see more secondary particle production. How do these secondary particles contribute to the dose magnitudes and distributions?

## 5.4 Preliminary questions

The following questions should help you prepare for the lab:

1. The physical principle under study is simulation of physical interactions of different particles in matter, relevant for radiotherapy. Write a 200 to 300 words description of this phenomenon and how it relates to your measurements. This is preparation for your report, for which you will be expected to discuss those issues. Show your word count.

2. By inspecting Figure 30, where the green, blue and red lines represent the tracks of different particle types (by charge) resulting from the 1 MeV gamma beam interacting in a water phantom, what observations can be made about the nature of the interaction between the beam and the phantom?

3. Suppose a 200 keV gamma ray beam is attenuated by 10% by passing through the phantom. If you assume the full gamma energy of the lost photons is deposited in the phantom, how many beam photons must be incident to result in an absorbed dose of 50 Gray?

# References

[1] D.E. Gerber and T.A. Chan, Am. Fam. Physician, **78**, 1254-62 (2008).

[2] International Atomic Energy Agency, *Radiation Oncology Physics*, Non-serial Publications , IAEA, Vienna (2005). https://www.iaea.org/publications/7086/radiation-oncology-physics

# 6 Rutherford Scattering

This experiment aims to acquaint the student with the experimental procedures necessary to measure an experimental cross-section that can be directly compared to a theoretical prediction. It is based on measuring the Rutherford scattering cross-section in $\alpha$+Au elastic scattering using the Edwards Accelerator Laboratory at Ohio University.

## 6.1 Physical concept

*The following paragraphs are copied from "Particles and Nuclei: An Introduction to the Physical Concepts" by B. Povh et al. (ISBN: 978-3-540-79367-0).*
Scattering experiments are an important tool of nuclear and particle physics. They are used both to study details of the interactions between different particles and to obtain information about the internal structure of atomic nuclei and their constituents. These experiments will therefore be discussed at length in the following. In a typical scattering experiment, the object to be studied (the target) is bombarded with a beam of particles with a (mostly) well-defined energy. Occasionally, a reaction of the form

$$a + b \rightarrow c + d \tag{19}$$

between the projectile and the target occurs. Here, $a$ and $b$ denote the beam and target particles, and $c$ and $d$ denote the products of the reaction. In inelastic reactions, the number of reaction products may be larger than two. The reaction products' rate, energies, masses, and their angles relative to the beam direction may be determined with suitable detector systems. It is nowadays possible to produce beams of a broad variety of particles (electrons, protons, neutrons, heavy ions, . . .). The beam energies available vary between $10^{-3}$ eV for "cold" neutrons up to $10^{12}$ eV for protons. It is even possible to produce beams of secondary particles that have been produced in high-energy reactions. Some such beams are short-lived, such as muons, $\pi^-$, or K-mesons. Solid, liquid, or gaseous targets may be used as scattering material or, in storage ring experiments, another beam of particles may serve as the target.

### 6.1.1 Scattering cross-section

The reaction rates measured in scattering experiments and the energy spectra and angular distributions of the reaction products yield information about the dynamics of the interaction between the projectile and the target, i.e., about the shape of the interaction potential and the coupling strength. The most important quantity for the description and interpretation of these reactions is the so-called cross-section $\sigma$, which is a yardstick of the probability of a reaction between the two colliding particles.

**Geometric reaction cross-section.** We consider an idealized experiment to elucidate this concept. Imagine a thin scattering target of thickness $d$ with $N_b$ scattering centers $b$ and a

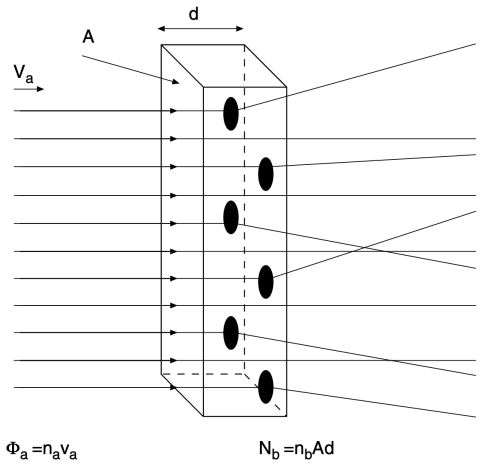



Figure 31: Measurement of the geometric reaction cross-section. The particle beam, $a$, coming from the left with velocity $v_a$ and density $n_a$, corresponds to a particle flux $\Phi_a = n_a v_a$. It hits a (macroscopic) target of thickness $d$ and cross-sectional area $A$. Some beam particles are scattered by the target's scattering centers, i.e., deflected from their original trajectory. The frequency of this process is a measure of the cross-sectional area of the scattering particles.

particle density $n_b$. Each target particle has a cross-sectional area $\sigma_b$, to be determined by the experiment. We bombard the target with a mono-energetic beam of point-like particles a. A reaction occurs whenever a beam particle hits a target particle, and we assume that the beam particle is then removed from the beam. We do not distinguish between the final target states, i.e., whether the reaction is elastic or inelastic. The total reaction rate $\dot{N}$, i.e., the total number of reactions per unit of time, is given by the difference in the beam particle rate $\dot{N}_a$ upstream and downstream of the target. This is a direct measure of the cross-sectional area $\sigma_b$ (Fig. 31). We further assume that the beam has a cross-sectional area $A$ and particle density $n_a$. The number of projectiles hitting the target per unit area and per unit of time is called the flux $\Phi_a$. This is just the product of the particle density and the particle velocity $v_a$:

$$\Phi_a = \frac{\dot{N}}{A} = n_a \cdot v_a \tag{20}$$

and has dimensions $[(\text{area}\times\text{time})^{-1}]$.

The total number of target particles within the beam area is $N_b = n_b \cdot A \cdot d$. Hence the reaction rate $\dot{N}$ is given by the product of the incoming flux and the total cross-sectional area seen by the particles:

$$\dot{N} = \Phi_a \cdot N_b \cdot \sigma_b. \tag{21}$$

This formula is valid if the scattering centers do not overlap and particles are only scattered off individual scattering centers. The area presented by a single scattering center to the

incoming projectile $a$ will be called the *geometric reaction cross-section* in what follows:

$$\sigma_b = \frac{\dot{N}}{\Phi_a . N_b} = \frac{\text{number of reactions per unit time}}{\text{beam particles per unit time per unit area} \times \text{scattering centers}} \tag{22}$$

This definition assumes a homogeneous, constant beam (e.g., neutrons from a reactor). In experiments with particle accelerators, the formula used is:

$$\sigma_b = \frac{\text{number of reactions per unit time}}{\text{beam particles per unit time} \times \text{scattering centres per unit area}}, \tag{23}$$

since the beam is then generally not homogeneous, but the area density of the scattering centers is.

**cross-sections.** This naive description of the geometric reaction cross-section as the effective cross-sectional area of the target particles, (if necessary convoluted with the cross-sectional area of the beam particles) is in many cases a good approximation to the true reaction cross-section. An example is high-energy proton-proton scattering, where the geometric extent of the particles is comparable to their interaction range. The reaction probability for two particles is, however, generally very different from what these geometric considerations would imply. Furthermore, a strong energy dependence is also observed. The reaction rate for the capture of thermal neutrons by uranium, for example, varies by several orders of magnitude within a small energy range. The reaction rate for scattering of (point-like) neutrinos, which only feel the weak interaction, is much smaller than that for the scattering of (also point-like) electrons which feel the electromagnetic interaction. The shape, strength, and range of the interaction potential, and not the geometric forms involved in the scattering process, primarily determine the effective cross-sectional area. The interaction can be determined from the reaction rate if the flux of the incoming beam particles, and the area density of the scattering centers are known, just as in the model above. The total cross-section is defined analogously to the geometric one:

$$\sigma_{\text{tot}} = \frac{\text{number of reactions per unit time}}{\text{beam particles per unit time} \times \text{scattering centres per unit area}}, \tag{24}$$

The cross-section is a physical quantity with dimensions of [area], and is independent of the specific experimental design. A commonly used unit is the *barn*, which is defined as:

$$1 \text{ barn} = 1 \text{ b} = 10^{-28} \text{ m}^2$$
$$1 \text{ millibarn} = 1 \text{ mb} = 10^{-31} \text{ m}^2$$
$$\text{etc...}$$

Typical total cross-sections at a beam energy of 10 GeV, for example, are

$$\sigma_{pp}(10\,\text{GeV}) \approx 40\,mb \tag{25}$$

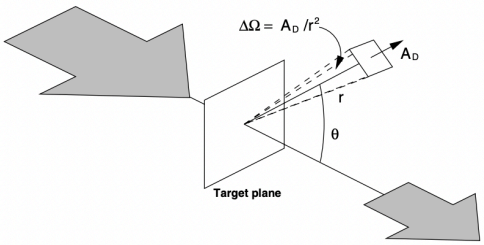



Figure 32: Description of the differential cross-section. Only particles scattered into the small solid angle $\Delta\Omega$ are recorded by the detector of cross-sectional area $A_D$.

for proton-proton scattering; and

$$\sigma_{\nu p}(10\,\mathrm{GeV}) \approx 7 \cdot 10^{-14}\, b = 70\, fb \tag{26}$$

for neutrino-proton scattering.

**Luminosity.** The quantity

$$\mathcal{L} = \Phi_a \cdot N_b \tag{27}$$

is called luminosity. Like the flux, it has dimensions of $[(\mathrm{area}\times\mathrm{time})^{-1}]$. From Equation 20 and $N_b = n_b \cdot d \cdot A$ we have

$$\mathcal{L} = \Phi_a \cdot N_b = \dot{N}_a \cdot n_b \cdot d = n_a \cdot v_a \cdot N_b. \tag{28}$$

Hence the luminosity is the product of the number of incoming beam particles per unit time $\dot{N}_a$, the target particle density in the scattering material $n_b$, and the target's thickness $d$; or the beam particle density $n_a$, their velocity $v_a$ and the number of target particles $N_b$ exposed to the beam.

**Differential cross-sections.** In practice, only a fraction of all the reactions is measured. A detector of area $A_D$ is placed at a distance $r$ and at an angle $\theta$ with respect to the beam direction, covering a solid angle $\Delta\Omega = A_D/r^2$ (Fig. 32). The rate of reactions seen by this detector is then proportional to the differential cross-section $d\sigma(E,\theta)/d\Omega$:

$$\dot{N}(E,\theta,\Delta\Omega) = \mathcal{L} \cdot \frac{d\sigma(E,\theta)}{d\Omega}\Delta\Omega \tag{29}$$

If the detector can determine the energy E of the scattered particles, then one can measure the *doubly differential* cross-section $d^2\sigma(E,E,\theta)/d\Omega dE$. The total cross-section $\sigma$ is then the integral over the total solid angle and overall scattering energies:

$$\sigma_{tot}(E) = \int_0^{E'_{\max}} \int_{4\pi} \frac{d^2\sigma(E,E',\theta)}{d\Omega dE'} d\Omega dE' \tag{30}$$

### 6.1.2 Rutherford Scattering Physics

The Rutherford scattering formula may be derived classically (or quantum-mechanically!) assuming the Coulomb potential between two particles (labeled 0 and 1):

$$U(r) = \frac{Z_0 Z_1 e^2}{4\pi\epsilon_0 r}, \tag{31}$$

where $r$ is the distance between the particles, $Z_0 e$ is the charge of particle 0, $Z_1 e$ is the charge of particle 1, and $\epsilon_0$ is usual electrostatics constant ($\epsilon_0 = 8.854187... \times 10^{-12}$ F/m). The resulting differential cross-section is

$$\frac{d\sigma}{d\Omega_{\text{c.m.}}} = \frac{Z_0^2 Z_1^2 e^4}{16(4\pi\epsilon_0)^2 E_{\text{c.m.}}^2} \frac{1}{\sin^4(\theta_{\text{c.m.}}/2)},$$

where $E_{\text{c.m.}}$ and $\theta_{\text{c.m.}}$ are the center-of-mass kinetic energy and scattering angle, respectively. Note that the Rutherford differential cross-section diverges as $\theta_{\text{c.m.}} \to 0$ and the total cross-section is in fact, infinite.

Since we will not be experimenting the center-of-mass coordinate system; there is more work to do. The algebra of changing coordinate systems for the Rutherford formula is non-trivial and is seldom discussed in textbooks. The formulas below are taken from a paper by Sargood [1]. Let's now assume that label 0 refers to the projectile and label 1 refers to the target which is at rest. First, we will define

$$K = \left( \frac{M_0 \cos\theta + \sqrt{M_1^2 - M_0^2 \sin^2\theta}}{M_0 + M_1} \right)^2, \tag{32}$$

where $M_i$ are the masses and $\theta$ is the laboratory scattering angle, where 0° is by definition the direction of the incident projectile. We will also define the parameter

$$a = \sqrt{1 - \left( \frac{M_0 \sin\theta}{M_1} \right)^2}. \tag{33}$$

The differential cross-section in the laboratory is given by:

$$\frac{d\sigma}{d\Omega} = \frac{Z_0^2 Z_1^2 e^4}{4a(4\pi\epsilon_0)^2 E_0^2} \frac{(a + \cos\theta)^2}{\sin^4\theta}, \tag{34}$$

where $E_0$ is the kinetic energy of the incident projectile. The energy of the scattered projectile is given by $E_0' = KE_0$. Since $K$ is dependent on both the target mass and the scattering angle, particles scattered from different target nuclei will have different energies; Figure 33 shows the $K$ values for several nuclei as a function of laboratory angle. Note that in the limit $M_0/M_1 \to 0$ ("light" projectile and "heavy" target) we have $K \to 1$ and $a \to 1$ and

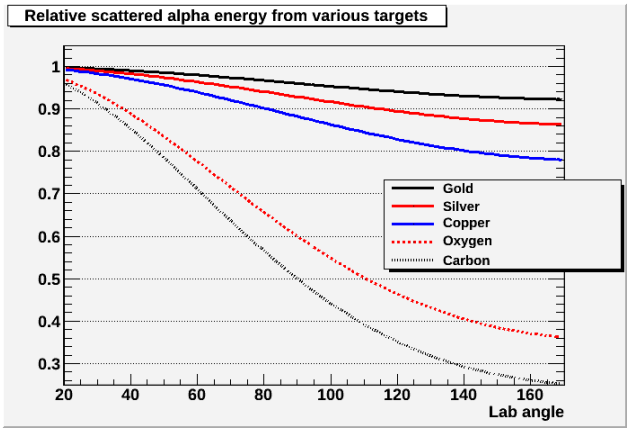



Figure 33: Relative scattered energies ($K$) for several nuclei present in the target foils as a function of laboratory angle.

the lab system is also the center-of-mass system – one can also verify that the differential cross-section formulas are equivalent in this limit using the trig identity

$$\frac{1+\cos\theta}{\sin^2\theta} = \frac{1}{2\sin^2(\theta/2)}. \tag{35}$$

It is convenient to expand Eqs. (33) and (34) in powers of $M_0/M_1$; the result is

$$\frac{d\sigma}{d\Omega} = \frac{Z_0^2 Z_1^2 e^4}{16(4\pi\epsilon_0)^2 E_0^2}\left[\frac{1}{\sin^4(\theta/2)} - 2\left(\frac{M_0}{M_1}\right)^2\right] + O\left[\left(\frac{M_0}{M_1}\right)^4\right]. \tag{36}$$

The fundamental constants can be inserted as follows

$$\frac{e^4}{(4\pi\epsilon_0)^2} = \left(\frac{e^2}{4\pi\epsilon_0\hbar c}\right)^2 (\hbar c)^2 \tag{37}$$

$$= \left(\frac{1}{137.036}\right)^2 \times (197.327 \text{ MeV-fm})^2 \times \left(\frac{10 \text{ mb}}{\text{fm}^2}\right) \tag{38}$$

$$= 20.735 \text{ mb-MeV}^2 \tag{39}$$

to yield "nuclear physics" units. The final formula is

$$\frac{d\sigma}{d\Omega} = 1.296\left(\frac{\text{mb-MeV}^2}{\text{sr}}\right)\left(\frac{Z_0 Z_1}{E_0}\right)^2\left[\frac{1}{\sin^4(\theta/2)} - 2\left(\frac{M_0}{M_1}\right)^2\right] \tag{40}$$

which can be used with $E_0$ in MeV to calculate the differential cross-section in mb/sr.

### 6.1.3 Rutherford Scattering Experiment

The experiment will acquire data at several scattering angles on either side of the incident beam direction. You will record the counts in each of our two silicon detectors. The naming convention is to call the fixed detector RBS and the movable detector PRD. In addition to summing up the counts in the ADC spectra, you will have to record the system live time (LT) for each measurement, as well as the real-time duration (RT) of each run and the charge integrator value. The counts in the ADC spectra must be multiplied by the ratio RT/LT to determine the true number of particles which struck the detector during the real-time duration of each run.

In defining $E_0$ for this experiment, use of the incident $\alpha$ beam energy is only an approximation. In general, a better approximation is to use the average alpha energy in the target given by:

$$E_{\text{avg}} = \frac{E_i + E_f}{2}, \tag{41}$$

$E_i$ is the incident alpha energy, and $E_f$ is the alpha energy after passing through the gold foil. It would be relevant to compare the sensitivity of the final result with and without this correction. Stopping Power is the average energy dissipated by ionizing radiation in a medium per unit path length of travel of the radiation in the medium. You may use this tabulation of Stopping powers to estimate the effect for $\alpha$ beams of a few MeV passing through targets of a few $\mu$m.

Based on the previous discussion, the measured differential scattering cross-section in units of cm$^2$/sr is given by:

$$\frac{d\sigma}{d\Omega} = \frac{N_s}{N_0 \cdot \Delta\Omega \cdot n_1} \tag{42}$$

where:

- $N_s$ = the number of detected particles, corrected by the ratio RT/LT.
- $N_0$ = the number of beam particles that impinged on the scattering foil. This quantity is determined by the charge integrator. Remember that we are using a $He^{2+}$ beam when converting the measured number of $\mu$C to the number of particles.
- $\Delta\Omega$ = the solid angle of the detector (sr). The solid angle is given by $A_c/R^2$, where $A_c$ is the area of the detector (or the defining aperture), and $R$ is the distance of separation between the aperture and the target.
- $n_1$ = the number of target nuclei per cm$^2$.

If the target is a compound material, the total number of detected particles is

$$N_s = N_0 \cdot \Delta\Omega \cdot \sum_i \left( \frac{d\sigma_i}{d\Omega} \cdot n_i \right) \tag{43}$$

where $d\sigma_i/d\Omega$ is the cross-section of a nucleus $i$ and $n_i$ is the target density of that nucleus.

### 6.1.4 To go further

- Cross-Section can be measured experimentally and computed using quantum mechanics, how?
- Rutherford made 3 conclusions from his experiment. Which ones?

## 6.2 Experimental setup

### 6.2.1 Pre-Experiment Approach and Details

One of our graduate students will be assisting us in setting up the experiment and collecting the data. Your main work will be analyzing the data once it is in hand and assisting in positioning the silicon detector for each run.
♢ Do not be surprised if the experiment does not go smoothly or takes more than 2 hours to complete. Experimental nuclear physicists seem to spend a lot of time waiting around for accelerator personnel to deliver the beam or tune the beam to desired levels.

### 6.2.2 Pre-Experiment Checkout

1. After performing the essential pre-lab preparations, begin to understand the experimental setup. This includes understanding the purpose of each part of the system. The basic layout of the Rutherford scattering experiment is shown in Fig. 34.

2. Go through the set-up step by step to understand how the electronics are laid out in both the experimental area and the control room. This basic layout and the purpose of each electronics module should be clear given the experience you have gained in the first part of this course. The manuals for each of the electronics used are available for review. A complete diagram is contained in Fig. 36.

3. Make sure you know how to set the angle of the movable silicon detector to the precise angle you want and that the angle readback makes sense to you. Note that the movable detector is limited to angles in the range from 25° to 150° on one side of the beam and from 250° to 335° on the other. Figure 35 suggests a procedure to understand the angular reading of the detector.

4. Take measurements of the radial positions of the two silicon detectors. This is required to compute the detector's solid angle.

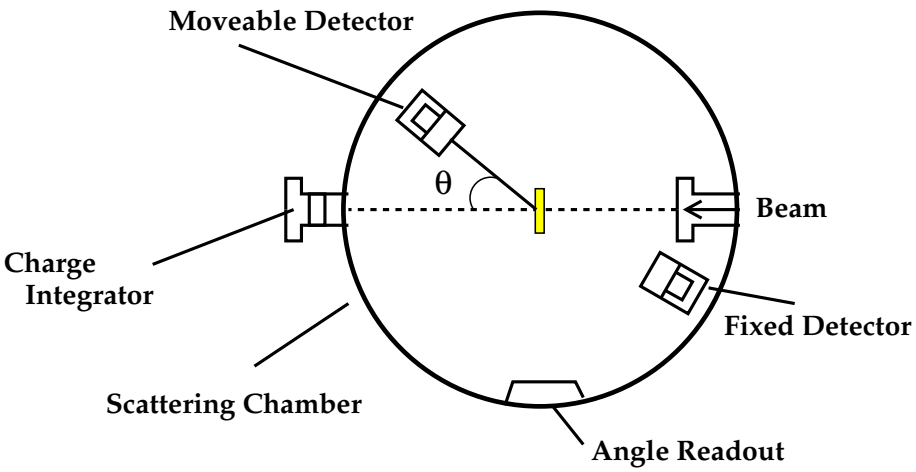
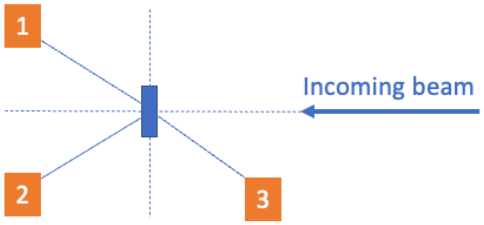



Figure 34: Major components of the Rutherford scattering experiment for this laboratory.




| Detector position | Angle Marker |
|---|---|
| 1 | |
| 2 | |
| 3 | |

| Detector distance from target | |
|---|---|

Figure 35: Suggested procedure to understand the angular reading of the detector.

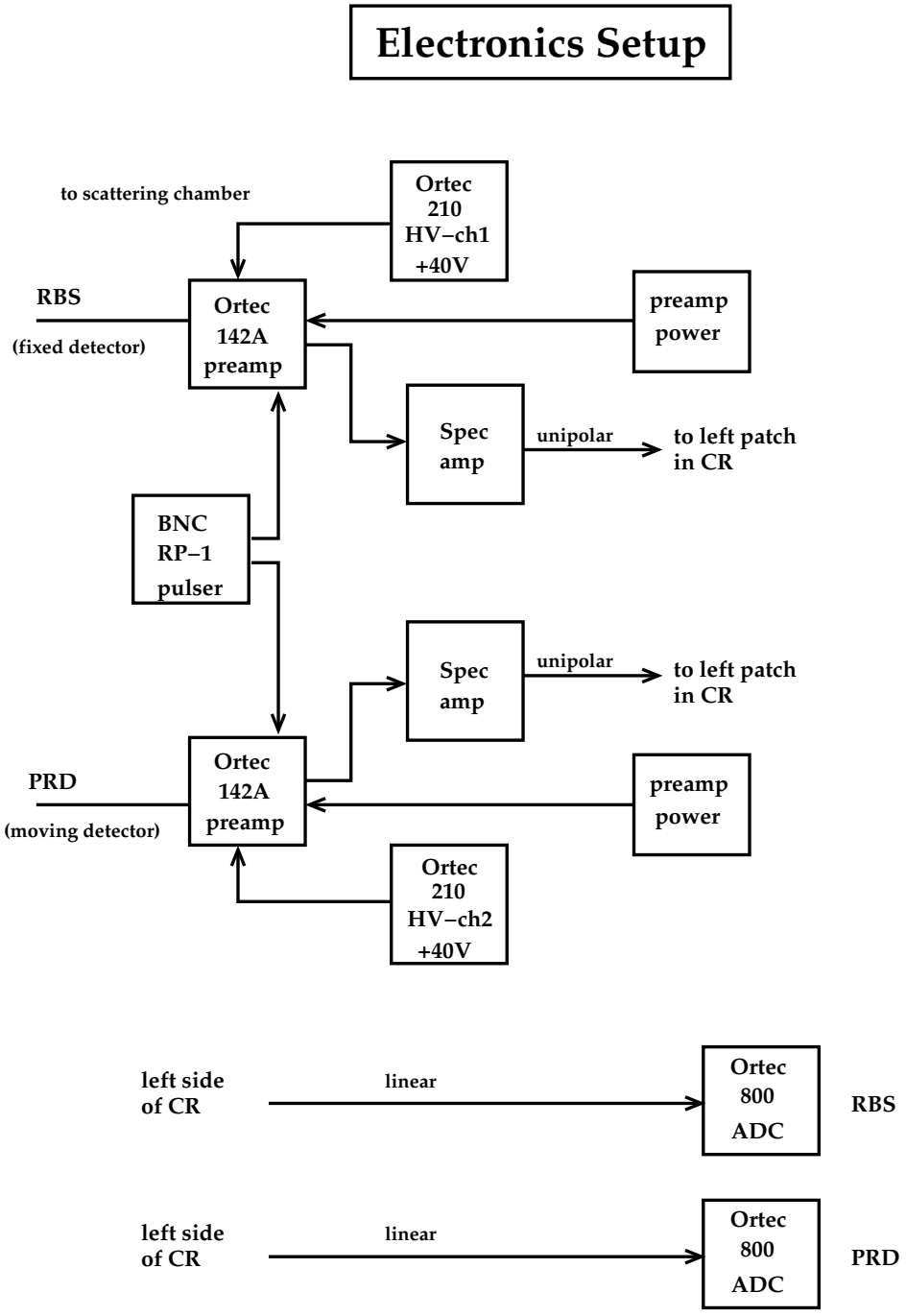



Figure 36: Electronics layout for the Rutherford scattering experiment.

5. Make sure you record the size of the apertures from the two silicon detectors. These are required for solid angle computation.

6. Make sure you understand how to read out the charge integrator for the alpha beam. This is required to determine the number of alphas incident on our gold foil.

7. Make sure you record the thickness of the gold foil (relevant units are $\mu$g/cm$^2$). This is necessary to compute the cross-section.

8. Make sure you know how to set the angle of the target (the gold foil). In this experiment, we will want to minimize the material traversed by the scattered alpha particle in the gold foil. To accomplish this, we will set the angle of the target to be $\theta_{scatt}/2$ for angles $\theta_\alpha \leq 90°$. Here $\theta_{scatt}$ is the defined angle of the silicon detector. For angles larger than 90°, we will set the target angle to be at 0° (i.e., normal to the incident beam flux).

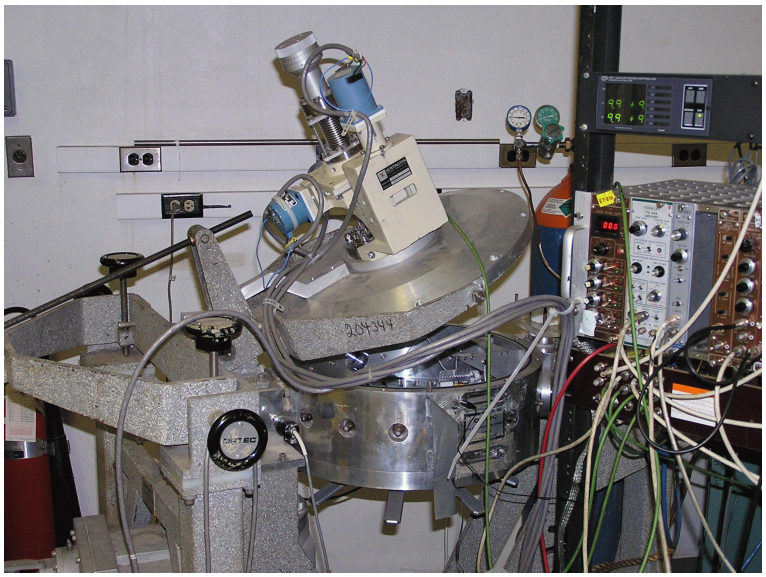
### 6.2.3 Data Acquisition operation

*This section was authored by Andrea Richard in 2016*

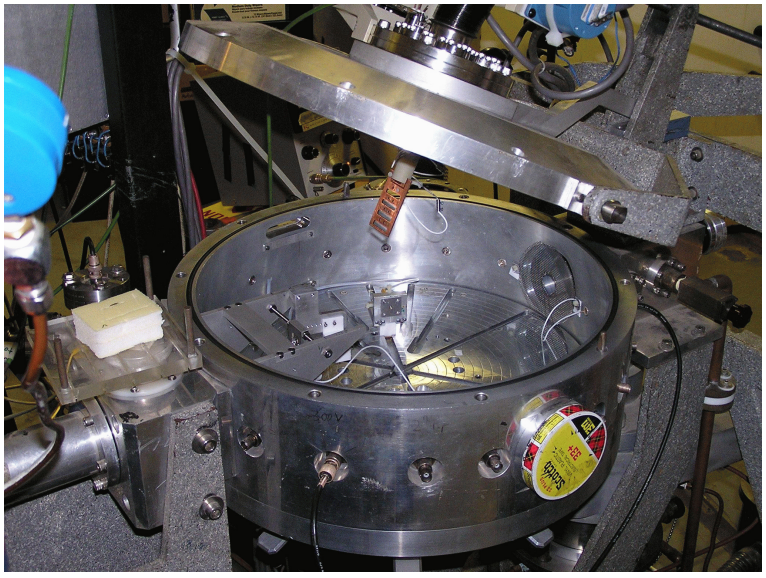
**Do this only at the start of the experiment**

- To initialize the DAQ
  Double-click on the DDAQ_panel icon.
  Click on *"control+"* (green box).
  Click on *"start gonidaq"* (green box).
  Type *"scaninit"*, then enter.
  Type *"4"*, then enter.
  Type *"bci"*, then enter. This allows us to start and stop the DAQ based on preset BCI values.
  Type *"disk #"*, then enter. # is a number that you choose. For example, you may choose DDMMYY or 010323 if you take data on March 1st, 2023.
  Type *"tag 0"*, then enter. This will start the tags at 0. The first data files will be saved under ddaq_ingram/data/disk#/tag0000.
  Type *"ugl"* to write the tags in dat format.

- To initialize the goniometer and move the target.
  Type *go* then enter
  Type *"y"* to initialize.
  Type the (x,y,z) coordinates as read from the goniometer in the Small Target Room.
  After turning, move the foil into place by typing *"goni15"* then enter.
  Type the (x,y,z) coordinates from the alignment.



**View of Scattering Chamber (External)**



**View of Scattering Chamber (Internal)**

**Do this for each run**

- To start a run
  Type *"go"* then enter.
  Type *"y"*, then enter. This clears the data.
  Type a number. This will set the duration of the run in terms of charge on target as measured by the BCI.
  The Beam Charge Integrator (BCI) converts a beam current $10^{-R}$ Amp into a 10 kHz signal (F=10,000). So if you received a $10^{-R}$ Amp beam current for 1 s, the BCI will report F=10,000 counts. Note that the BCI input signal goes through an attenuator P before being converted to a frequency. This means that if the beam on target is $10^{-R}$ Amp, the BCI input signal is only $10^{-R}/P$ Amp. Therefore, the charge on target is BCI $\times$ $10^{-\mathrm{R}}$ $\times$ P/$F$ in Coulomb[6]

  Type *"dt"* then enter. This allows you to check the dead time and run time.
  Type *"runfor"*, then enter to check how much longer the run will last.

- When a run ends
  The output will include:
  *"BCI"*: see comments above to interpret in terms of Coulombs.
  *"Deadtime"*- dl(t) is the PRD deadtime, dt(r) is the RBS detector deadtime.
  *"Runtime"*
  Click *"savetags"* (white box) to save the histograms. The data are saved under ddaq_ingram/data/disk#. Where # is the number you choose at the beginning of the experiment.

**To get the files off the DAQ computer**

- Open a regular Terminal window.

- First, you need to decode the data to the format you can easily read: Type *"cd ddaq_ingram/data/disk#"*, then type *"dat2xy fffff.dat"* where ffff is the filename. Hint: mind the space between the words.

- To transfer the data file to the thumb drive,
  First, you want to mount the thumb drive to the computer: type
  *"sudo mount /dev/sda1 /mnt/flash"*. Note that "flash" might not be the good word, depending on which port you plug your thumb drive into. You might have to use something like "usbmnt" for example. The password is "edwards".
  Next, you want to transfer the files to the thumb drive. You may use the icons from the X-display or type *"cd ddaq_ingram/data/disk#"* where # is the disk number you choose earlier, then type

[6] If $R = -7$ and $P = 100$, a 1nA beam reports 1 BCI counts per second.

*"sudo cp ffff.xy /mnt/flash/."* where ffff is the filename.
Finally, you want to un-mount the thumb drive, type *"sudo umount /dev/sda1"*.

## 6.3 Experiment goals and law under study

The Rutherford scattering cross-section varies with the beam energy, the scattering angle, the target nucleus, and the projectile nucleus. For this experiment, you will work with a foil of total number density $8 \times 10^{17}$ atoms/cm$^2$. The target you will be using is roughly mostly gold with a bit of Silver and traces of Copper.

Perform measurements of the number of $\alpha$ particles detected (yield) as a function of the scattering angle or the beam energy with both silicon detectors (LRD and RBS). Measure the yields to at least 5% statistical precision. The class might be split into two groups: one measures the variation of the cross-section versus angle using the largest beam energy available, while the other measures the cross-section versus beam energy.

1. Notice how the position of the ADC peak varies with angle or beam energy. Perform an energy calibration of the ADC using these peak positions and verify that the peak positions appear where the kinematic relationship says they should be. Make sure to use your backward Gold, Silver, and Copper peaks in plus of your peaks versus energy or angle.

2. Using your Rutherford Back Scattering measurement at high energy and assuming that you indeed measure Rutherford scattering, confirm the composition of the foil in terms of gold, silver, and copper.

3. Measure the Rutherford cross-section versus the angle or energy and compare it to the theoretical one. You will have to decide how to deal with the Gold-Silver-Copper composition.

## 6.4 Preliminary questions

The following questions should help you prepare for this lab:

1. The physical principle under study is "Measurement of the Rutherford Scattering cross-section." Write a 200-300 words description of this concept and how it relates to your measurements. This is in preparation for your report/presentation, for which you will be expected to discuss this issue.

2. What is the kinetic energy of alpha recoiling off Gold, Silver, or Copper atoms if the kinetic energy of the incident alpha is 2 or 6 MeV and the scattering angle is 45 or 135 degrees? You have to compute 3 atoms $\times$ 2 beam energy $\times$ 2 angles = 12 values.

3. Using a computer graphing software, draw the cross-section variation for a pure Gold target for a constant angle of 60 degrees and beam energy varying between 2 and 6 MeV. Draw the cross-section variation for a pure Gold target for a constant beam energy of 4.5 MeV and an angle varying between 25 and 155 degrees.

4. Suppose you place a Silicon detector of area $1\text{mm}^2$, 10 cm away from a pure Gold target of number density is $8 \times 10^{17}$ atoms/$\text{cm}^2$ and at an angle of 60 degrees concerning the beam direction. Suppose you bombard that target with a 4.5 MeV alpha beam of $1nA$ for 1 minute. How many alphas will you detect in the Si detector?

5. How does your previous result change if you are working with a target foil made up of 80% Gold, 15% Silver, and 5% Copper?

## Reference

# References

[1] D.G. Sargood, Physics Reports **93**, No. 2, 61-116 (1982).

# A Edwards Accelerator Data Acquisition System – Background Information

The Data Acquisition (DAQ) used for this lab was developed by engineer D. Carter of OHIO University. The DAQ runs on Raspberry Pi computers that are labeled "EAL PIDAQ". The full documentation of the DAQ system is available at daqlinux.com

- To **start the DAQ system:**
  1. Click on the PiDAQ icon. A long and thin GUI[7] opens.
  2. On the GUI, click on "control+", and select "start daq". A daq terminal window opens. From there, you may run the DAQ using the GUI or issuing commands in the daq terminal window.
  3. On the GUI, click "acquire" on the GUI or type "a" in the daq terminal window.
- To **look at the data** while they are acquired:
  1. On the GUI, open the pull-down menu titled "plot+". The selection of plots available depends on the version of the DAQ system you are running and the need of your experiment. You may also type "plot ABCD" in the daq terminal window. "ABCD" is the name of the histogram you want to look at.
  2. On the daq terminal window, type "q" for the list of defined histograms. This list might be more expansive than what is listed in the "plot+" pull-down menu.
  3. On the newly open window plot, you should see the histogram grow and the number of events "Sum" on the top right of the window increase. Note that at any time, only one histogram at a time is updated in real-time. An histogram not being updated real-time bear the label "STATIC".
- To **save the data to disk**:
  1. Stop the data acquisition. Click "Stop" on the GUI or type "s" in the daq terminal window.
  2. Specify the disk(folder) where you want to save the data. On the daq terminal window, type "disk ####", where "####" is a 4-digit number.
  3. Specify the run tag (file name) you want to save the data into. On the daq terminal window, type "tag ####", where "####" is a 4-digit number. From now on, each time you will save the data, the tag number will automatically be incremented by 1.

[7] Graphical User Interface

4. Save the data on disk. Click "SaveTag" on the GUI or type "sav" in the daq terminal window.
5. Your data are saved in the ∼/Desktop/ugl1/data/dsk####/tag$$$$.dat file[8]. "####" is the disk number you chose in the previous steps and "$$$$" is the tag number.
6. If your choice of disk and tag is such that you are trying to overwrite a file, the daq system will warn you with an error message in the daq terminal window. Instead of saving your data in file tag$$$$.dat , the data will be saved in tag$$$$.dat.1 (or .2, .3..).
7. The file ∼/Desktop/ugl1/data/dsk####/daq.log documents the saving operation. Specifically, it lists which histograms are saved in which tag file.
8. You need to convert the data from a daq format to a two-column wise file. In a regular terminal window[9], navigate to the directory where your data files are saved. Type " cd ∼/Desktop/ugl1/data/dsk####" where #### is the number you chose previously. Once this command was accepted, in the regular terminal, type "dat2xy tag####.dat" where "####" is the tag number. The new file is "tag####.xy".

- To **run the daq for a specific amount of time**:.

1. Make sure the daq is not taking data, and the histograms have been cleared. Click "Stop" and "Clear" in the GUI or type "s" and "clear *" in the daq terminal.
2. Type "time XXX" in the daq terminal window with XXX being the duration of your data taking in seconds.
3. Click or type "Run".
4. If you change your mind and want to stop the data taking that was launched for a certain duration, click or type "Abort" to stop the computer timing.

[8]To make this fun, it seems that not all machines are using "ugl1" so for the machine you work with it might be "ugl3" or some other numbers...

[9]This regular terminal window has a black background. The DAQ terminal has a white background.

# B General Characteristic of Detectors

*This summary was put together by Prof. Krauss from the University of Durham in the UK. It follows closely Chapter 5 of the excellent book by W.R. Leo (ISBN 3-540-17386-2)*

Detectors used in elementary particle and nuclear physics are based on the principle of transferring radiation energy to detector mass. Charged particles transfer their energy through collisions to atomic electrons leading to excitation and ionization. In most cases, neutral particles have to produce charged particles first inside the detector volume, which in turn transfer their energy by excitation or ionization to the detector. All these interaction processes are random processes. There is no detector that is perfectly suited for all possible applications. Depending on the application, detectors and complex detector systems are designed according to several criteria. There are several characteristics specifying the features of a detector:

## B.1 Sensitivity

The first question arises whether the detector is sensitive to the type of radiation and energy range under consideration. The detector sensitivity depends on the following factors:

- the interaction cross-section with the detector material,
- the detector mass,
- the detector noise,
- the protection material around the detector

Ex: If the interaction cross section is small (e.g. for neutrinos scattering on a detector material) one needs a large detector mass to reach reasonable sensitivity if the incoming particle flux can not be significantly increased.

## B.2 Detector Response Function

Related to sensitivity is the detector response (function) to the radiation under study. Usually, the output signal of an electrical detector is a current pulse where the time integral of the signal corresponds to the amount of ionization produced by the particle-detector interaction. If the shape of the signal does not depend on the amount of ionization, the amplitude of the signal is a measure of the radiation energy deposited in the detector. If this relation is linear, the response of the detector is called to be linear.

A particle of defined energy may lead to a spectrum of signal amplitudes. This is called the detector response function. For example, photons with definite energy may interact with the detector material by Compton scattering, resulting in a broad spectrum of deposited energies due to the subsequent interaction of the recoil electrons. In contrast, charged

particles with definite energies that are stopped within the detector material will rather lead to a Gaussian distribution of signal amplitudes.

## B.3 Energy resolution

Suppose the detector is designed to measure the energy of a particle. Usually, due to fluctuations in the number of excitations and ionizations in the detector material, one observes a Gaussian-like peak for a monoenergetic particle beam instead of an ideal delta-function peak. The width of this peak determines the capability to distinguish particles with different energies. The energy resolution $\Delta E$ is given by the full-width-at-half-maximum (FWHM) of the signal peak. For a Gaussian distribution with standard deviation $\sigma$, we have

$$\text{FWHM} = 2.35\sigma$$

Energies closer than this resolution can not be separated. The relative energy resolution is given by

$$\Delta E/E$$

Generally, the average energy needed to produce an ionization is a constant and only depends on the material. Consequently, the average number of ionizations increases with the deposited energy. As the number of ionizations fluctuates according to a Poissonian distribution the relative energy resolution scales like

$$\frac{\Delta E}{E} \propto \frac{1}{\sqrt{E}}$$

## B.4 Detector Efficiency

The total efficiency of a detector is defined by the fraction of events registered at the detector with respect to the number of events emitted by a radiation source:

$$\epsilon_{tot}$$

If the mean free path for interaction with the detector material is much smaller than the actual detector length, then the total efficiency can be written as the product:

$$\epsilon_{tot} = \epsilon_{int} \times \epsilon_{acc}, \tag{44}$$

between the intrinsic efficiency and the geometrical acceptance of the detector. The intrinsic efficiency is given by the fraction of events registered by the detector with respect to the number of events hitting the detector. The geometrical acceptance generally depends on the polar and azimuthal angle of the emitted particle. If a source emits particle isotropically and a detector of active surface area $S$ is placed at a distance $R$ of the source, then the detector efficiency is

$$\epsilon_{acc} = \frac{S}{4\pi R^2}$$

It may also depend on the particle's momentum (charged particles below a certain threshold momentum might be stopped due to energy loss before reaching the detector). The acceptance can also differ between particle types: the detector device used to reconstruct charged particles may cover an angular range different from the detector device used to detect neutral particles. In many cases, the intrinsic efficiency as well as the geometrical acceptance, are determined by a Monte Carlo simulation.

## B.5 Response Time

This is the time between the arrival of the radiation and the formation of an output signal. If the signal is formed on a very short time scale with a fast-rising flank, the signal can mark a precise moment in time. This characteristic is of importance if timing information is crucial, e.g., in Time-Of-Flight measurements with scintillators or in space determinations using a drift chamber through drift time measurements. The duration of the signal is also important if dead time or a pile-up of subsequent signals may become important, as discussed in the next subsection.

## B.6 Dead time

Related to efficiency is the dead time of a detector. The energy (charge) deposition process and the 'readout' of the information takes a finite time in which the detector and its associated electronics cannot register a subsequent signal. Depending on the type of detector and the rate of particle interactions with the detector material, the issue of dead time can be rather important.

These events are lost if the detector is insensitive to other events during the 'readout' time. If the detector is sensitive to additional events during the 'readout' time, these events may pile-up resulting in a signal distortion.

In experiments with large particle fluxes, dead time can be reduced by designing the detector with a high granularity such that the occupancy per detector cell is reasonably small. The occupancy measures the number of particles traversing a detector cell per event. This will, however, increase the costs of building the detector and its readout electronics.

Pile-up can be reduced by shaping the detector signal such that a very short and as a consequence, a small signal is formed. This, however, is limited by the inherent noise of the detector.

# C Counting statistics

*This section is copied from this website.*

You will observe that when counting random events, the actual number of counts recorded ($N$) is often very different from the average number of counts recorded ($\bar{N}$). Approximately 1/3 of the time the actual number of counts $N$ will differ from the average number of counts N by more than the standard deviation $\sigma = \sqrt{N}$. The relative standard deviation $\sigma/N = 1/\sqrt{N}$ decreases as $N$ increases. This means that when counting a large number of random events the percentage difference between the actual number of counts $N$ and the average number of counts $\bar{N}$ is likely to be much smaller and the measurement will have a much higher statistical significance than when counting a small number of random events.

## C.1 Random events

There are many types of events that occur at a well-defined average rate, but for which we cannot predict the outcome of any particular measurement. For example, a light bulb has an average lifetime. But we cannot predict how long a particular light bulb will last. This is a characteristic of a random process. A random variable is a function that associates a unique numerical value with every outcome of an experiment. The value of the random variable will vary from trial to trial as the experiment is repeated. For the light bulb, the experiment is measuring its lifetime, and the random variable is the number of days it produces light before it burns out.

The average number of rainy days in May in Nashville is 11. We cannot predict how many rainy days Nashville will have next year in May and on which days it will rain. If we record the number of rain days next May, the new number will most likely differ from that of the preceding May. The number we will record cannot be predicted using the knowledge of a preceding observation nor can it be used to predict the following observation.

Radioactive decay is a random process. We cannot predict exactly when a certain unstable nucleus will decay, we can only predict the probability that the nucleus will decay in a certain time interval. Similarly, we cannot predict exactly how many decays will take place in a particular radioactive sample in a particular time interval, but only the average number of decays. If we actually count the number of decays in several time intervals, we get a different number of counts each time, but they have a more or less definite mean.

## C.2 Statistical distributions

Assume we count the number of times N a given random event occurs in a certain time interval. If we repeat our experiment $m$ times, we will find the numbers $N_1$, $N_2$, $N_3$, ..., $N_m$. We can calculate the average number of counts

$$\bar{N} = \left( \sum_{1}^{m} N_i \right) /m. \tag{45}$$

The variance of these measurements is

$$\sigma^2 = \frac{\sum_1^m (N_i - \bar{N})^2}{m} \tag{46}$$

Variance is the square of the Standard deviation. Physicists also call the Standard deviation the Root Mean Square (RMS). Note that this denomination is specific for Physical scientists; see this article.

The actual number of counts will be distributed around this average value. Numbers close to the average will be recorded frequently, and numbers very different from the average will be recorded infrequently. If we want to know how the actual number of events counted in a time interval $\Delta t$ is distributed around the average number of events counted in $\Delta t$, we can make a histogram of the number of times a certain number $N$ appears.

### C.2.1 Rare event distribution: Poisson statistics

For a large number of rare events, we find that the probability of recording a particular number $N$ is given by the **Poisson distribution**:

$$P(N) = \bar{N}^N e^{-N}/N! \tag{47}$$

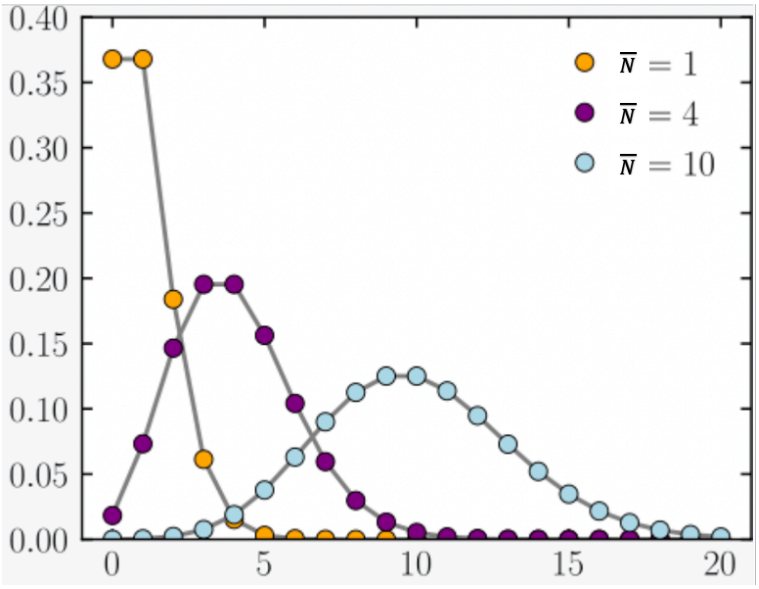



Figure 37: *Poisson distributions. Each graph shows the normalized distribution of counts when the average number of events is 1 (orange), 4 (purple) or 10 (blue).*

The Poisson distribution describes a wide range of phenomena in the sciences. It describes the probabilities of random occurrences and it can be applied to "intervals" on the space or time axes.

### C.2.2 Frequent events: Gaussian statistics

For a large number of frequent events, the probability of recording a particular number N is given by the Gaussian distribution:

$$P(N) = \frac{1}{\sqrt{2\pi}N} e^{-(N-\bar{N})^2/2N} \tag{48}$$

For a large $\bar{N}$ the Poisson distribution approaches the Gaussian distribution.

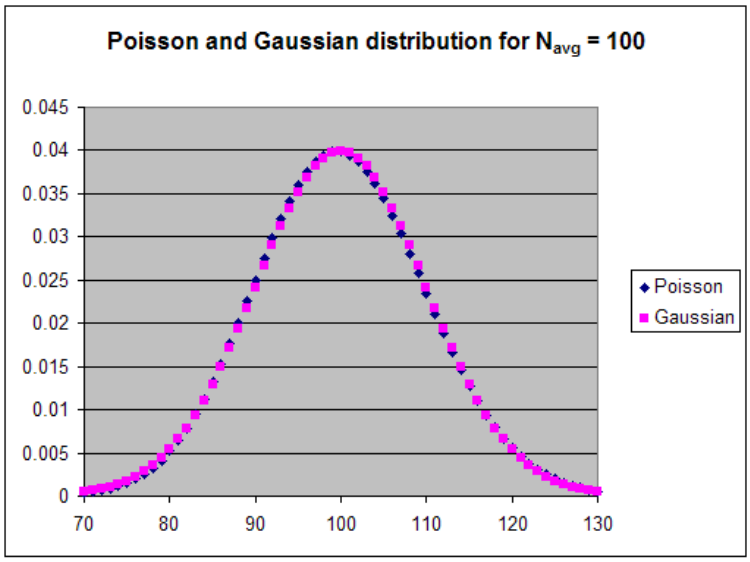



Figure 38: *Gaussian versus Poisson distributions. These histograms show the normalized distribution of counts when the average number of events is 100. Both distributions are very close to each other.*

Statistical theory predicts that the standard deviation of the Gaussian distribution is given by $\sigma = \sqrt{N}$. The standard deviation $\sigma$ is a measure of the width of the distribution. Approximately 1/3 of the counts will lie outside the interval $N - \sigma$ to $N + \sigma$.

For a series of $m$ measurement $N_i$, the standard deviation is calculated using

# D Silicon Detectors

Semiconductor charged particle detectors have been used extensively in experimental nuclear physics research for over 30 years, and have revolutionized nuclear particle detection. Silicon detectors can be used to measure a wide range of charged particles. This range includes protons and electrons as low as 20 keV up to fission fragments of energy over 100 MeV. The inherent resolution of ion-implanted and surface barrier detectors is surpassed only by magnetic spectrometers. The detector output pulses rise rapidly, hence they are well suited for fast ($\sim$1 ns) timing with coincidence circuitry or time-to-amplitude converters.

The efficiency of silicon charged particle detectors for their active volume is essentially 100%, and their energy vs. pulse height curves are linear over a rather impressive range. They also have good long-term pulse height stability. This is particularly noticed when they are contrasted with scintillation counters, gas proportional counters, or ionization chambers. The high efficiency of these detectors allows for straightforward analysis of cross-sections via the extraction of detected particle yields from the accumulated spectra (provided data acquisition/computer dead times are accounted for accurately).

Solid-state detectors can be thought of as ionization detectors. When a charged particle enters the depletion region of the detector, it loses energy primarily by making electron-hole pairs in the silicon. For each electron-hole pair that is made, the initial charged particle must lose 3.6 eV. In this experiment, we will be measuring alpha particles that have an energy of less than 3 MeV. If a 3 MeV alpha particle enters the detector, roughly $8\times10^5$ electron-hole pairs will be produced. The detector is reverse-biased and these electron-hole pairs are collected to produce the output pulse of the detector. Since a large number of charge carriers is produced, the statistical variation in the number collected is small, and hence very good energy resolution is possible. The "partially depleted" silicon detector used in this experiment has a resolution of roughly 20 keV.

The three main parameters that define a silicon surface-barrier detector are its resolution, active area, and depletion depth. The shape of a typical detector is a circular disk. Thus the active area is simply the area of the face of this disk (provided it is placed normal to the incident flux of radiation). The depletion depth of a detector is synonymous with the sensitive depth of the detector. For any given experiment, this depth must be sufficient to completely stop all the incident charged particles that are to be measured. The detector's ability to do this is dependent upon both the energy and the particle type. Fig. 39 is a range-energy curve for five of the more common charged particles. From it, the maximum depth can be determined for the maximum energy of a particle type. From Fig. 39, note that a 3 MeV alpha particle is completely stopped with about 20 $\mu$m of silicon.

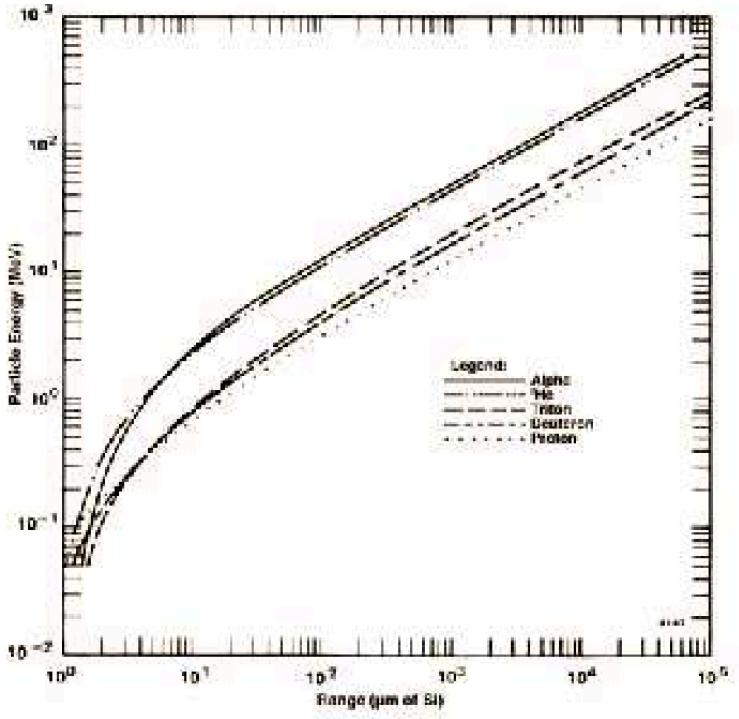



Figure 39: *Range of different charged particles traveling in Silicon. This picture is taken from C.F. Williamson et al., "Range of Stopping Power of Chemical Elements for Charged Particles of 0.5 to 500 MeV", CEA-R-3042, July 1966.*

# E Germanium Detectors

*This section is copied from the MIRION article on High-Resolution Gamma-Ray Spectroscopy with HPGE Detectors*

For semiconductor detectors, the thickness of the depletion region (depletion depth) d is given by the following equation:

$$d = \left(\frac{2\epsilon V}{eN}\right)^{1/2}$$

where: $\epsilon$ is the dielectric constant.
$V$ is the reverse bias voltage.
$N$ is the net impurity concentration in the semiconductor material
$e$ is the electron charge.

For gamma-ray spectroscopy, a large depletion region is required for complete charge collection and increased efficiency. Here for a given reverse bias voltage, the value of N needs to be reduced as much as possible. The high-purity germanium detectors that are being manufactured today can achieve net impurity concentration as low as $0.8 \times 10^{10}\text{cm}^{-3}$.

In insulators, the band gap between the valence band (fully occupied) and conduction band (completely empty) is high, and no electric field or temperature rise can provide enough energy for electrons in the valence band to cross the gap and reach the conduction band (Figure 40). However, in semiconductors, the band gap is small, and increased temperature, incident charged particles, or applied electric field can impart enough energy to electrons to move them from the valence band to the conduction band. When an electron moves from the valence band to the conduction band, a hole is created in its place in the valence band. Holes behave like particles with positive charge, and they contribute to the overall conductivity of the semiconductor.

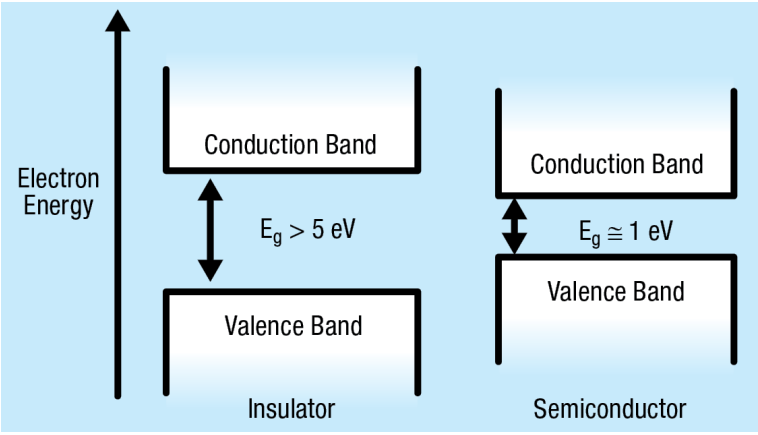



Figure 40: *Band structure in insulators and semiconductors. Eg is the band gap energy.*

**High-purity Germanium (HPGe) detectors** Germanium detectors are semiconductor diodes having a p-i-n structure (P-type contact, intrinsic layer, and N-type contact), in which the intrinsic (i) region is sensitive to ionizing radiation, particularly x-rays and gamma rays. Under reverse bias, an electric field extends across the intrinsic or depleted region. When photons interact with the material within the depleted volume of a detector, charge carriers (holes and electrons) are produced and are swept by the electric field to the p and n electrodes. This charge, which is in proportion to the energy deposited in the detector by the incoming photon, is converted into a voltage pulse by an integral charge-sensitive preamplifier. For a coaxial detector (Figure 41) the rectifying contact that forms the semiconductor junction is typically placed at the outer surface of the crystal. Thus, the outer contact for an n-type HPGe will be p+ and the inner surface will be n+. The depletion layer for such a detector grows inwards as the voltage is increased.

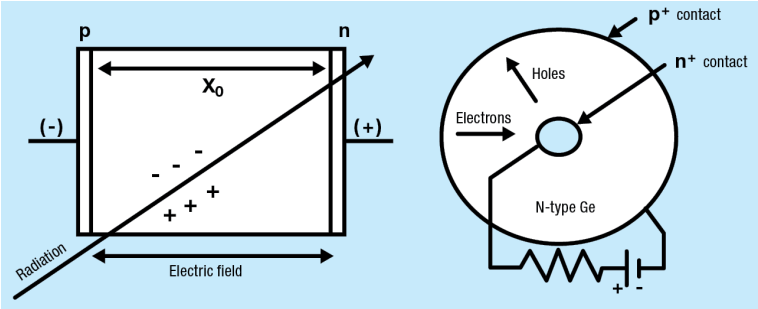



Figure 41: *A typical p-i-n junction detector (left) with reverse bias. A cross-section of an n-type coaxial detector perpendicular to the cylindrical axis (right).*

Since Germanium has a relatively low band gap, these detectors must be cooled to reduce the thermal generation of charge carriers (thus reverse leakage current) to an acceptable level. Otherwise, leakage-current-induced noise destroys the energy resolution of the detector. Liquid nitrogen, which has a temperature of 77K, has traditionally been the cooling medium for such detectors. Many modern detectors are supplied with an electrical cooling system, thus removing the need for liquid nitrogen. There are two disadvantages to semiconductor detectors. First, high-purity or lithium-compensated crystals cannot be made as large as NaI, so it is not possible to make detectors with the same efficiency. Second, the semiconductor detector must be cooled to near-liquid-nitrogen temperature when in use, which adds to the cost and complexity of the system.

**Energy resolution** The energy resolution of a germanium detector can be described as:

$$FWHM = \sqrt{w_d^2 + w_e^2} \tag{49}$$

where:
$w_d$ is the peak width due to detector effects, and

$w_e$ is the peak width due to electronic effects.

The width $w_d$ is energy-dependent. A simplified expression of $w_d$ reduced to :

$$w_d \text{ (in keV)} = 1.44\sqrt{E(\text{MeV})}$$

where:
$E$ is the energy of the photon incoming onto the detector. In the case of a Germanium detector, a large number of electron-hole pairs is produced, which results in good statistics from the charge collection and, therefore a good energy resolution.

The width $w_e$ is dependent on both the detector capacitance, which in turn depends on the size of the detector and the bias voltage. Overall the resolution improves with decreasing capacitance.

In a typical broad-energy detector, the resolution is 0.4 keV and 2.0 keV at 5.9 keV and 1332 keV, respectively.

# F Scintillators

One method for the detection of ionizing radiation is based on materials (known as *scintillators*) that convert the ionization energy into visible light. Scintillators are as common as the video display, where the image is a result of an electron beam striking "phosphors" on the inside face of the CRT. For particle detection, and particularly detection with energy measurement, the scintillators of choice are certain inorganic crystals doped with small amounts of "activators" in a strategy vaguely reminiscent of the use of dopants in semiconductors. In this laboratory, we use NaI(Tl), sodium-iodide doped with thallium.

In the pure crystal, electron-hole pairs are created as the ionization promotes electrons from the valence to the conduction band. The decay of the electrons back to the valence band is inefficient. The role of the activator is to create sites in the crystal energy levels in the forbidden region, close to the conduction band. Electron holes in the valence band drift to the activators and ionize them. The free electrons find the ionized sites and drop in, creating excited activator states. If these decay quickly to visible photons ($t < 500$ ns) the result is called *fluorescence*; this is the most useful component of the scintillation. Other metastable states require additional perturbations, say via thermal energy to de-excite, and lead to a slower release of light, called *phosphorescence*. In other cases, the excited activator states decay without radiation, and the energy of the ionization is said to be *quenched*. The number of electron-hole pairs that result in a scintillation photon is called the efficiency of the scintillator. The frequency of the scintillation light should be in a regime where the scintillator is transparent and the photomultipliers are sensitive.

NaI(Tl) is commonly used because the large $Z$'s give a good cross-section, it has a large light yield, and is inexpensive to grow into large ingots, enabling large arrays for e.g. medical x-ray detection. It is hygroscopic (being a salt), so the crystals have to be contained in hermetically sealed containers. The benchmark performance specifications for NaI(Tl) are $5{\times}10^4$ ion pairs per MeV of incident particle energy, and a scintillation efficiency of approximately 12%, yielding $4{\times}10^4$ photons per MeV incident.

The large light yield leads to a good energy resolution, but the time constant for the signal in inorganic scintillators is rather long, $\mathcal{O}(200$ ns). In high-rate or timing applications, an alternative family of plastic scintillators is used to reduce pile-up effects and to achieve good timing resolution, respectively. In these materials, excitation and decay of molecular states happen with atomic time scales, uncomplicated by transport effects, and the scintillation pulse widths are of order nanoseconds, with fractional nanosecond resolution on the rise time. However, the light output is lower and the energy resolution is poor compared to the inorganics.

# G Photomultipliers

The photomultiplier (PMT) is a tube device that functions as a one channel image intensifier. Light falls onto the photo-cathode, and electrons are emitted via the photoelectric effect. Some simple electromagnetic optics accelerates these electrons and directs them onto the first of a series of surfaces called dynodes, chosen from materials with good secondary electron emission. The dynodes are arranged mechanically so that adjacent dynodes are within millimeters of each other, and a single high voltage power input is directed down a resistor chain that maintains a potential difference of 50-100 V between each pair of dynodes (see Fig. 42).

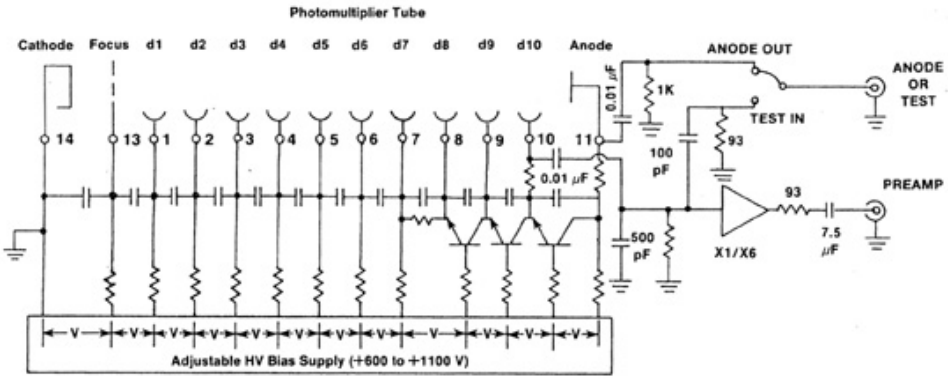



Figure 42: Typical PMT resistor chain in a grounded cathode system.

In this arrangement, the photo-electronics create a larger shower of electrons off the first dynode, and then a chain-reaction of electron emission flows down the tube, amplifying the size of the electron bunch at the dynode. Finally collected at an anode, amplifications of $\mathcal{O}(10^6)$ are typical, so if 1000 electrons leave the cathode, the final charge pulse is $\mathcal{O}(0.1$ nC), well within the input compliance of decent charge integrating amplifiers.

PMT's come in a variety of sizes and designs tuned to different applications. Recent developments include large light-collecting inputs, for use in detecting the Cherenkov radiation of solar and astrophysical neutrinos in huge underground water tanks, multi-channel tubes for processing position-sensitive signals from bundles of scintillating fibers, and tubes that can operate in large magnetic field environments. The main PMT that we use in our experiment is designed to provide appropriate amplification for the signals from the NaI scintillator. In these tubes the cathode is grounded and the anode is at positive high voltage.

The PMT gain is fixed by the dynode design and the applied high voltage (HV). Small changes in HV lead to significant effects in gain, so good HV regulation is important. With the fixed gain, the tube output depends entirely on the number of electrons at the first step, and one of the key performance issues for PMTs is the number of electrons produced per unit photon, this ratio is called the *quantum efficiency*.

Fig. 43 shows the quantum efficiency for several typical photocathode materials. Notice

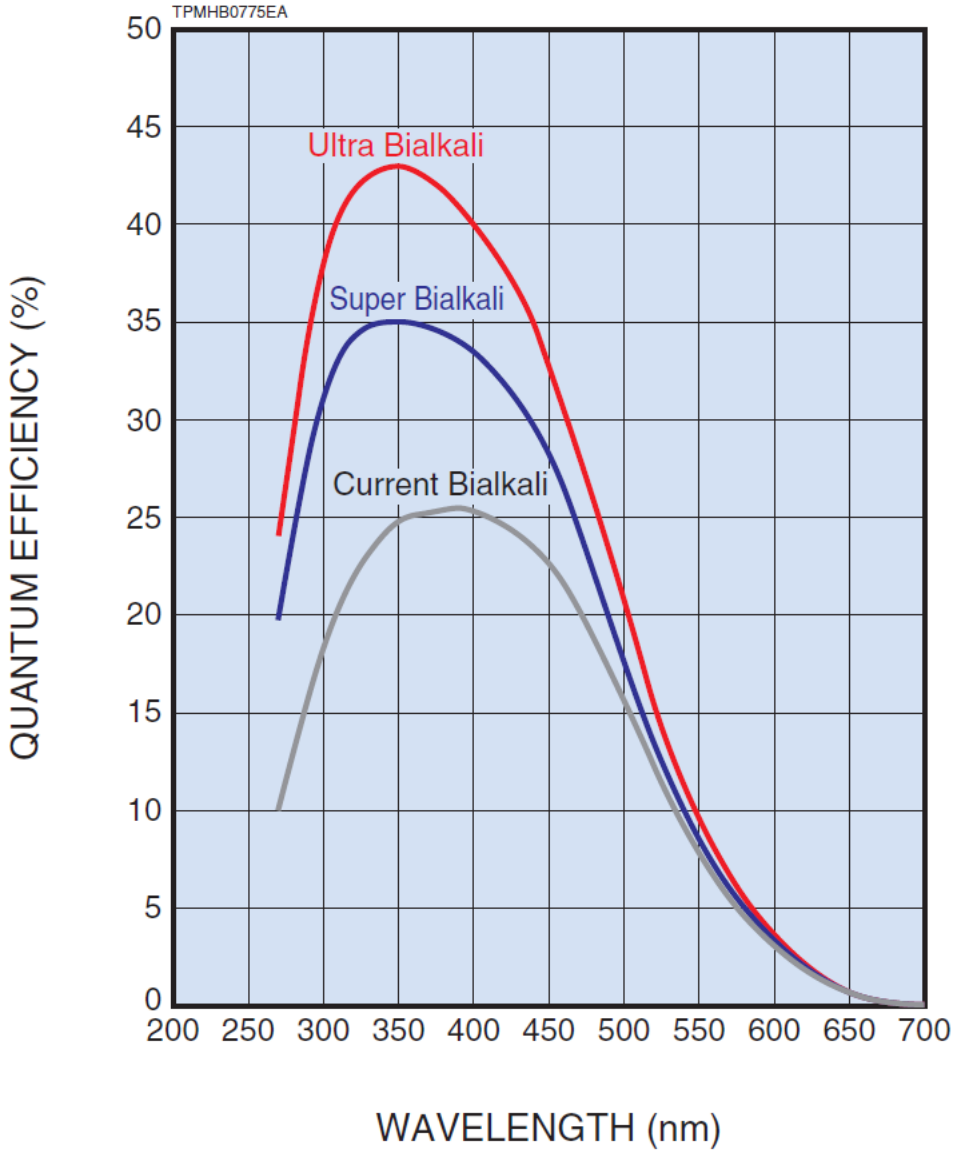



Figure 43: Quantum efficiency of some bialkali photocathodes. From the Hamamatsu Corporation.

that the quantum efficiency varies with frequency. The NaI(Tl) scintillation peaks at 415 nm, so the PMT's employed in this experiment (with bialkali photocathodes) are chosen to match this response. One expects good cathode materials to have quantum efficiencies greater than 20% at the frequencies of interest.